

AI 모델 성능 개선 결과

결론

- 관심 강우+MAE 가 가장 안정적
- 20년 이후 학습 모델: 최신성우수, 22년 이벤트 음수

72시간 예측

72시간 예측 후 뒷부분 날리고 앞 36시간만 예측

72시간 예측

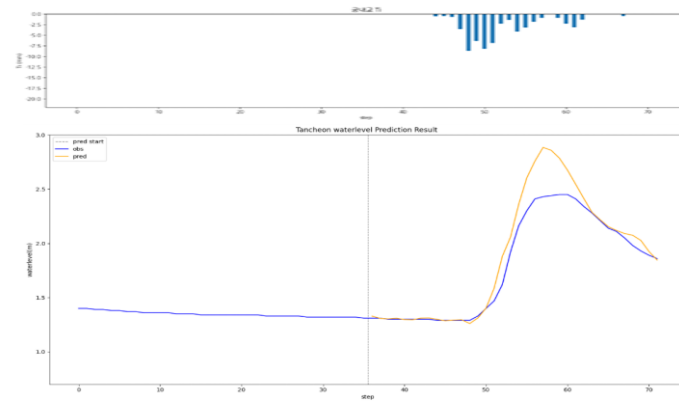
- » 음수 R^2 다수
- » MAE/RMSE 모두 가장 큼

기본 모델 대비 평균 R^2 : 궁내교 ▼저하 (- 62.3%) , 대곡교 ▼저하 (- 33.9%)

궁내교					대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.8759	0.0262	0.0960	0.1619	0.9934	0.0039	0.0495	0.0631
22.07.12	-1.9263	0.1161	0.3101	0.3407	-0.4222	0.1991	0.4156	0.4462
23.07.11	0.7968	0.0568	0.1515	0.2384	0.7761	0.2037	0.3916	0.4514
24.07.23	0.6043	0.0320	0.1075	0.1789	0.2256	0.0368	0.1646	0.1918
25.07.16	0.4520	0.0549	0.1735	0.2344	0.8822	0.0543	0.1855	0.2331
25.08.14	0.8114	0.0132	0.0951	0.1153	0.7106	0.1004	0.2621	0.3169

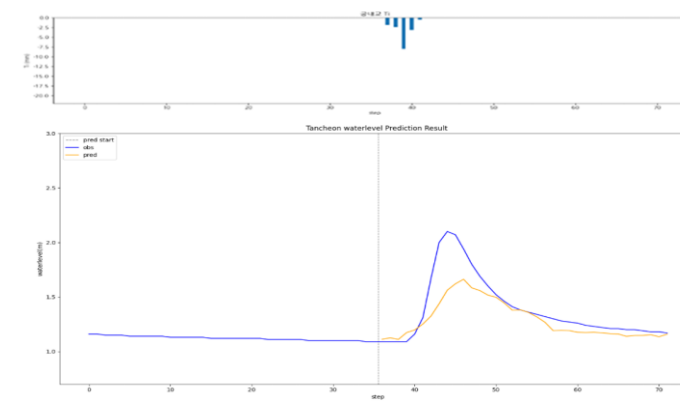
testdata_200802

R2 : 0.8759



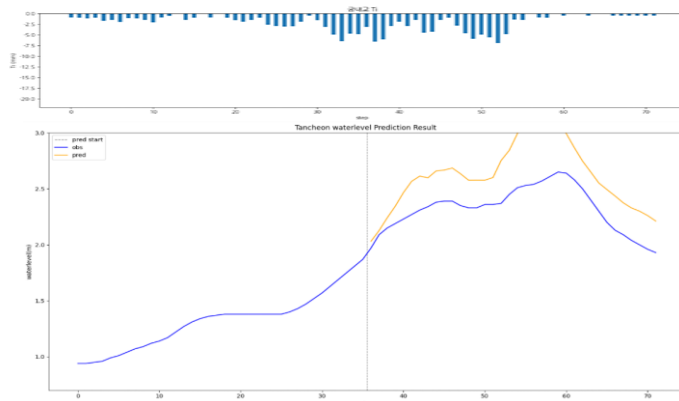
testdata_240723

R2 : 0.6043



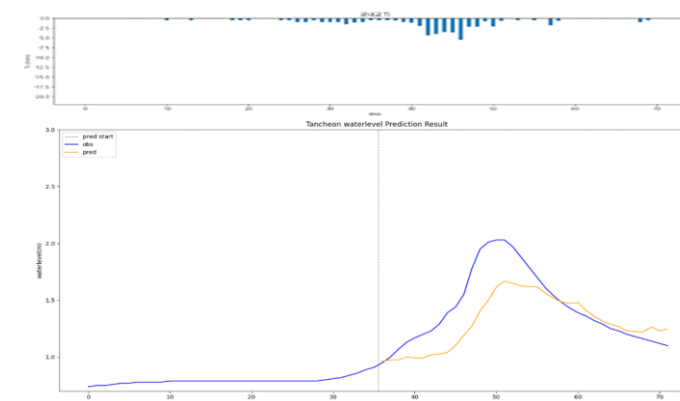
testdata_220712

R2 : -1.9263



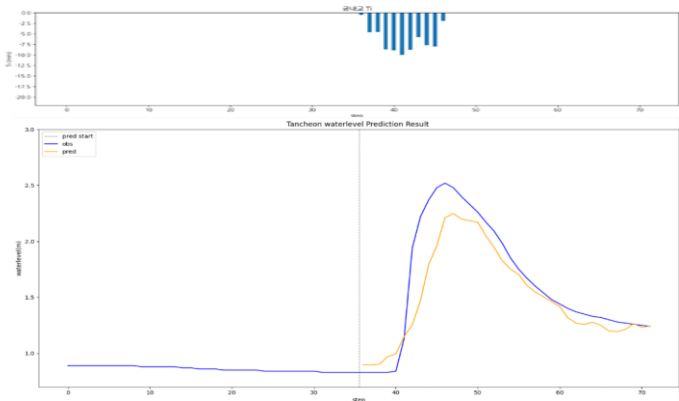
testdata_250716

R2 : 0.4520



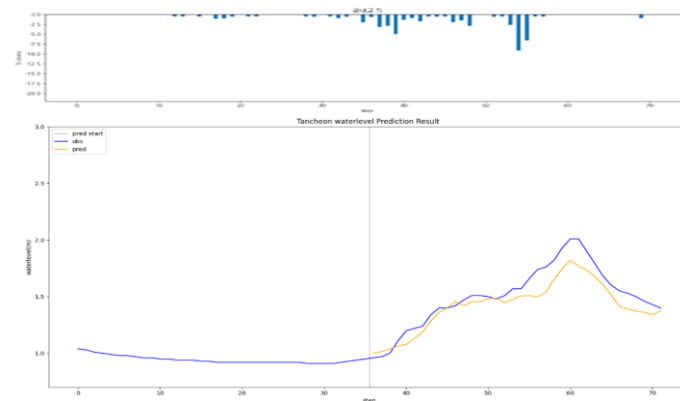
testdata_230711

R2 : 0.7968



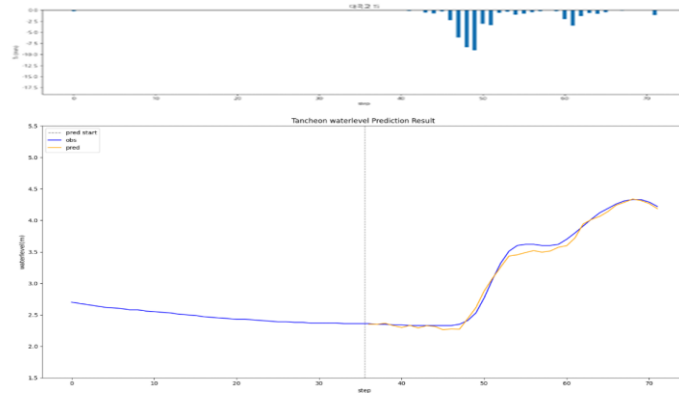
testdata_250814

R2 : 0.8114



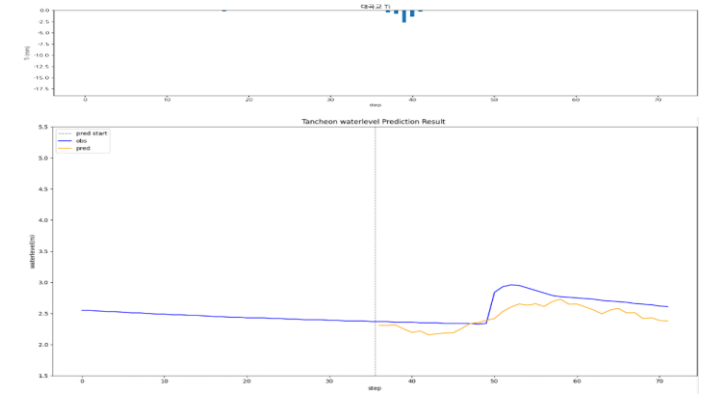
testdata_200802

R2 : 0.9934



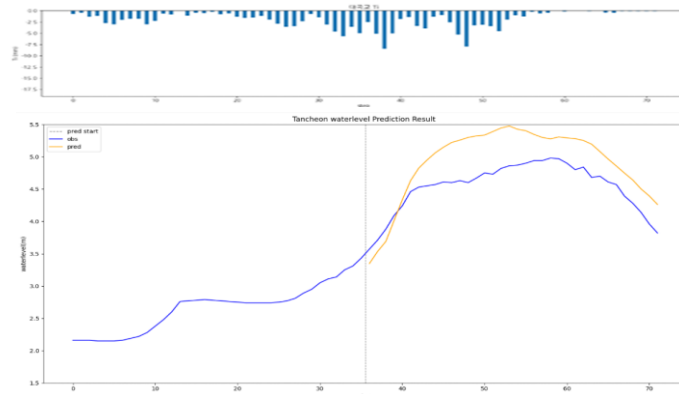
testdata_240723

R2 : 0.2256



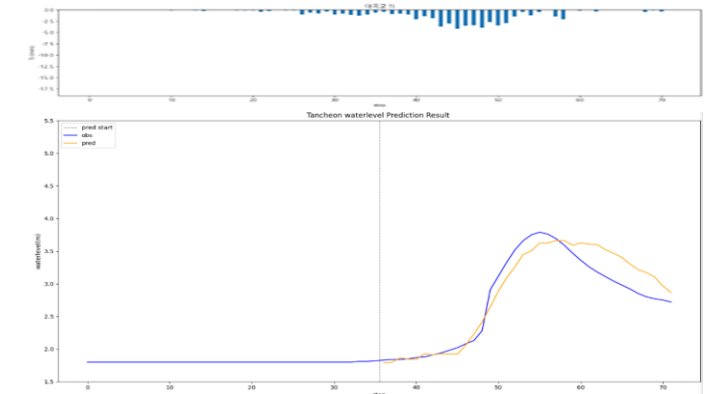
testdata_220712

R2 : -0.4222



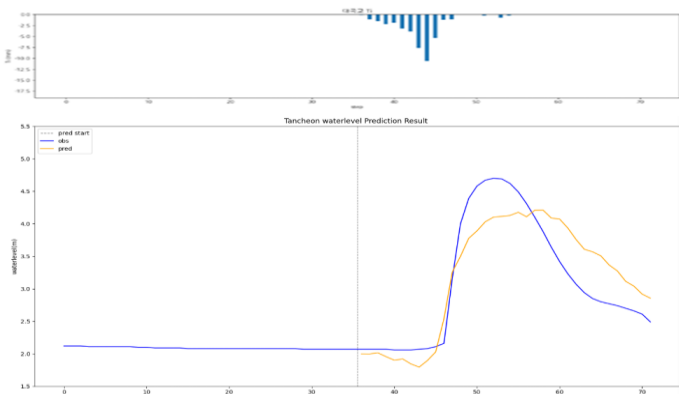
testdata_250716

R2 : 0.8822



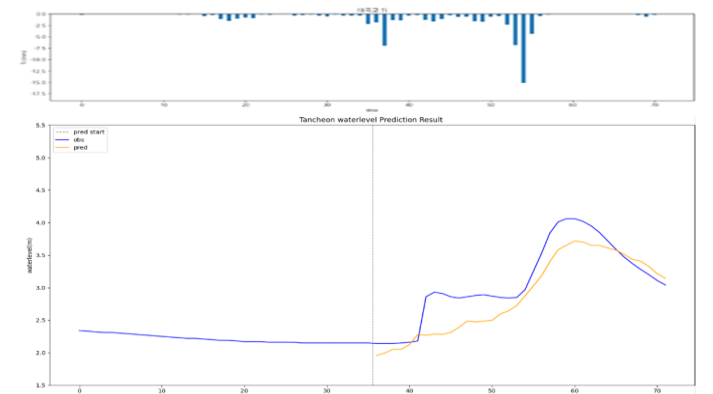
testdata_230711

R2 : 0.7761



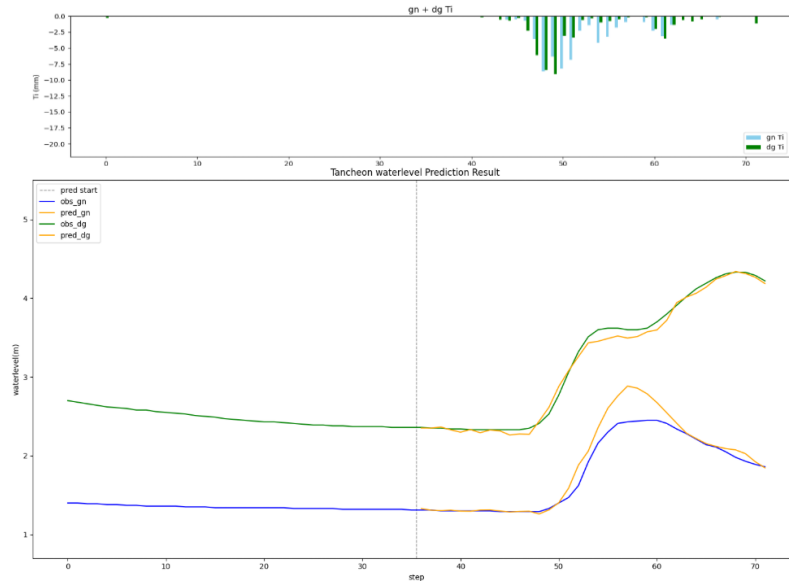
testdata_250814

R2 : 0.7106

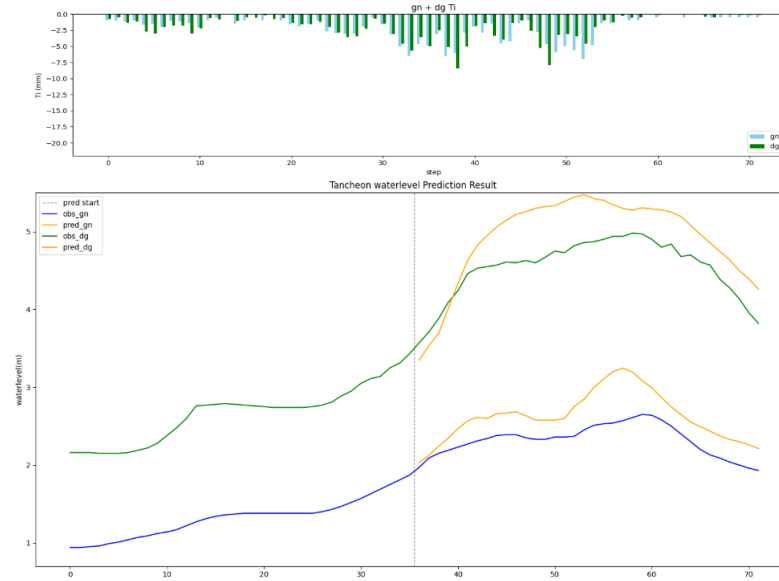


궁내교 + 대곡교

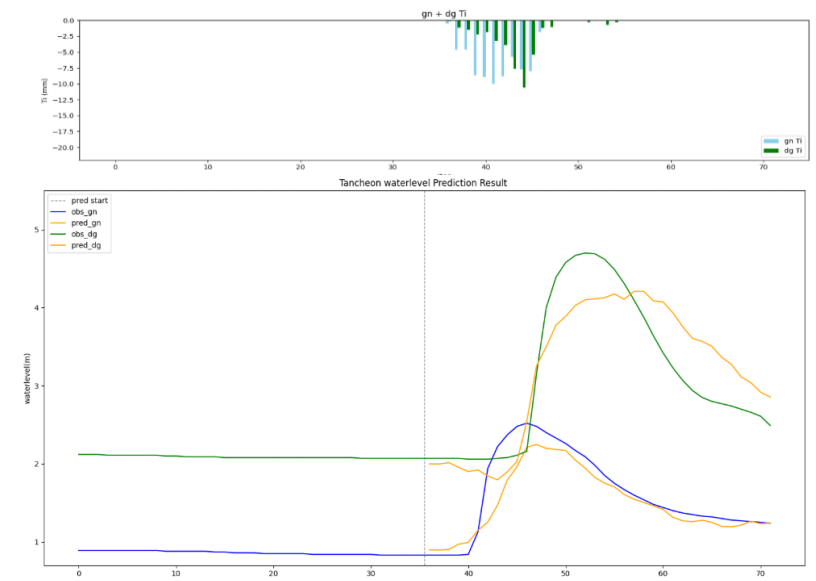
● 궁내교
● 대곡교 * model_sunnam_(gn/dg)72
● 예측



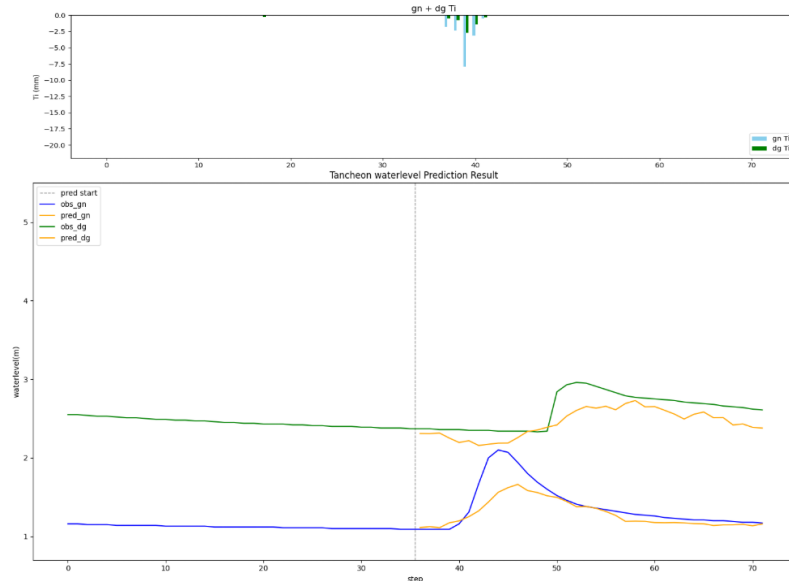
testdata_200802



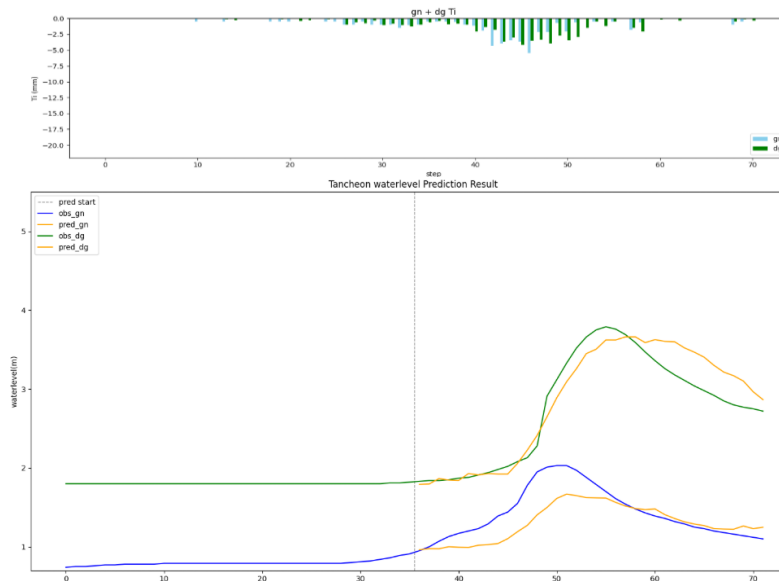
testdata_220712



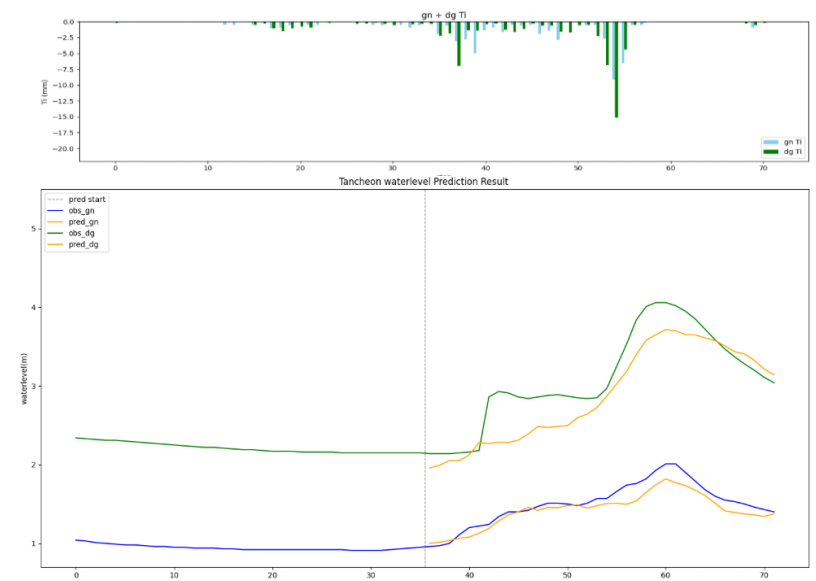
testdata_230711



testdata_240723



testdata_250716



testdata_250814

관심수위 + MAE

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0005
 - Loss_function = MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 67개 中 61개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

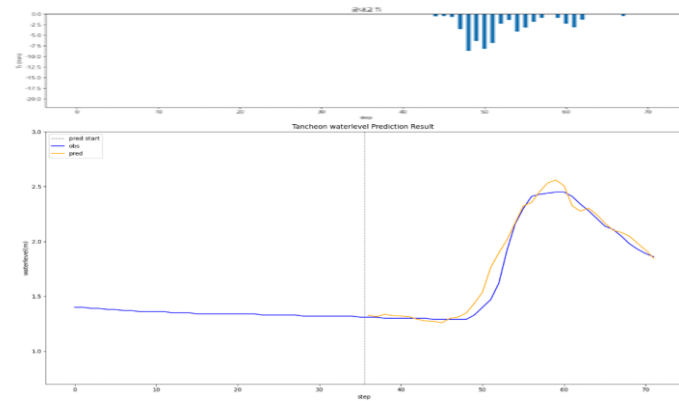
관심수위 + MAE

- » 피크 타이밍 & 크기 가장 정확
- » 기존 모든 실험 대비 가장 안정적
- » 이전 모델 대비 피크값 추적력, 고수위 예측 정확도 향상

궁내교					대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.9656	0.0072	0.0558	0.0852	0.8748	0.0763	0.1754	0.2763
22.07.12	0.4201	0.0230	0.1139	0.1516	0.5503	0.0629	0.2047	0.2509
23.07.11	0.8622	0.0385	0.1486	0.1963	0.8659	0.1219	0.2258	0.3492
24.07.23	0.7308	0.0217	0.0962	0.1475	0.6749	0.0154	0.0860	0.1244
25.07.16	0.7947	0.0205	0.1047	0.1435	0.9470	0.0244	0.1098	0.1563
25.08.14	0.5087	0.0346	0.1446	0.1861	0.8781	0.0423	0.1589	0.2056

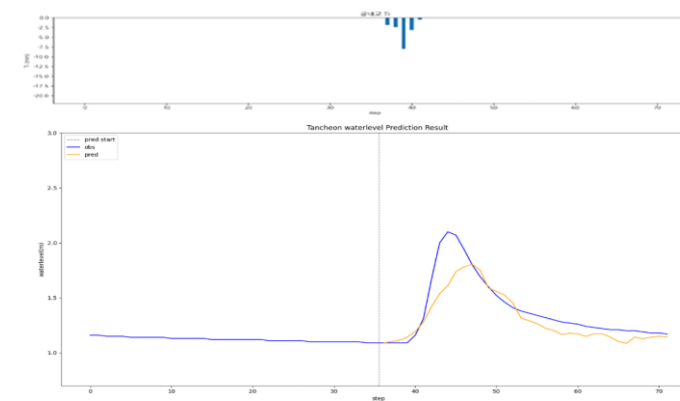
testdata_200802

R2 : 0.9656



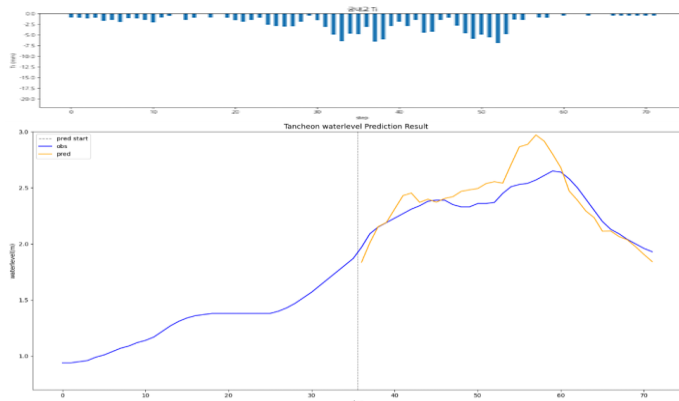
testdata_240723

R2 : 0.7308



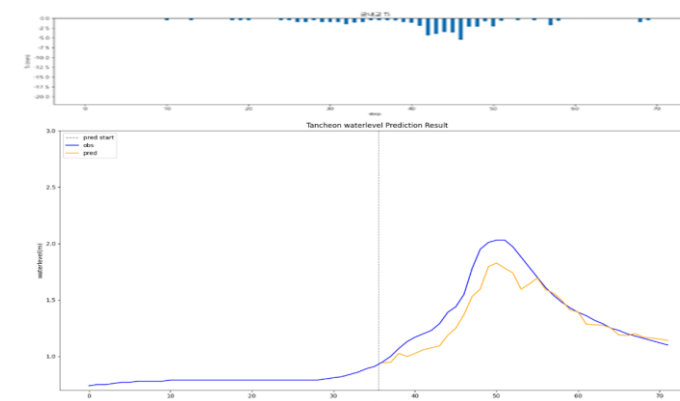
testdata_220712

R2 : 0.4201



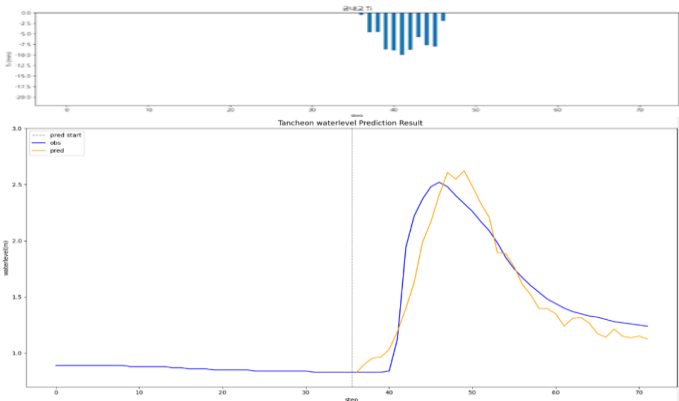
testdata_250716

R2 : 0.7947



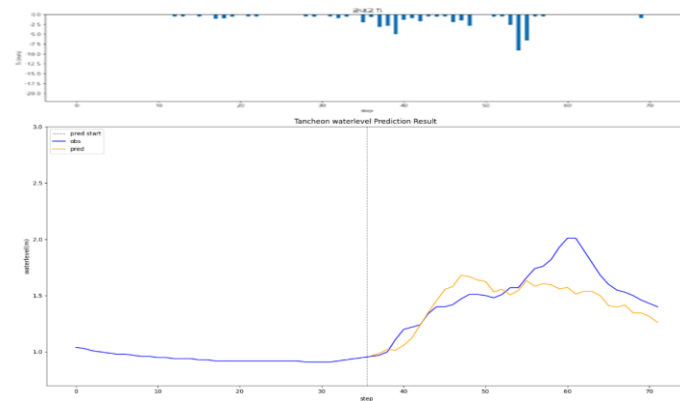
testdata_230711

R2 : 0.8622



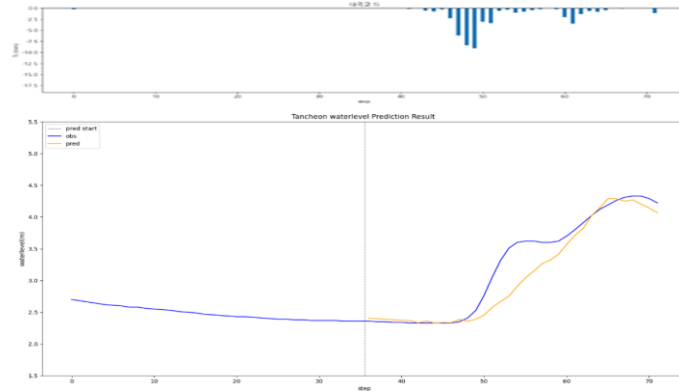
testdata_250814

R2 : 0.5087



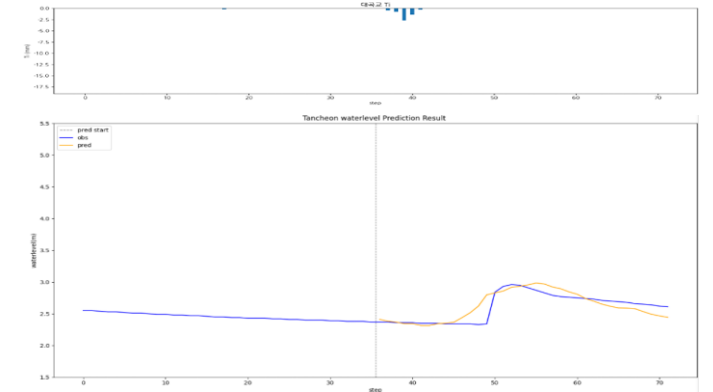
testdata_200802

R2 : 0.8748



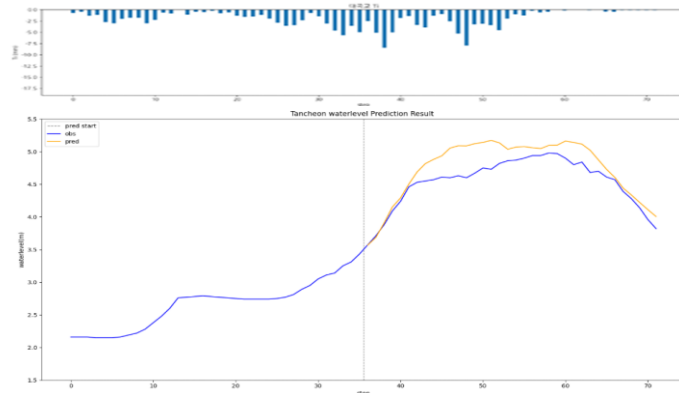
testdata_240723

R2 : 0.6749



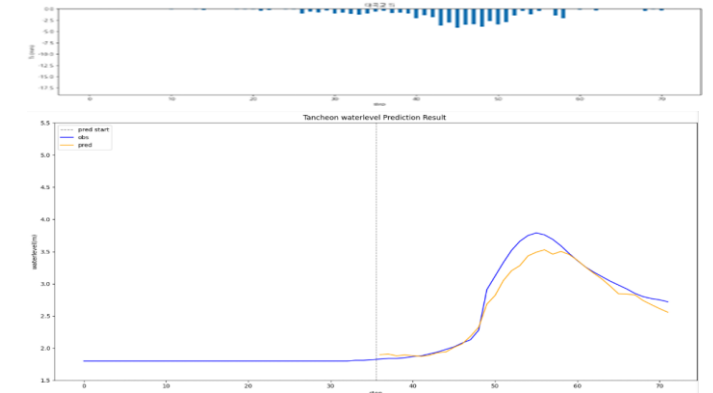
testdata_220712

R2 : 0.5503



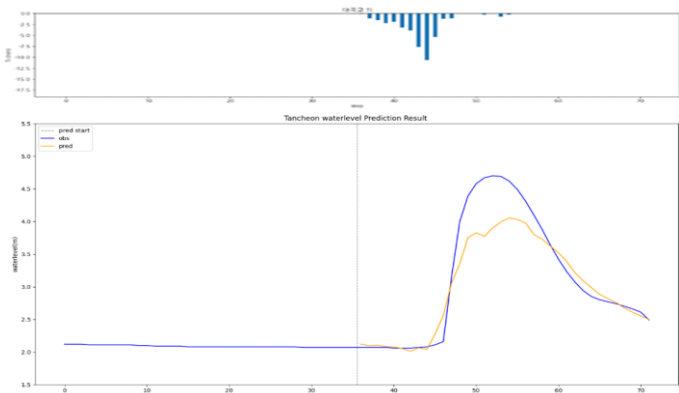
testdata_250716

R2 : 0.9470



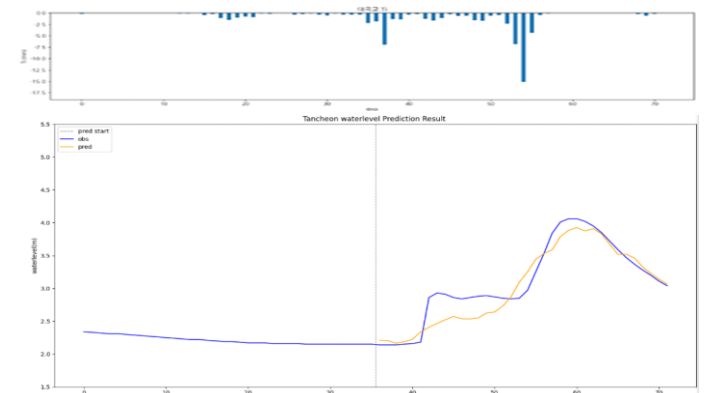
testdata_230711

R2 : 0.8659



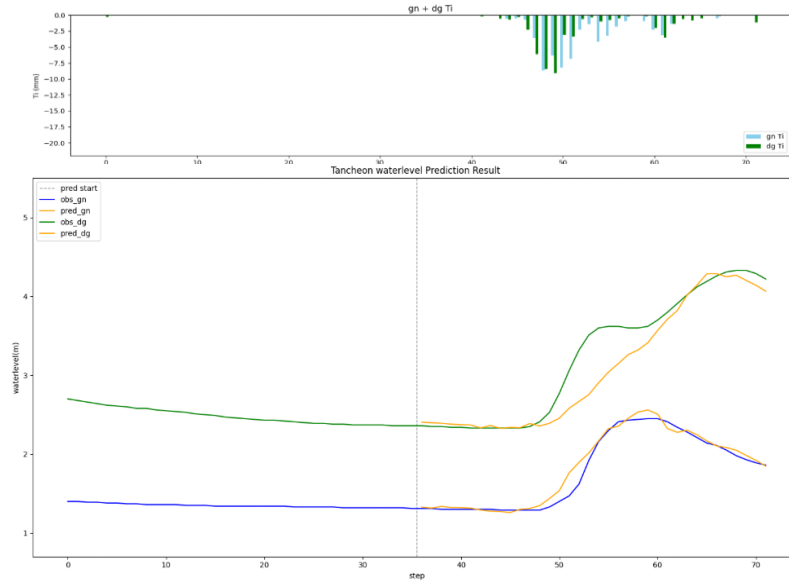
testdata_250814

R2 : 0.8781

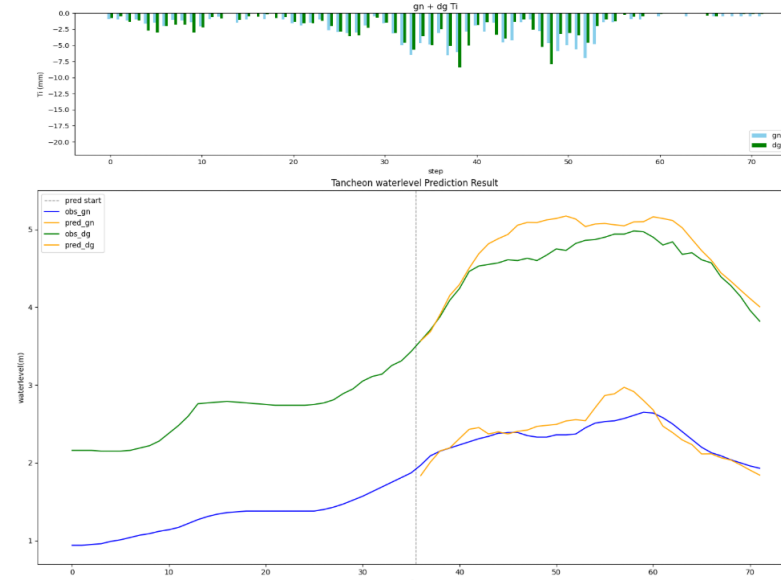


궁내교 + 대곡교

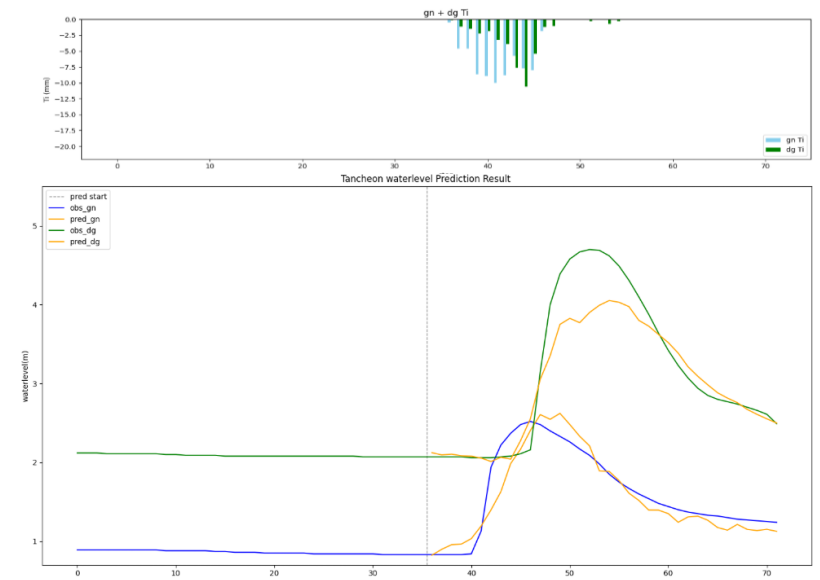
● 궁내교
● 대곡교 * model_(gn/dg)_wl18_3_2
● 예측



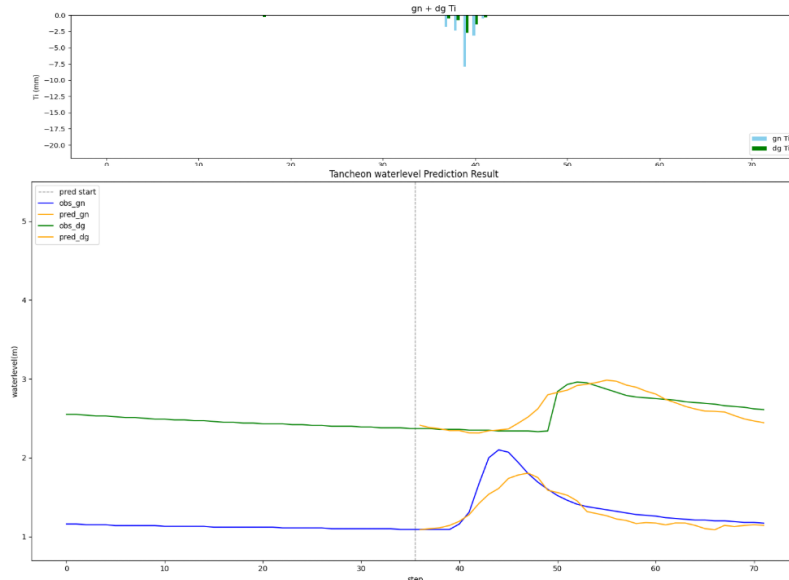
testdata_200802



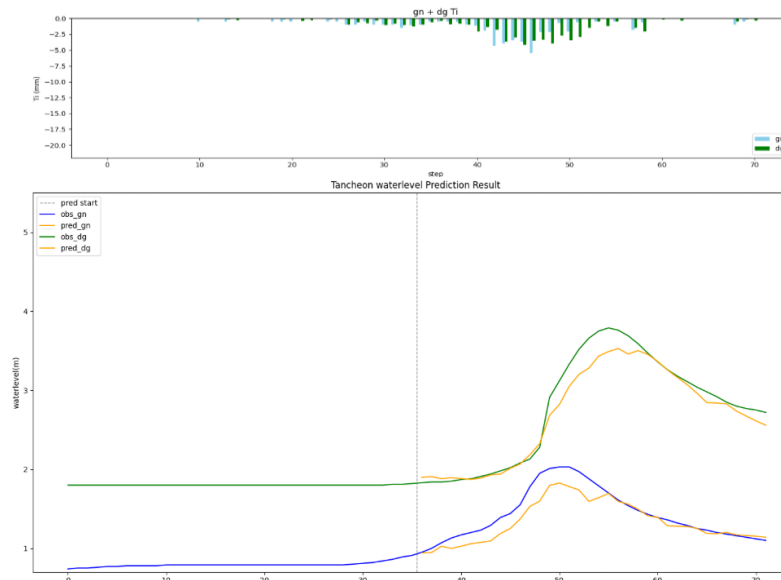
testdata_220712



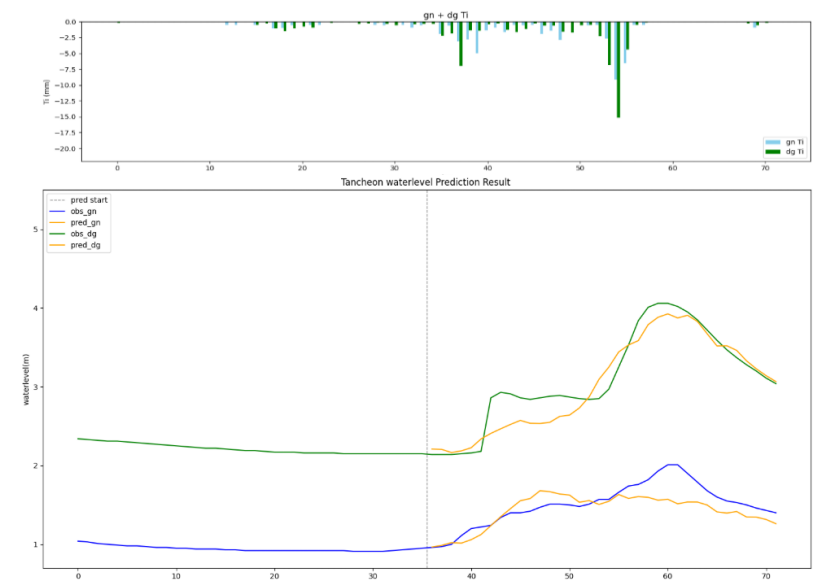
testdata_230711



testdata_240723



testdata_250716



testdata_250814

관심수위 + MAE
1시간 단위 예측

관심수위 + MAE 1시간 단위 예측

- » 궁내교 : Lead time sensitivity 낮음 (시간 이동에 강건)
- » 대곡교 : 궁내교보다 Lead-time sensitivity 큼 (특히 후행 1시간에서 성능 급락)
- » peak 위치가 $\pm 1h$ shift되어도 R^2 비슷하게 유지

		궁내교				대곡교			
Test data	time	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	01	0.9627	0.0085	0.0656	0.0656	0.8421	0.0716	0.1783	0.2676
	02	0.9656	0.0072	0.0558	0.0852	0.8748	0.0763	0.1754	0.2763
	03	0.8581	0.0236	0.1028	0.1538	0.8667	0.0633	0.2096	0.2517
22.07.12	15	0.8698	0.0108	0.0845	0.1041	0.9276	0.0271	0.1433	0.1648
	16	0.4201	0.0230	0.1139	0.1516	0.5503	0.0629	0.2047	0.2509
	17	0.7777	0.0148	0.0844	0.1218	0.3852	0.1563	0.3768	0.3953
23.07.11	06	0.8966	0.0368	0.1150	0.1920	0.8208	0.1877	0.2735	0.4333
	07	0.8622	0.0385	0.1486	0.1963	0.8659	0.1219	0.2258	0.3492
	08	0.8323	0.0351	0.1005	0.1874	0.9018	0.0779	0.2423	0.2791

관심수위 + MAE 1시간 단위 예측

- » 궁내교 : Lead time sensitivity 낮음 (시간 이동에 강건)
- » 대곡교 : 궁내교보다 Lead-time sensitivity 큼 (특히 후행 1시간에서 성능 급락)
- » peak 위치가 $\pm 1h$ shift되어도 R^2 비슷하게 유지

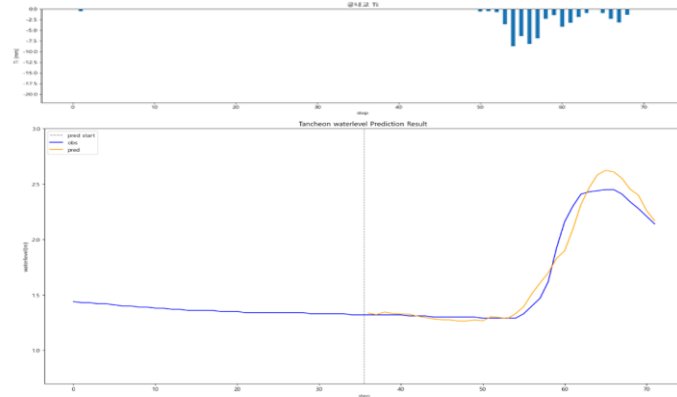
Test data	time	궁내교				대곡교			
		r2	mse	mae	rmse	r2	mse	mae	rmse
24.07.23	17	0.6247	0.0329	0.0925	0.1816	0.6779	0.0171	0.1038	0.1311
	18	0.7308	0.0217	0.0962	0.1475	0.6749	0.0154	0.0860	0.1244
	19	0.7209	0.0217	0.0892	0.1475	0.4809	0.0191	0.1026	0.1385
25.07.16	16	0.7968	0.0283	0.1263	0.1684	0.9713	0.0168	0.1071	0.1298
	17	0.7947	0.0205	0.1047	0.1435	0.9470	0.0244	0.1098	0.1563
	18	0.8285	0.0174	0.1056	0.1319	0.9366	0.0201	0.1147	0.1419
25.08.14	04	0.7630	0.0277	0.1195	0.1665	0.8931	0.0489	0.1846	0.2211
	05	0.5087	0.0346	0.1446	0.1861	0.8781	0.0423	0.1589	0.2056
	06	0.6803	0.0131	0.0906	0.1146	0.6361	0.0707	0.2163	0.2659

궁내교 1시간 단위

* model_(gn/dg)_wl18_3_2

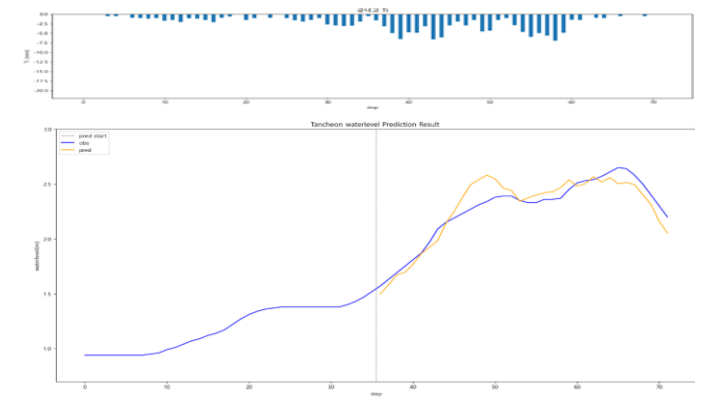
1시간 전 (01)

R2 : 0.9627



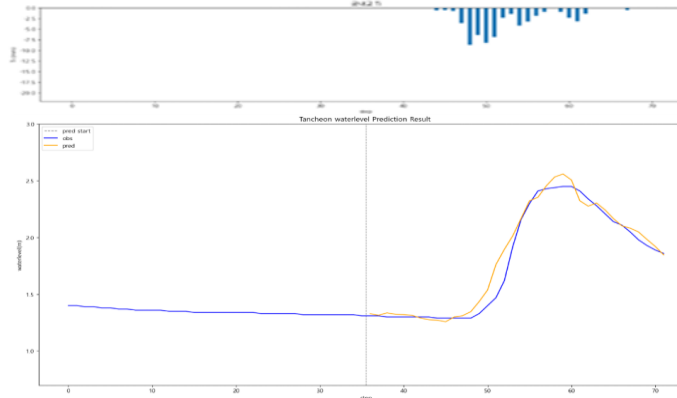
1시간 전 (15)

R2 : 0.8698



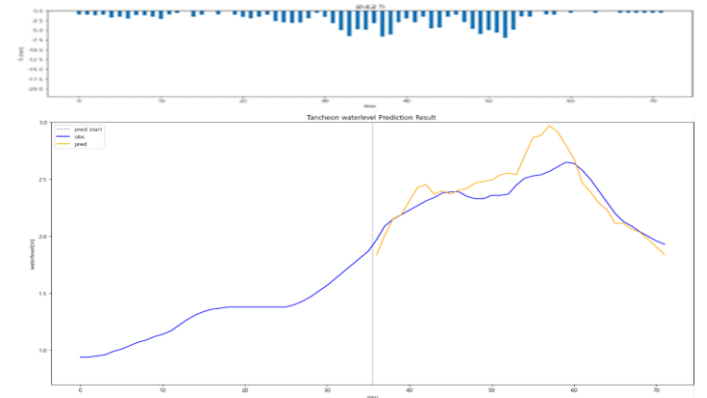
testdata_ 200802 (02)

R2 : 0.9656



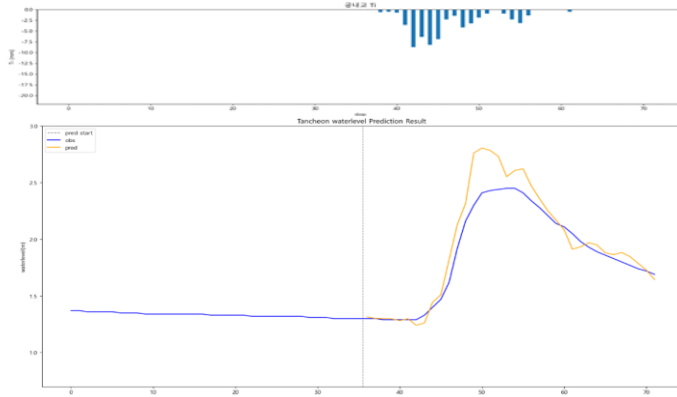
testdata_ 220712 (16)

R2 : 0.4201



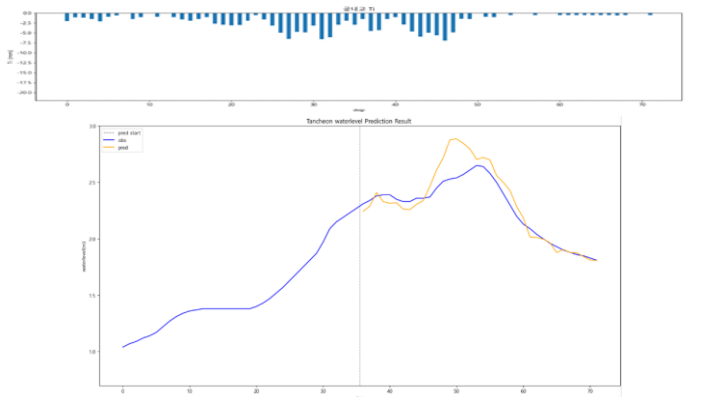
1시간 후 (03)

R2 : 0.8581



1시간 후 (17)

R2 : 0.7777

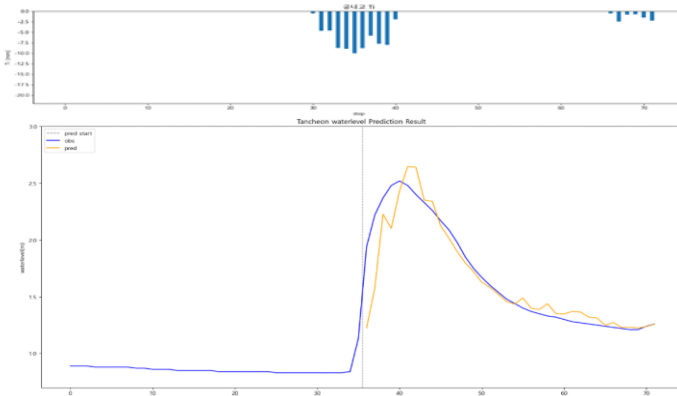
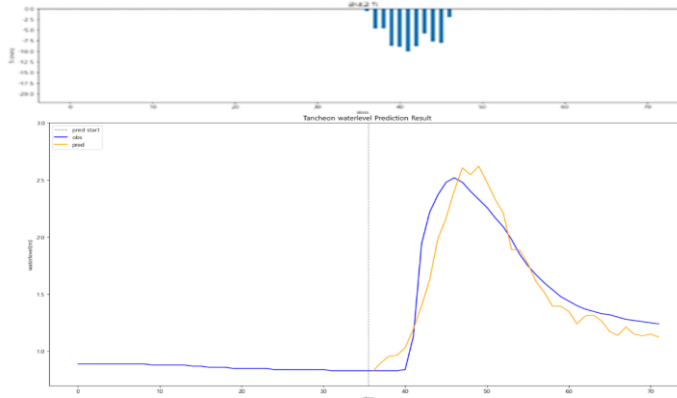
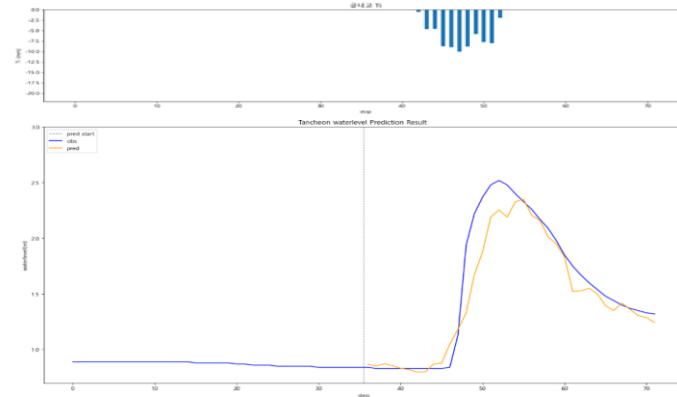


궁내교 1시간 단위

* model_(gn/dg)_wl18_3_2

1시간 전 (06)

R2 : 0.8622

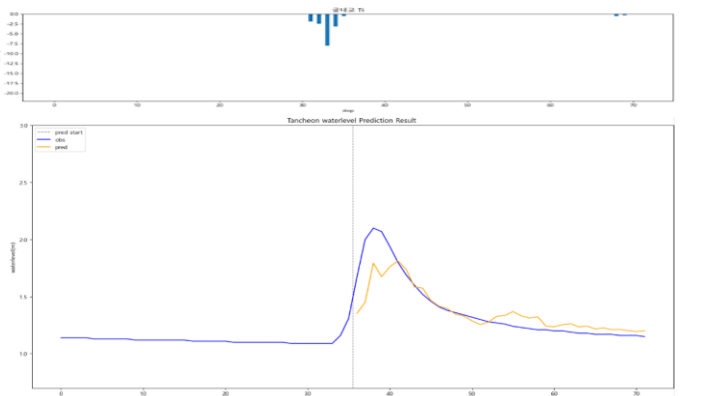
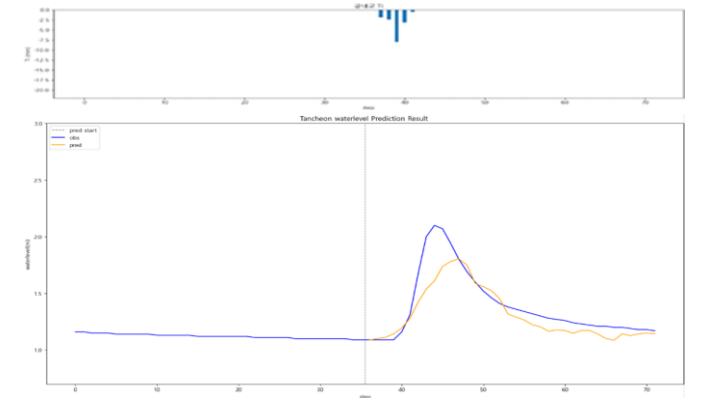
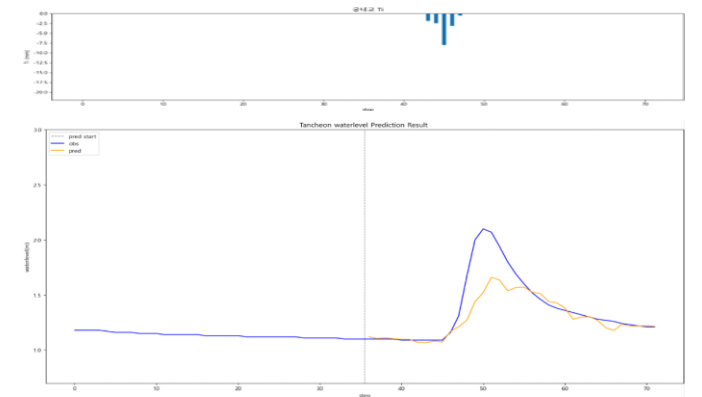


1시간 후 (08)

R2 : 0.8323

1시간 전 (17)

R2 : 0.6247



testdata_240723 (18)

R2 : 0.7308

1시간 후 (19)

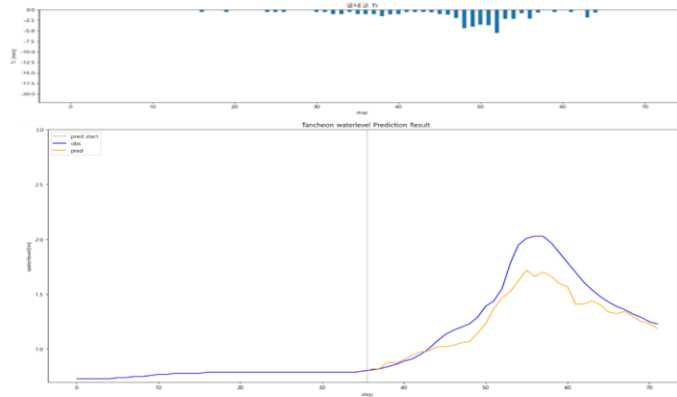
R2 : 0.7209

궁내교 1시간 단위

* model_(gn/dg)_wl18_3_2

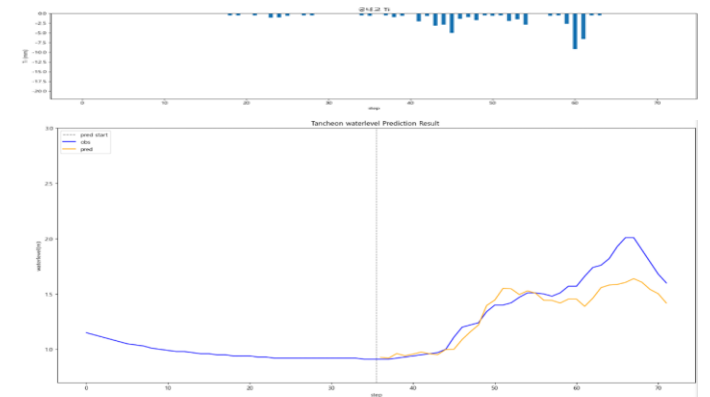
1시간 전 (16)

R2 : 0.7968



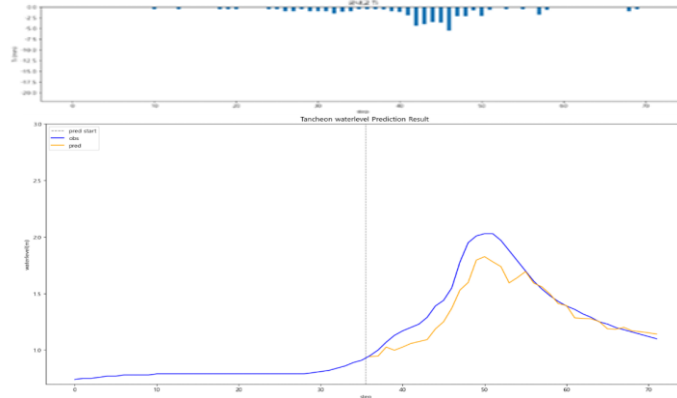
1시간 전 (04)

R2 : 0.7630



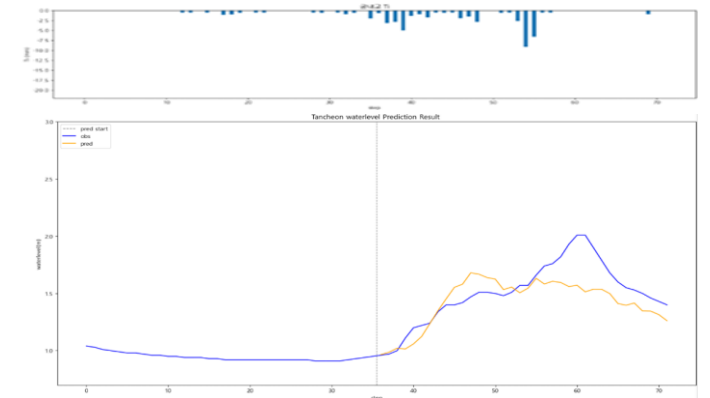
testdata_ 250716 (17)

R2 : 0.7947



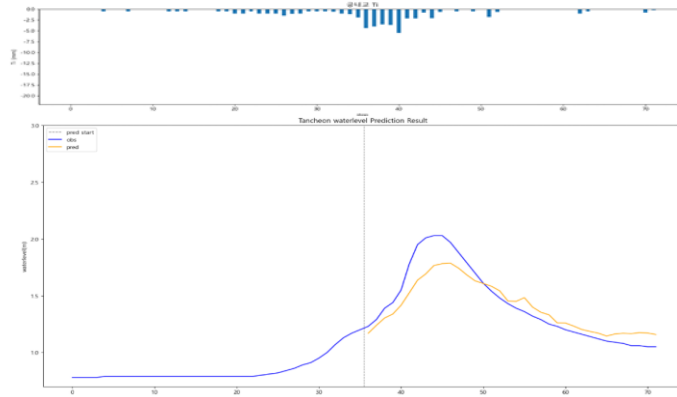
testdata_ 250814 (05)

R2 : 0.5087



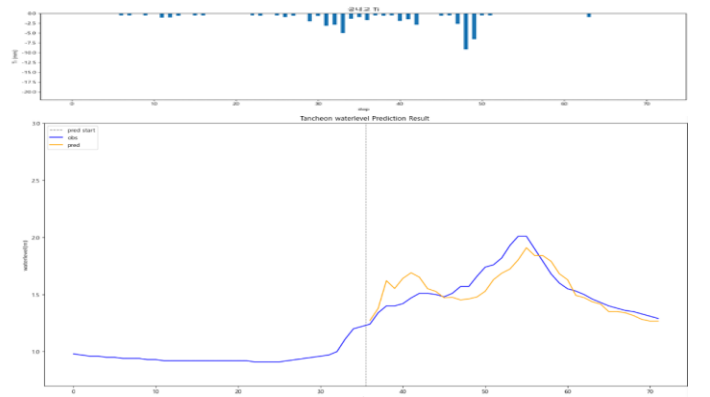
1시간 후 (18)

R2 : 0.8285



1시간 후 (06)

R2 : 0.6803

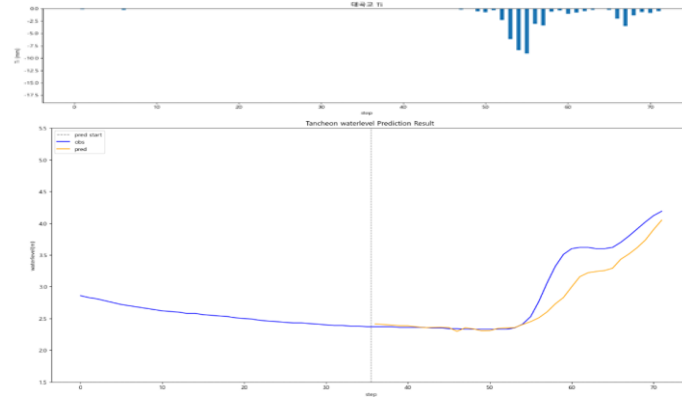


대곡교 - 1시간 단위

* model_(gn/dg)_wl18_3_2

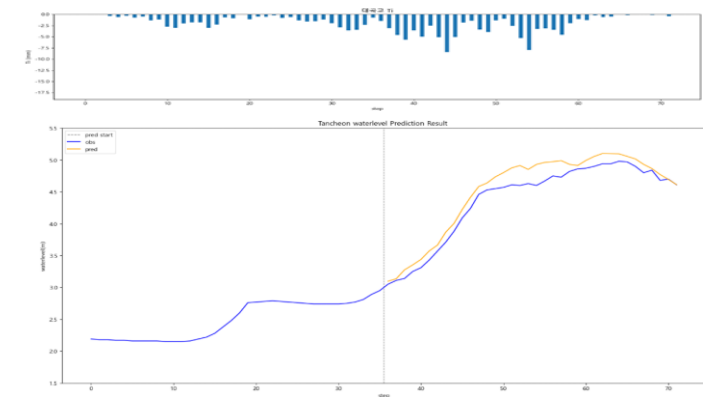
1시간 전 (01)

R2 : 0.8421



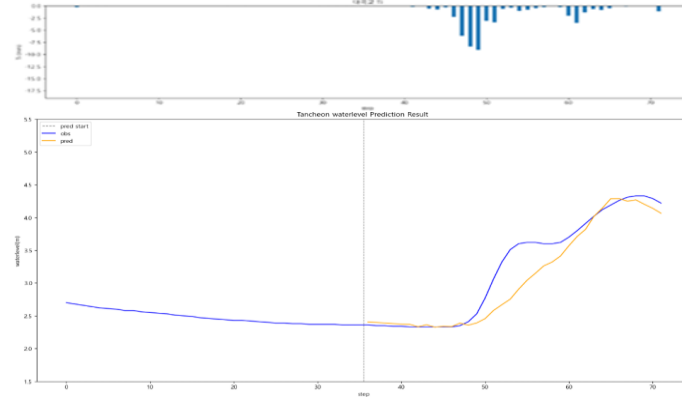
1시간 전 (15)

R2 : 0.9276



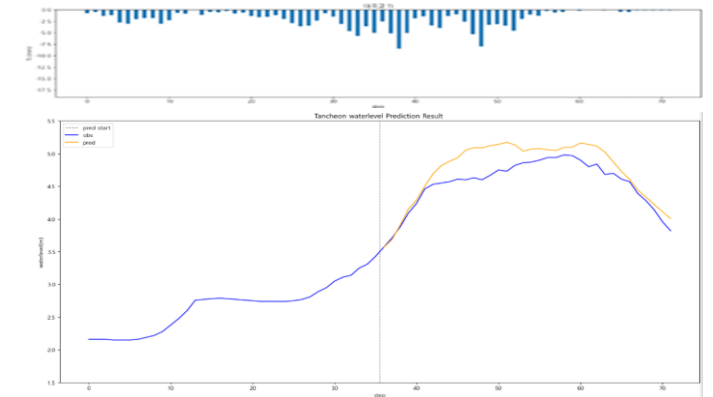
testdata_ 200802 (02)

R2 : 0.8748



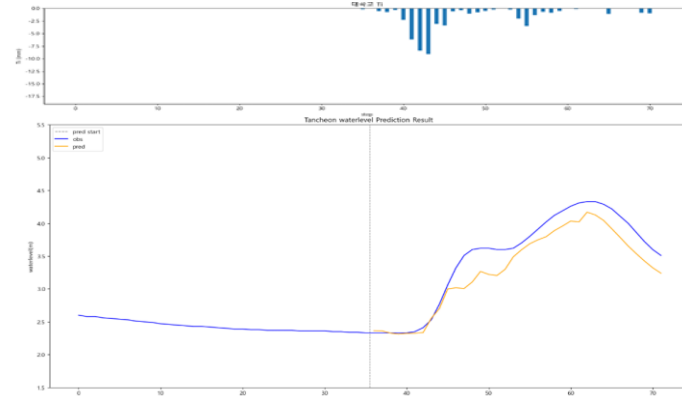
testdata_ 220712 (16)

R2 : 0.5503



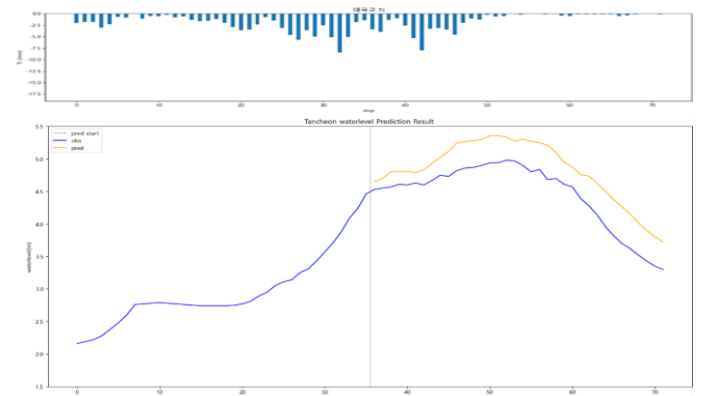
1시간 후 (03)

R2 : 0.8667



1시간 후 (17)

R2 : 0.3852

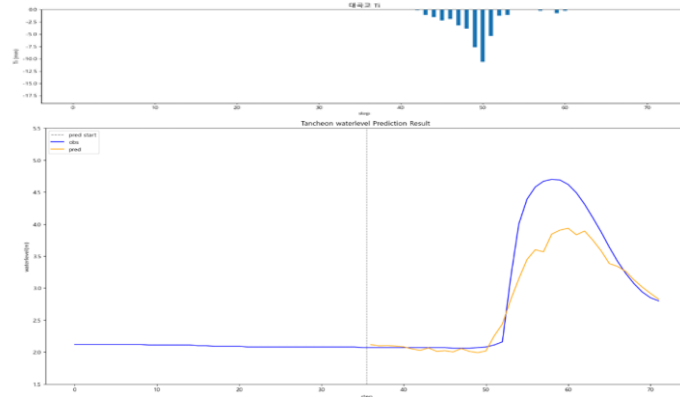


대곡교 - 1시간 단위

* model_(gn/dg)_wl18_3_2

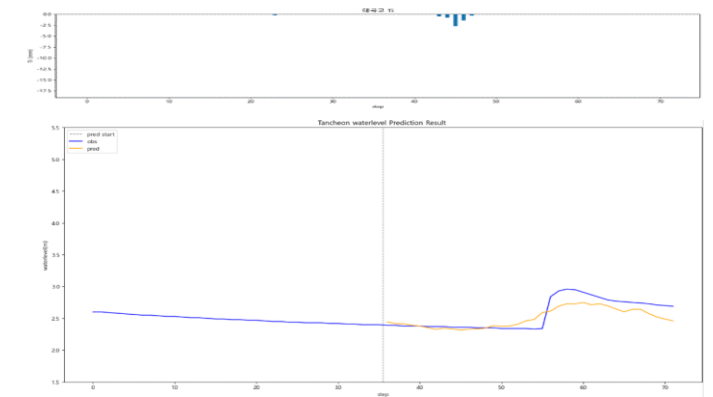
1시간 전 (06)

R2 : 0.8208



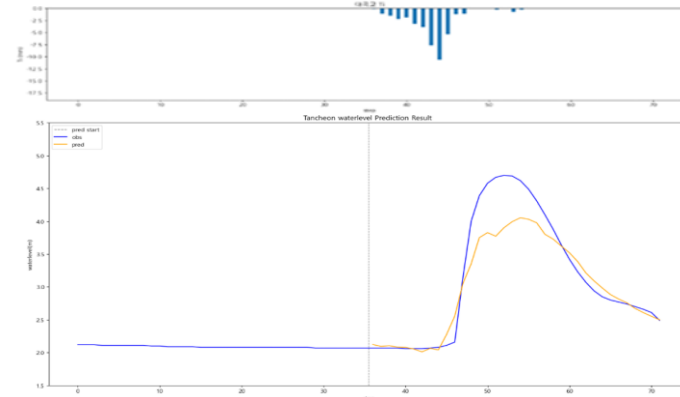
1시간 전 (17)

R2 : 0.6779



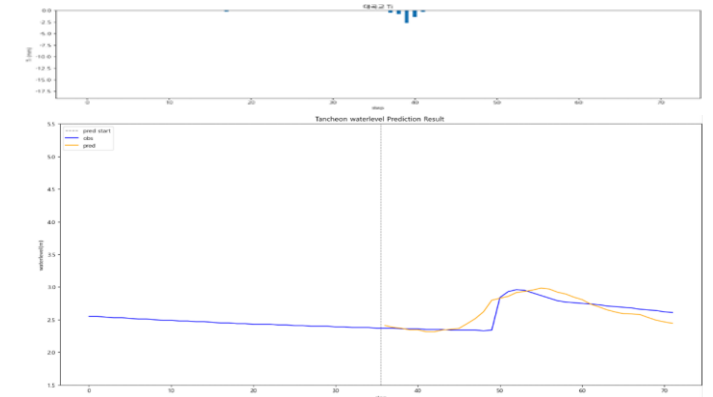
testdata_ 230711 (07)

R2 : 0.8659



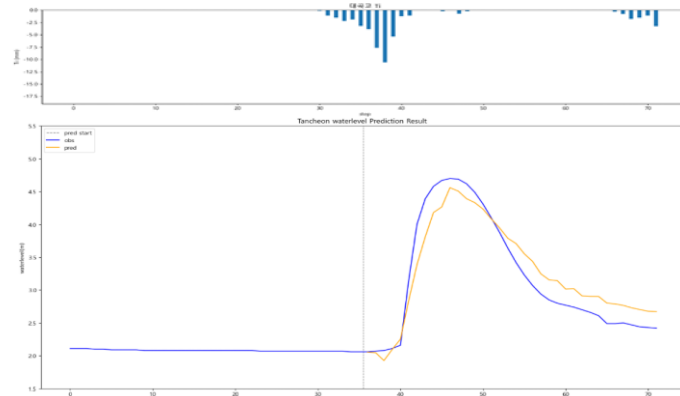
testdata_ 240723 (18)

R2 : 0.6749



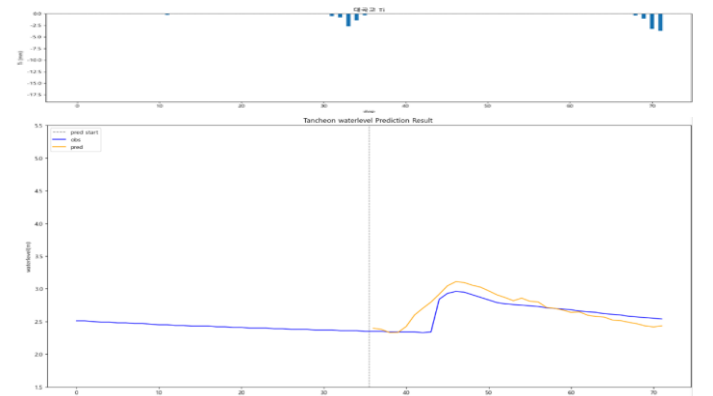
1시간 후 (08)

R2 : 0.9018



1시간 후 (19)

R2 : 0.4809

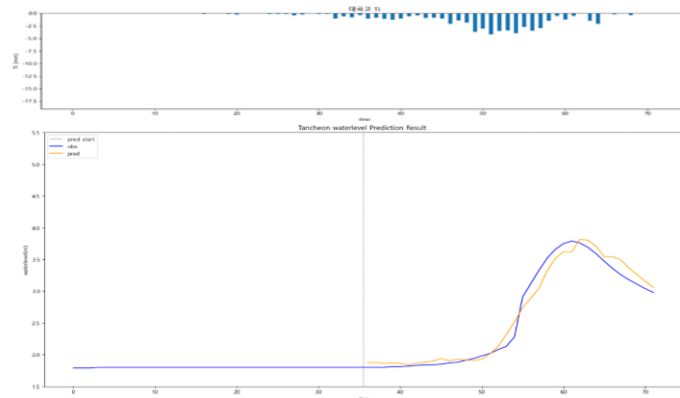


대곡교 - 1시간 단위

* model_(gn/dg)_wl18_3_2

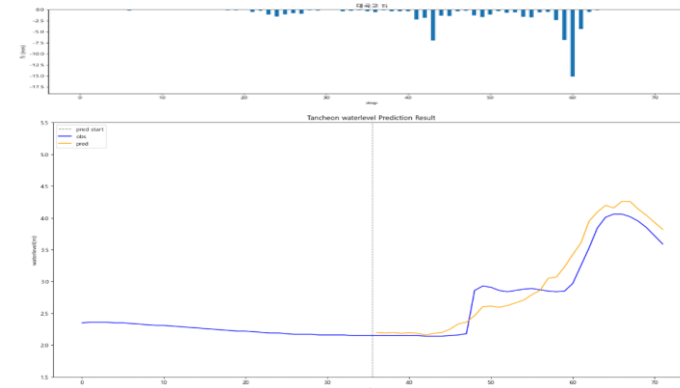
1시간 전 (16)

R2 : 0.9713



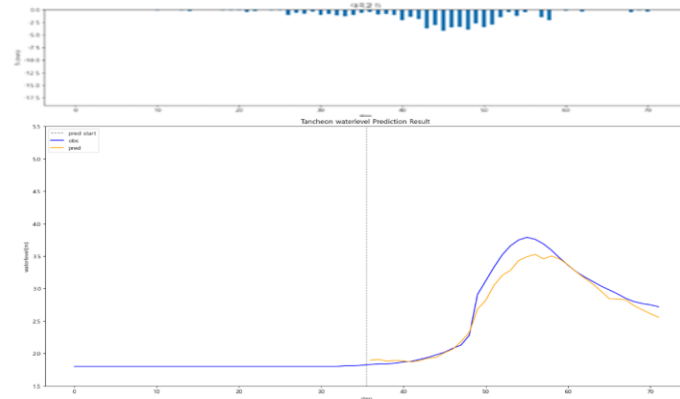
1시간 전 (04)

R2 : 0.8931



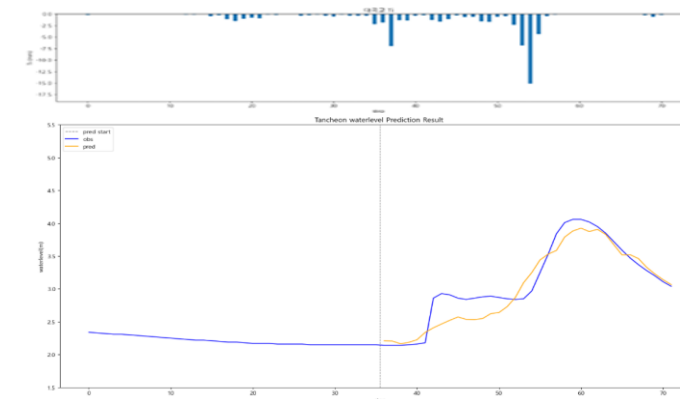
testdata_ 250716 (17)

R2 : 0.9470



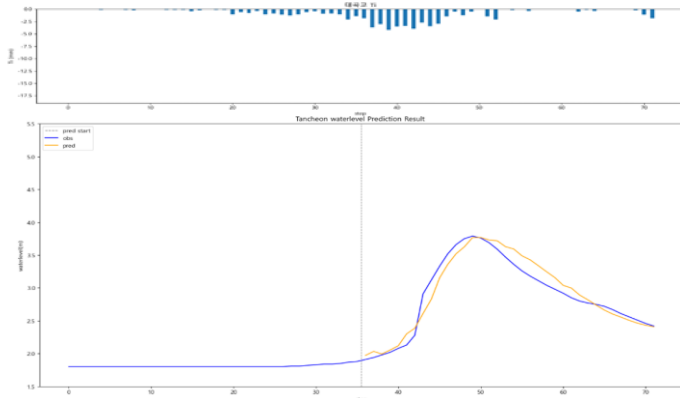
testdata_ 250814 (05)

R2 : 0.8781



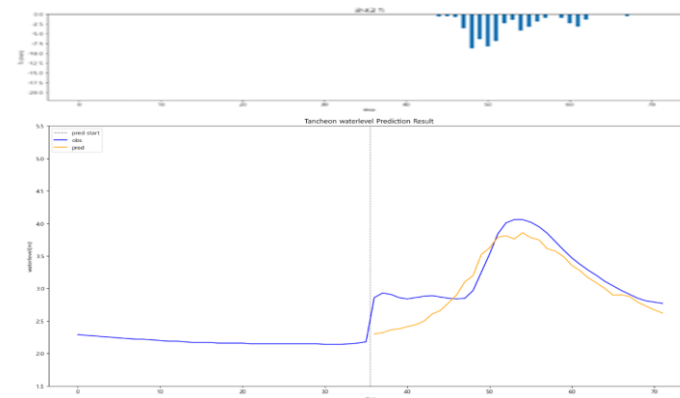
1시간 후 (18)

R2 : 0.9366



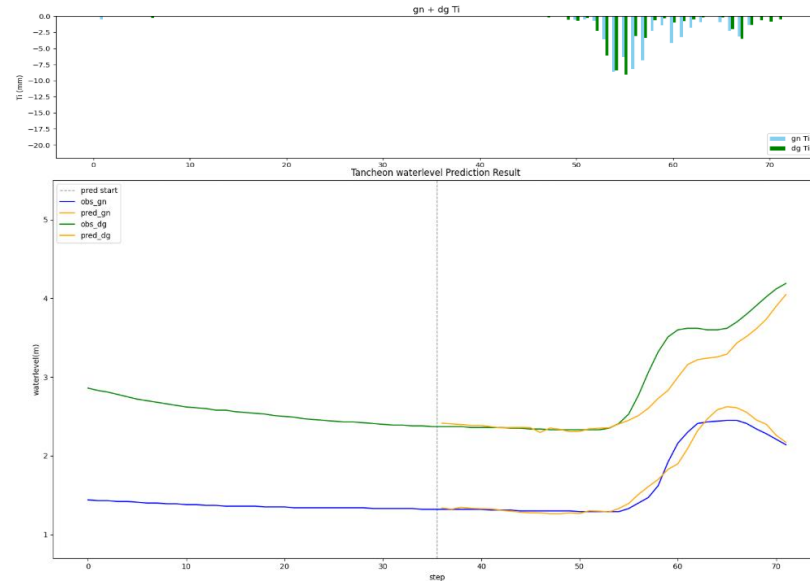
1시간 후 (06)

R2 : 0.6361

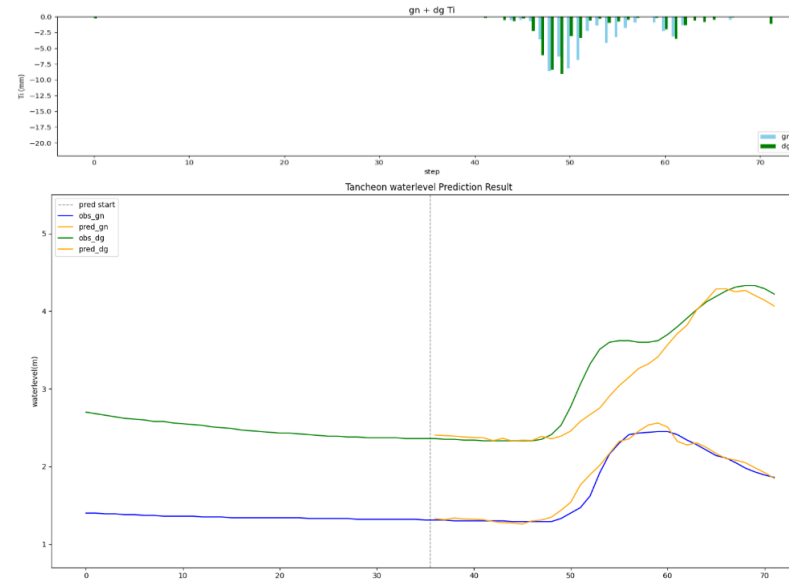


궁내교 + 대곡교

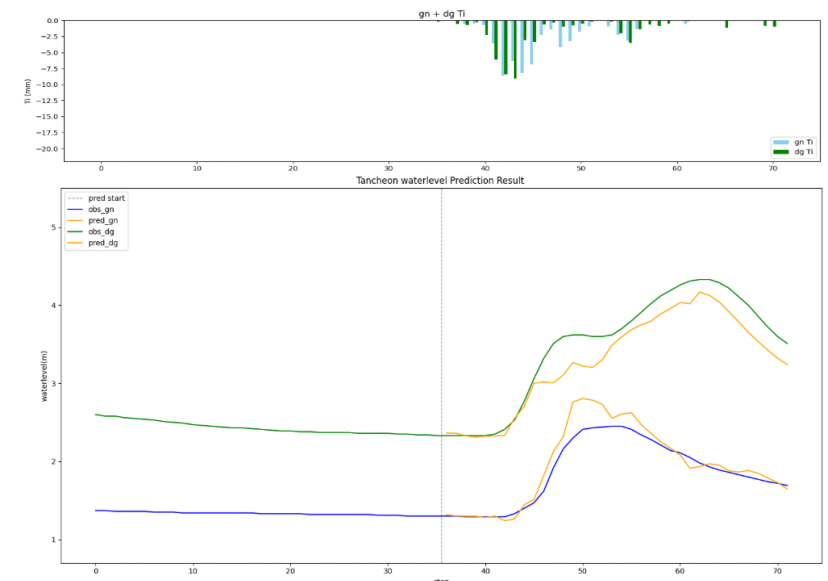
● 궁내교
● 대곡교
● 예측
* model_(gn/dg)_wl18_3_2



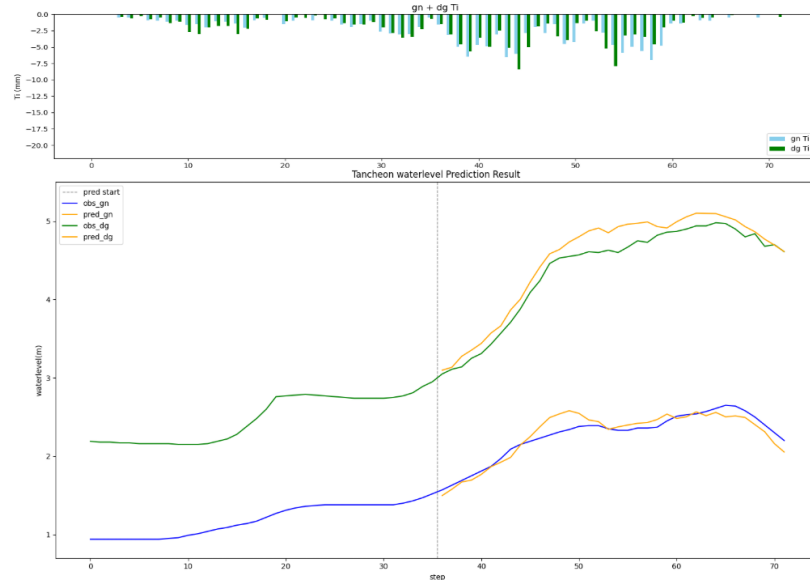
1시간 전 (01)



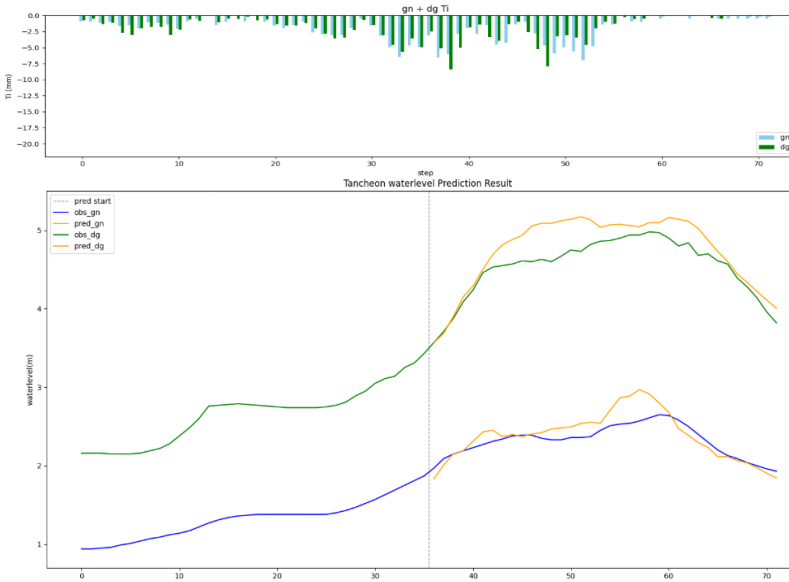
testdata_200802 (02)



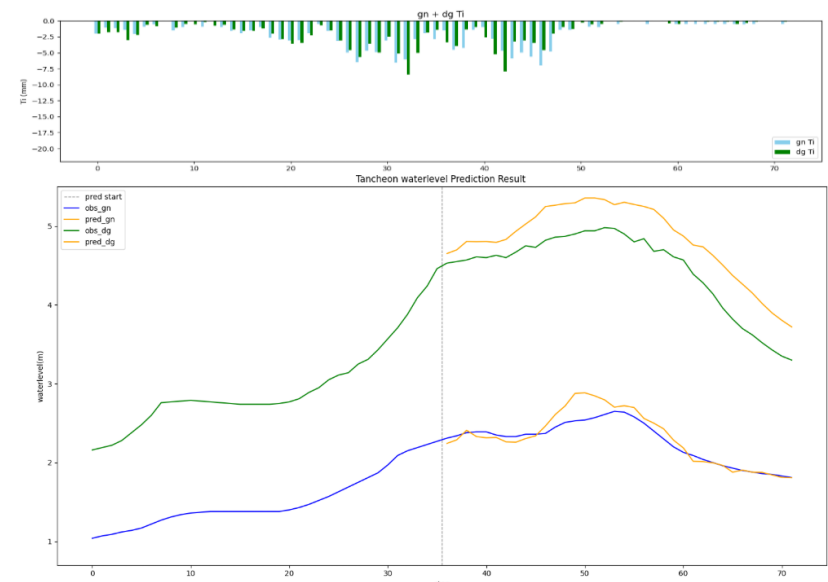
1시간 후 (03)



1시간 전 (15)



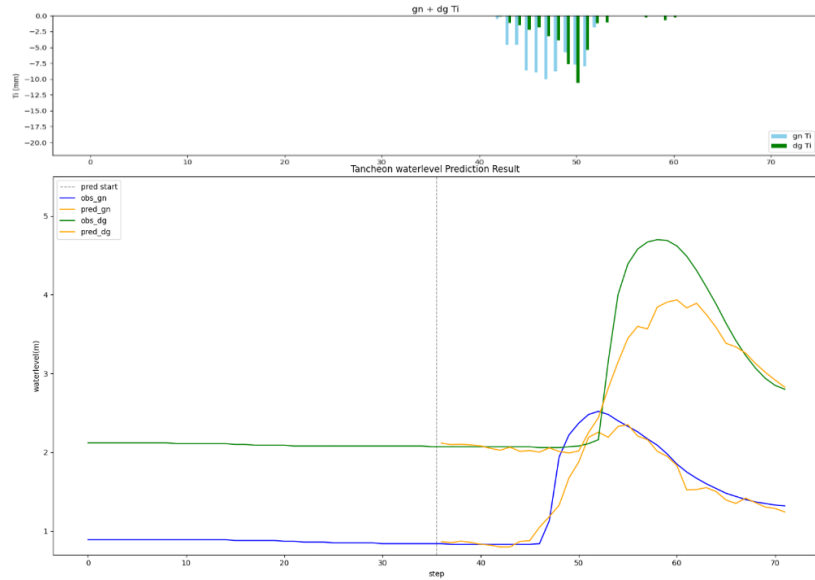
testdata_220712 (17)



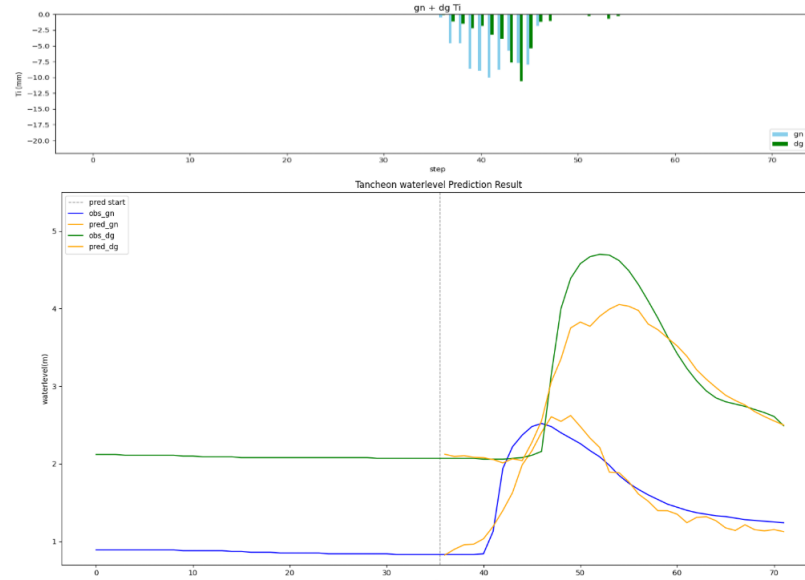
1시간 후 (17)

궁내교 + 대곡교

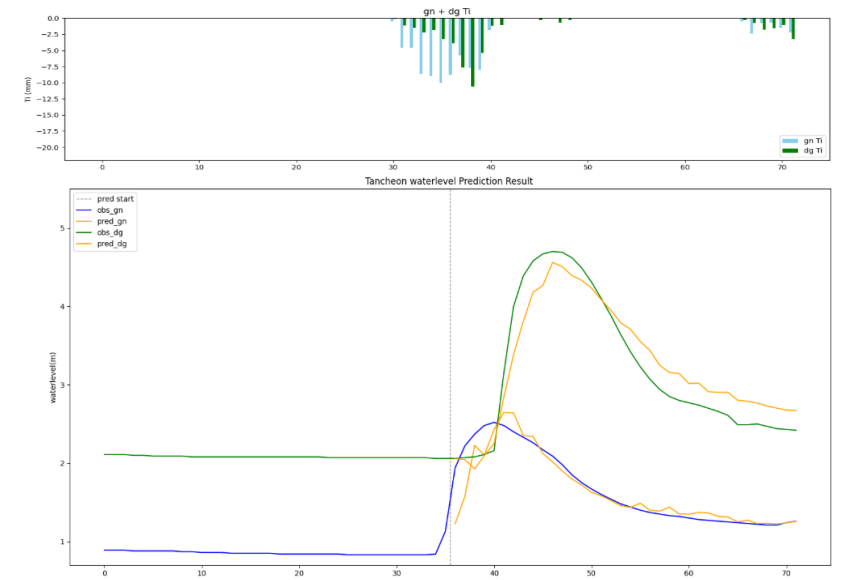
● 궁내교
● 대곡교
● 예측
* model_(gn/dg)_wl18_3_2



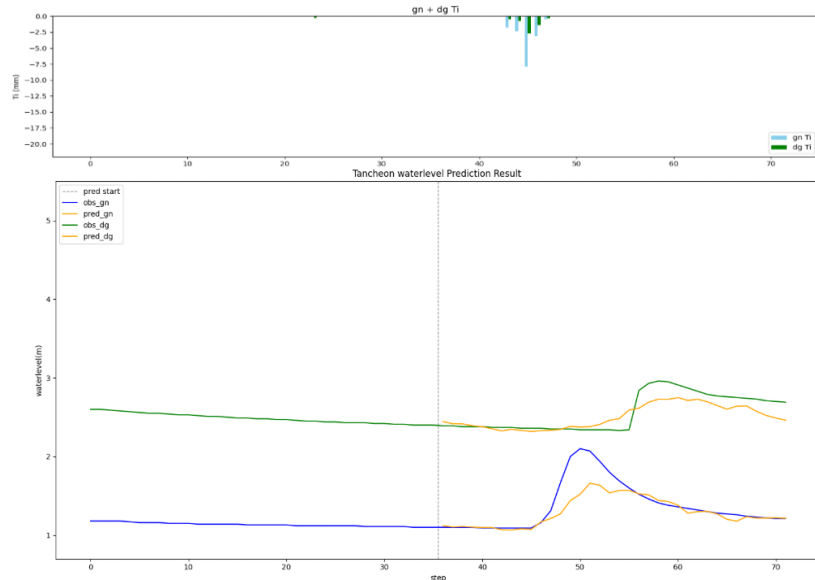
1시간 전 (06)



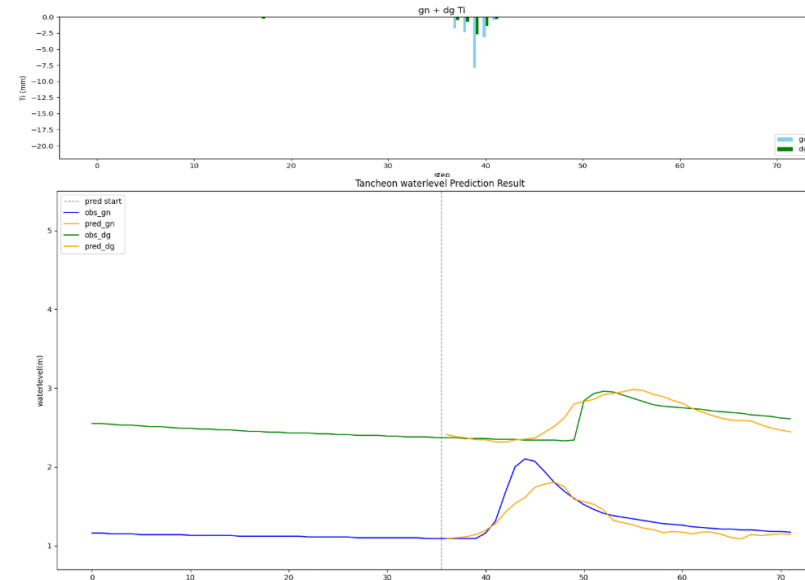
testdata_230711 (07)



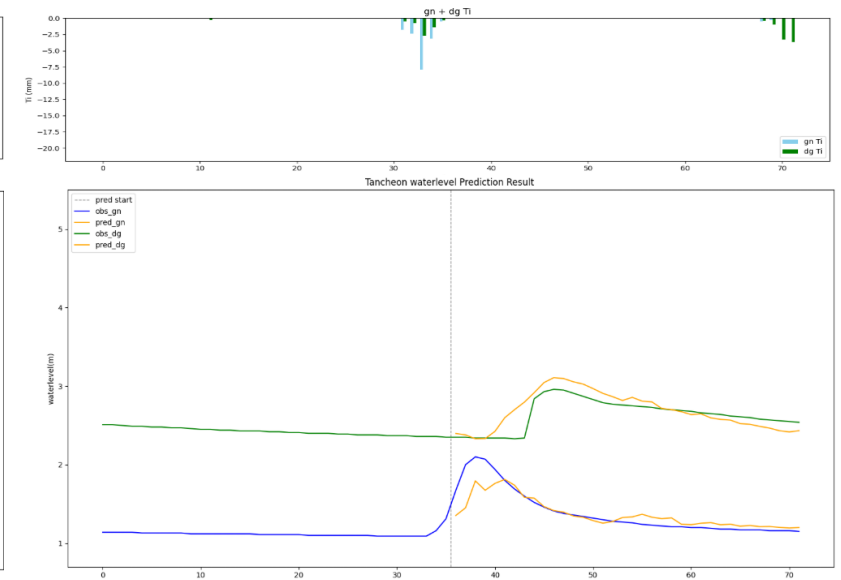
1시간 후 (08)



1시간 전 (17)



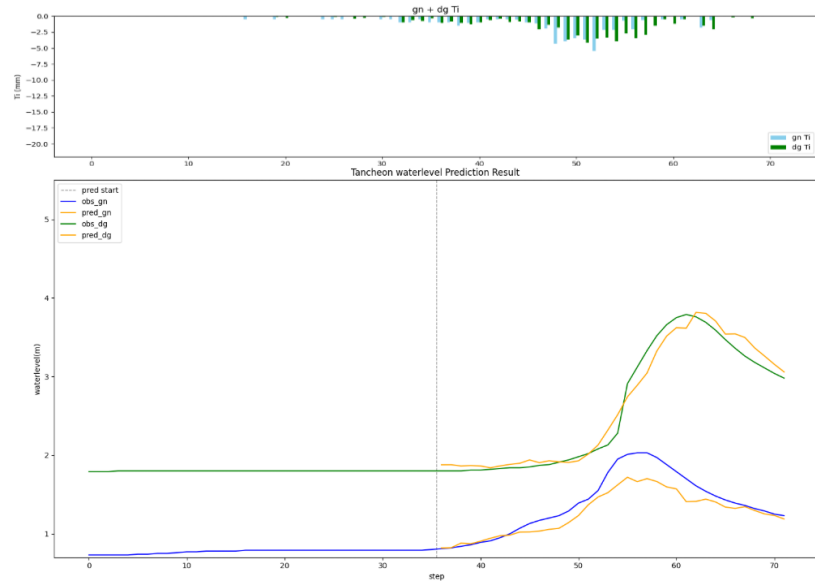
testdata_240723 (18)



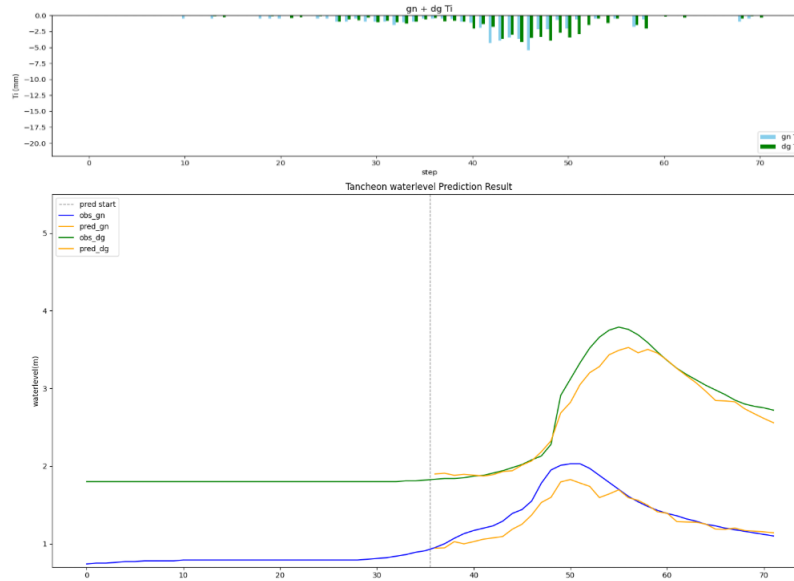
1시간 후 (19)

궁내교 + 대곡교

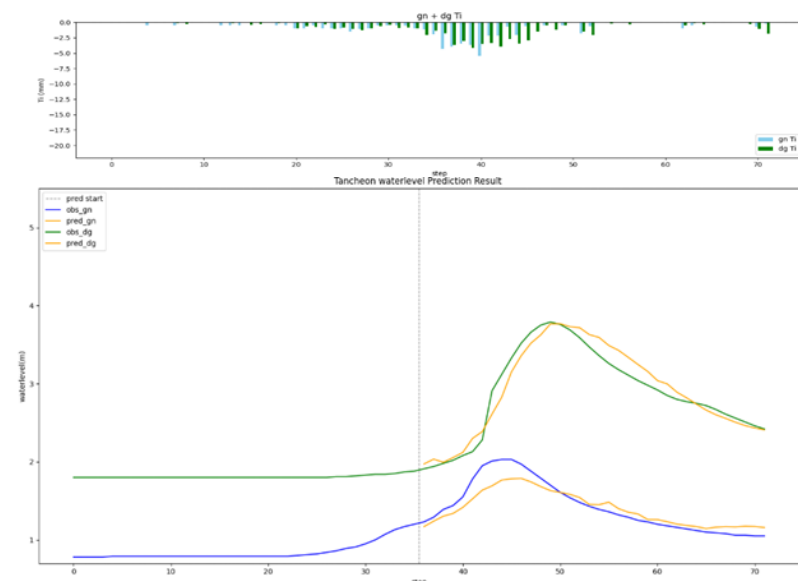
● 궁내교
● 대곡교
● 예측
* model_(gn/dg)_wl18_3_2



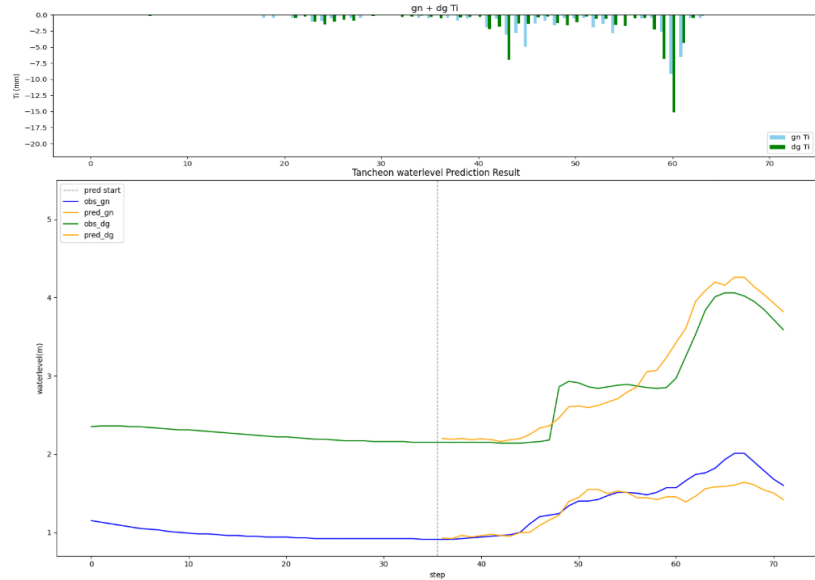
1시간 전 (16)



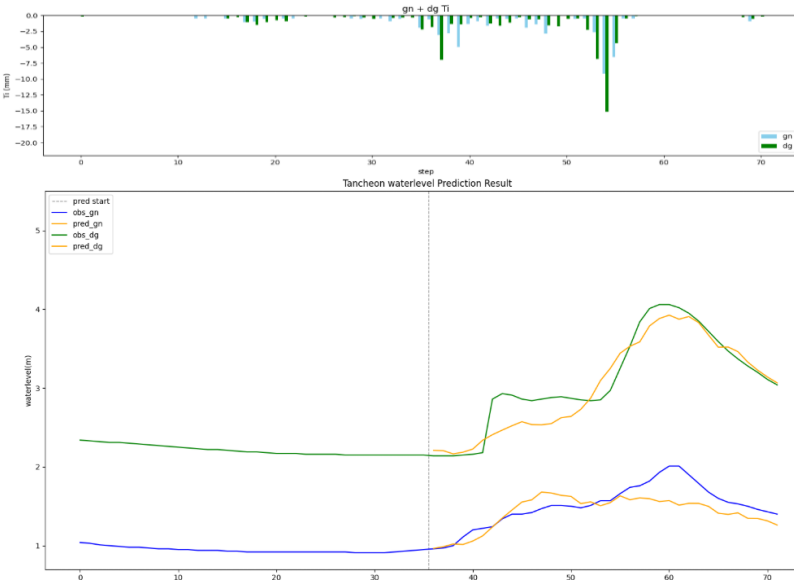
testdata_250716 (17)



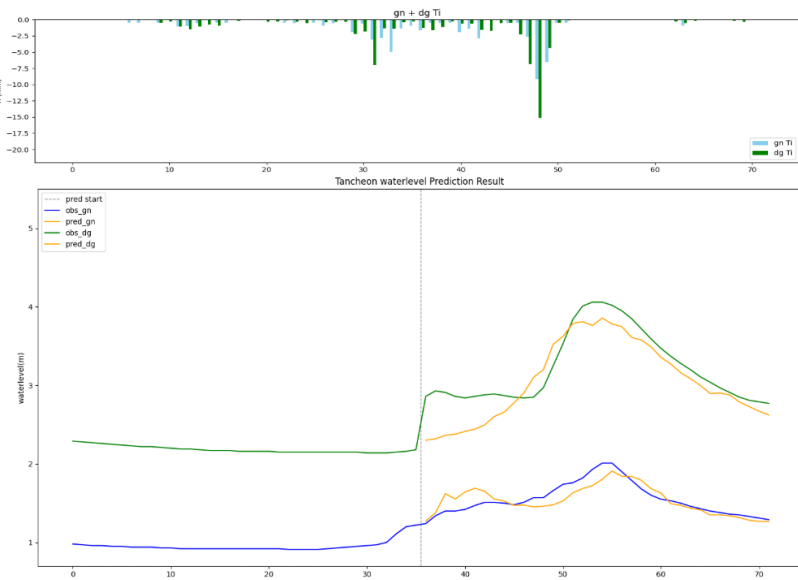
1시간 후 (18)



1시간 전 (04)



testdata_250814 (05)



1시간 후 (06)

유량 예측 모델

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 유량 그래프 스케일 고정 : 궁내교 (1.45 – 200.45) 대곡교 (3.93 – 712.55)
- 수위 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

-
- learning_rate = 0.0005
 - Loss_function = MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

-
- 유량 곡선식 : 23년 11월 이후 ~

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 67개 中 61개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

유량 예측 모델 (유량 비교)

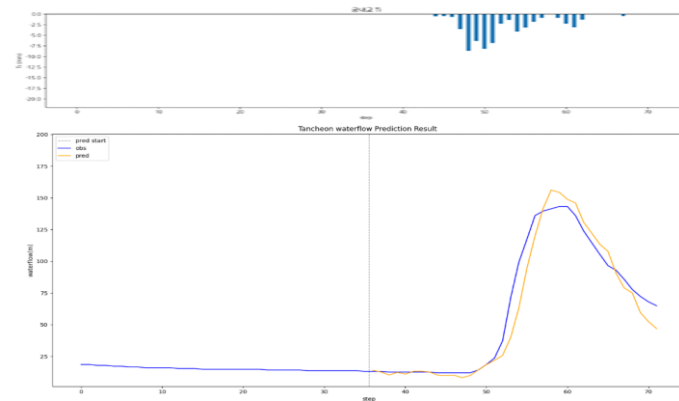
- » 평균적으로 안정적이나 특정 이벤트 성능 저하
- » 궁내교 : 22.07.12 가 유독 안맞음 (R² 음수)

기본 모델 대비 평균 R2 : 궁내교 ▼저하 (- 9%) , 대곡교 ▲개선 (+ 5.4%)

궁내교					대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.9470	133.2941	7.6087	11.5453	0.8104	2119.8319	31.4251	46.0416
22.07.12	-0.3192	1225.0217	27.96738	35.0003	0.9411	849.6534	22.2570	29.14881
23.07.11	0.9153	187.9712	10.4154	13.71025	0.7145	8592.75005	69.7216	92.6970
24.07.23	0.7854	99.3442	6.1220	9.9671	0.7932	92.4420	6.9991	9.6146
25.07.16	0.7869	113.8457	8.1333	10.6698	0.8734	916.8551	24.9128	30.2796
25.08.14	0.7481	88.8281	7.2954	9.4248	0.9158	666.7448	22.7927	25.8214

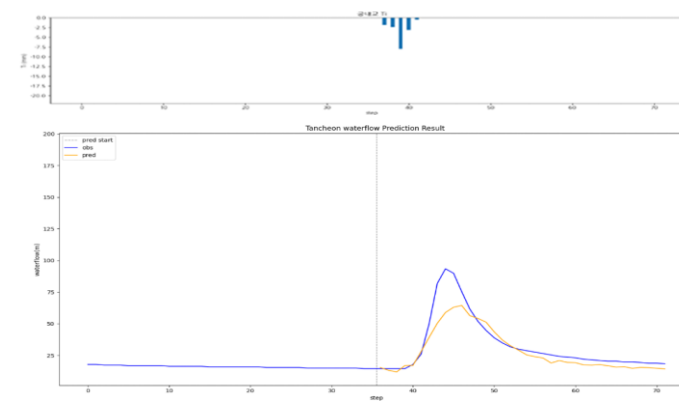
testdata_200802

R2 : 0.9470



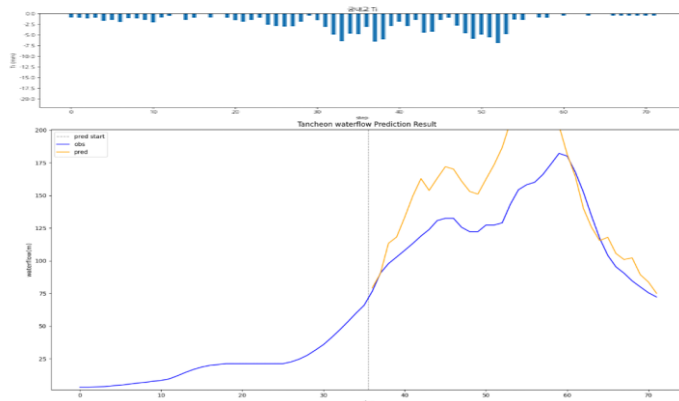
testdata_240723

R2 : 0.7854



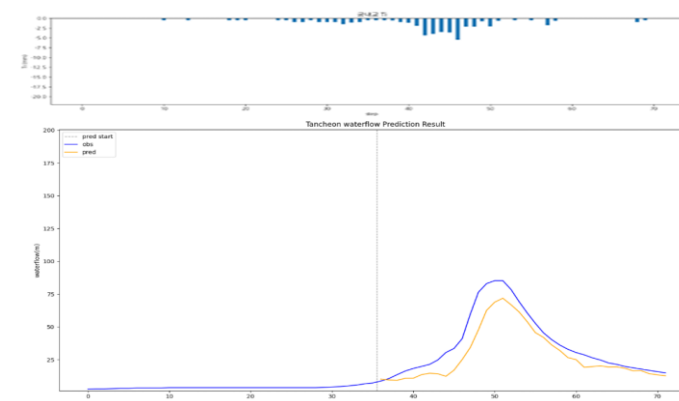
testdata_220712

R2 : -0.3192



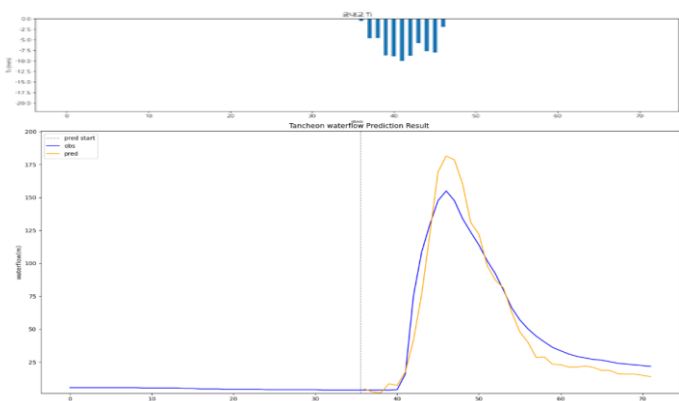
testdata_250716

R2 : 0.7869



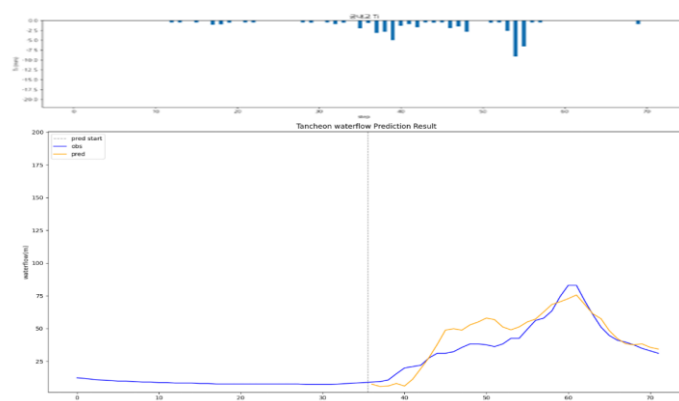
testdata_230711

R2 : 0.9153



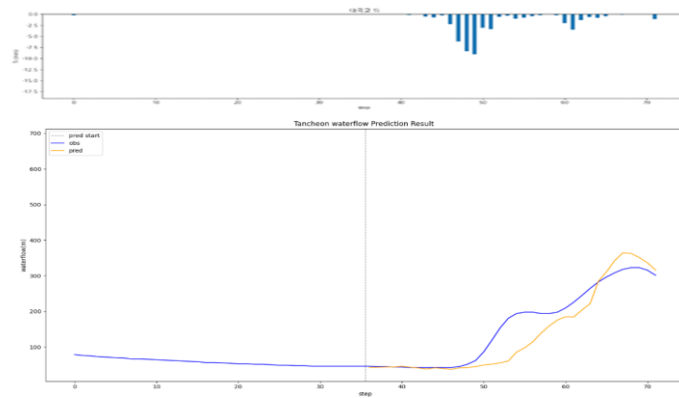
testdata_250814

R2 : 0.7481



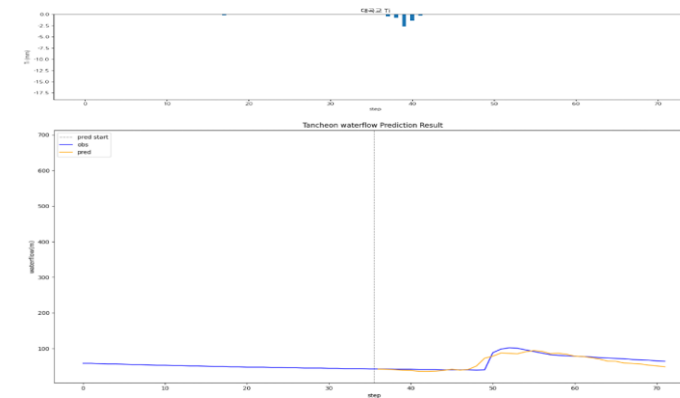
testdata_200802

R2 : 0.8104



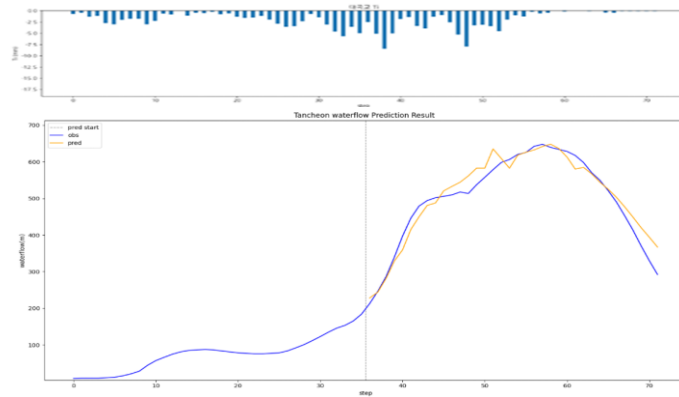
testdata_240723

R2 : 0.7932



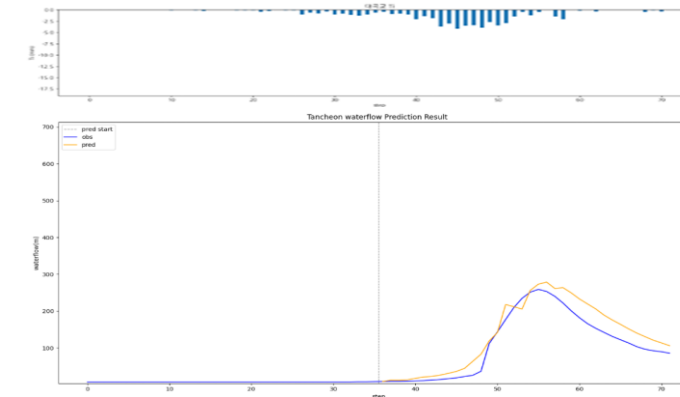
testdata_220712

R2 : 0.9411



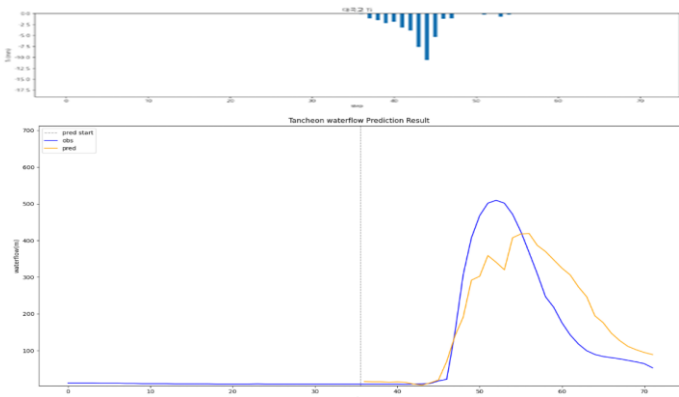
testdata_250716

R2 : 0.8734



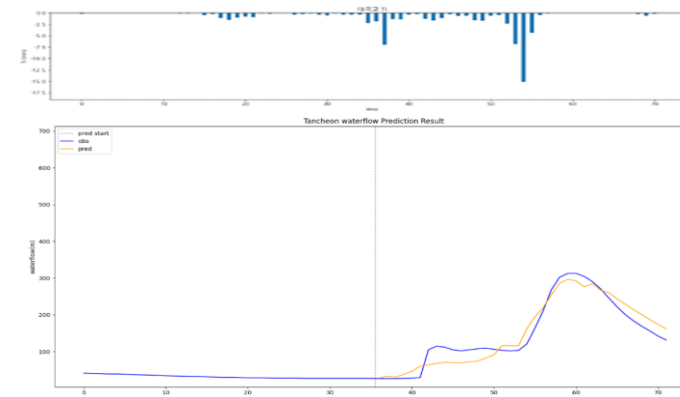
testdata_230711

R2 : 0.7145



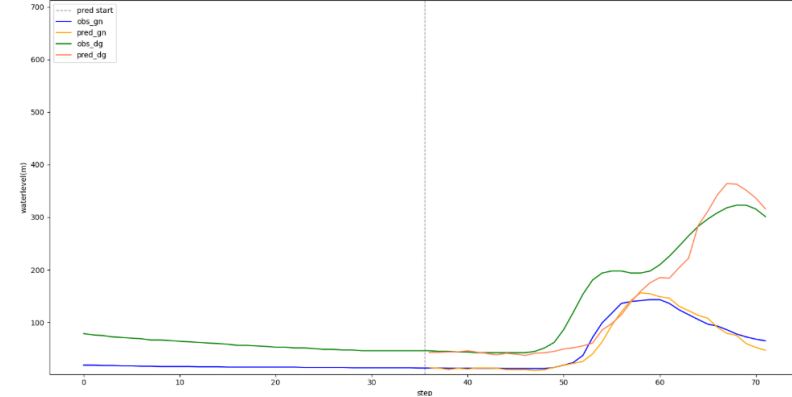
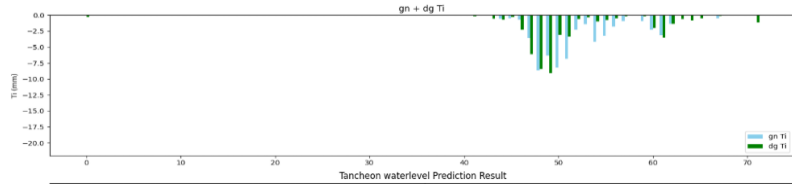
testdata_250814

R2 : 0.9158

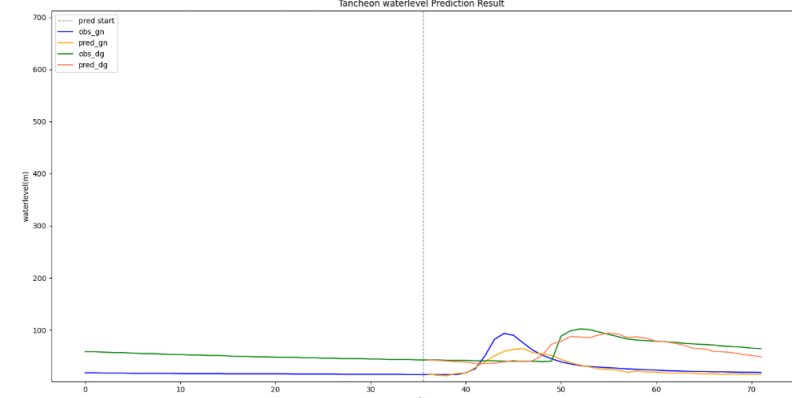
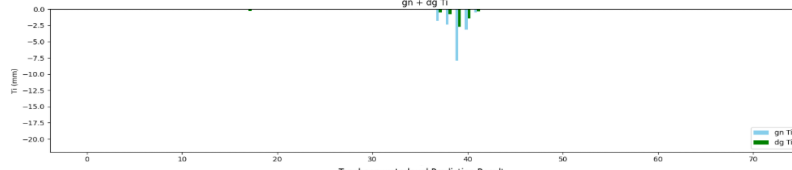


궁내교 + 대곡교

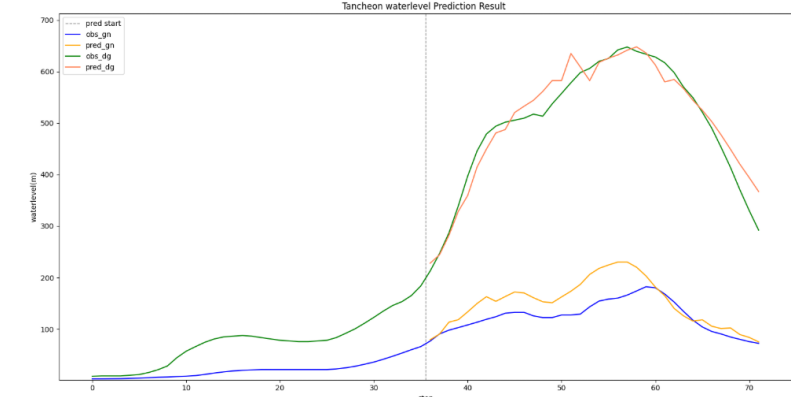
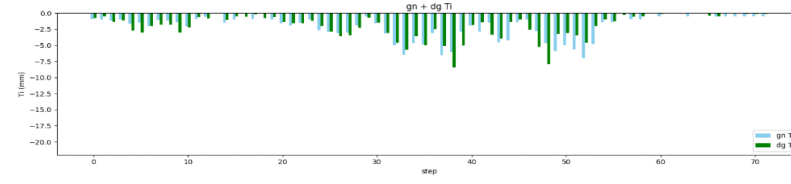
● 궁내교 ● 궁내교 예측
● 대곡교 ● 대곡교 예측 * model_(gn/dg)_flow



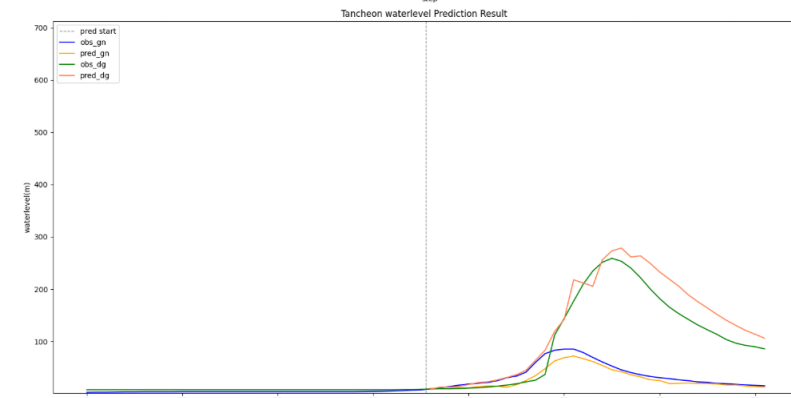
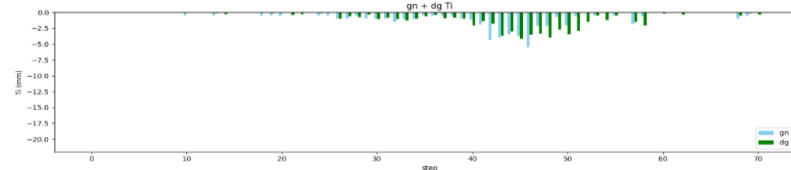
testdata_200802



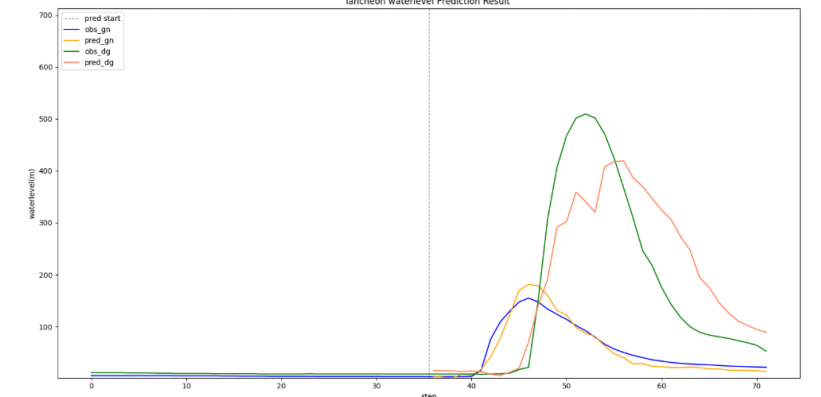
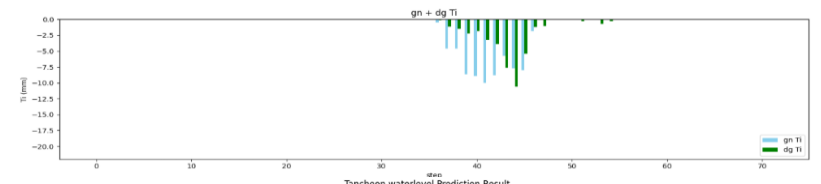
testdata_240723



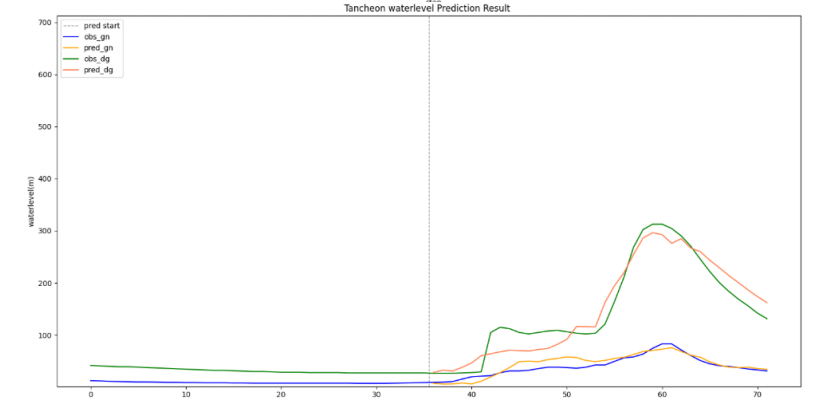
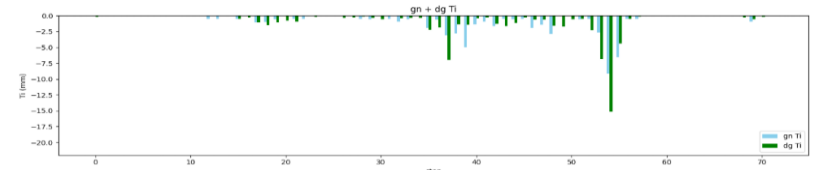
testdata_220712



testdata_250716



testdata_230711



testdata_250814

대곡교 컬럼 제외

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0005
 - Loss_function = MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	1.12	5.11	0.0

학습 , 검증 : 총 67개 中 60개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 7개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 55번 (42개)
- 검증 : 20230629 56번 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

대곡교 컬럼 제외

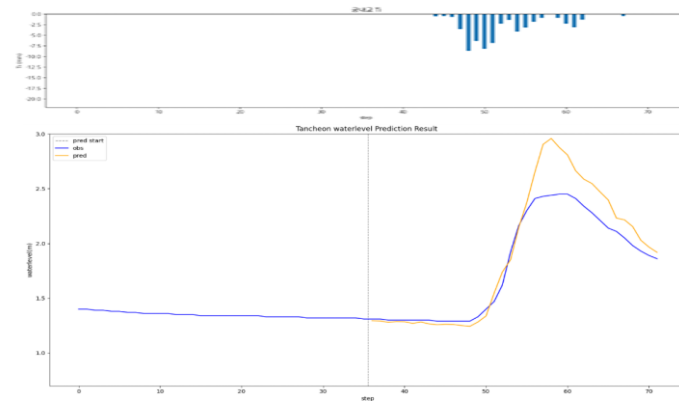
- » 대곡교 정보 제거 → 궁내교 예측에도 악영향
- » 상·하류 수위 상관성이 존재

기본 모델 대비 평균 R2 : 궁내교 ▼ **저하** (- 17.5%)

궁내교				
Test data	r2	mse	mae	rmse
20.08.02	0.8253	0.0369	0.1334	0.1922
22.07.12	-1.2836	0.0906	0.2571	0.3010
23.07.11	0.9267	0.0204	0.1150	0.1431
24.07.23	0.7720	0.0184	0.1050	0.1358
25.07.16	0.4997	0.0501	0.1578	0.2240
25.08.14	0.7909	0.0147	0.0918	0.1213

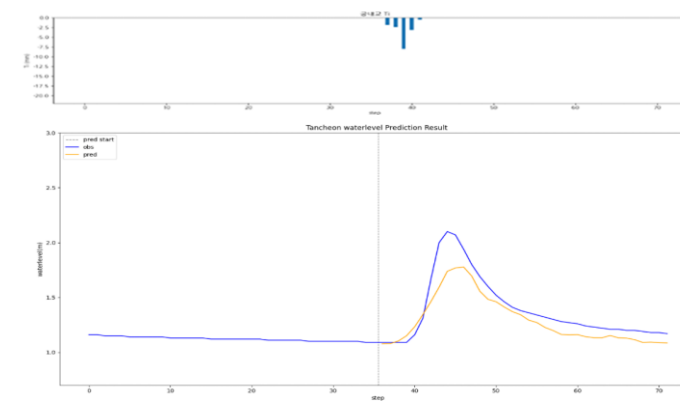
testdata_200802

R2 : 0.8253



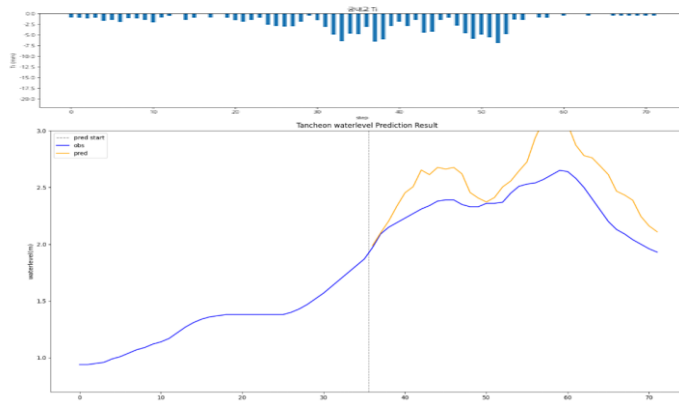
testdata_240723

R2 : 0.7720



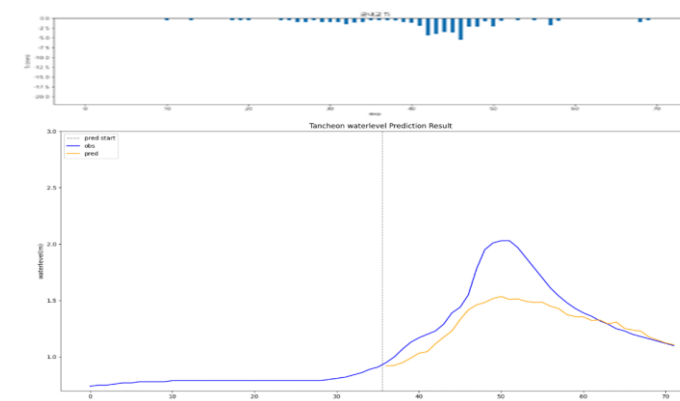
testdata_220712

R2 : -1.2836



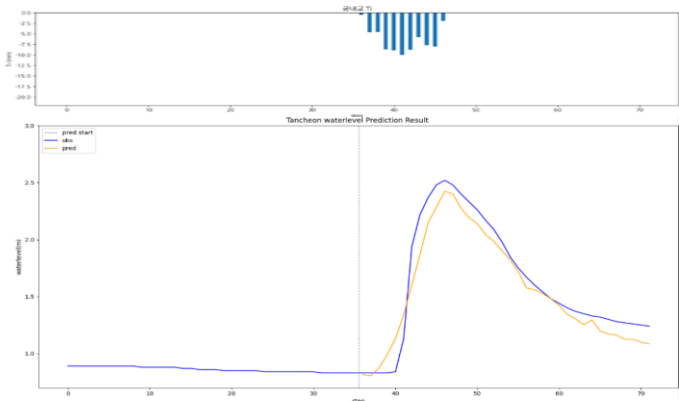
testdata_250716

R2 : 0.4997



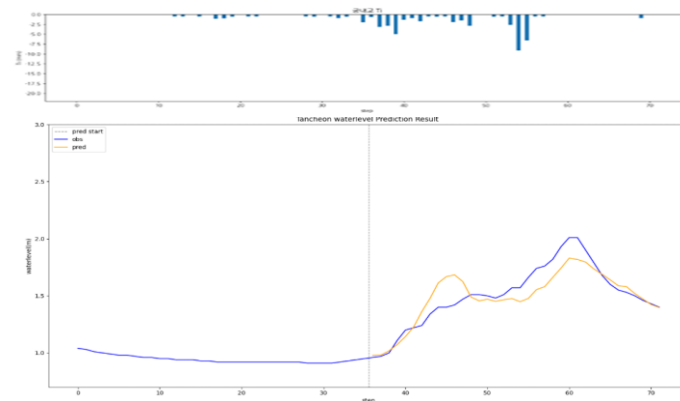
testdata_230711

R2 : 0.9267



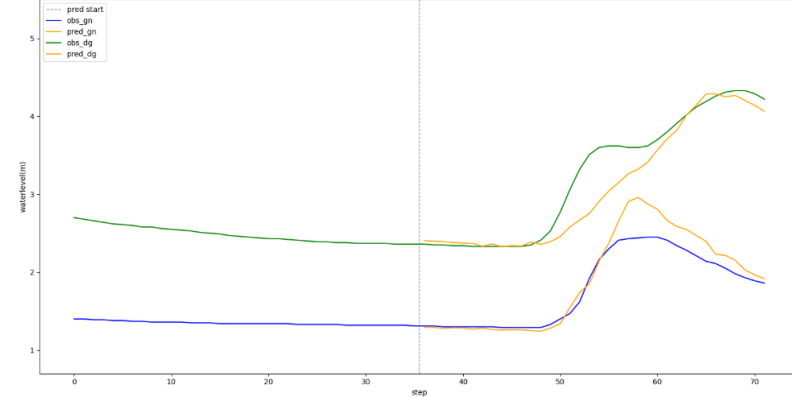
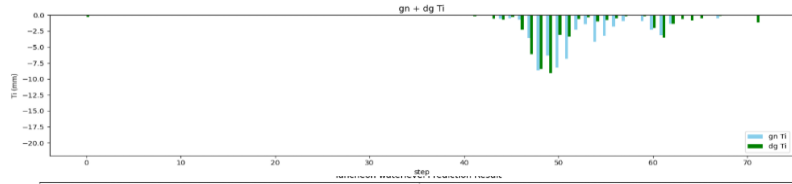
testdata_250814

R2 : 0.7909

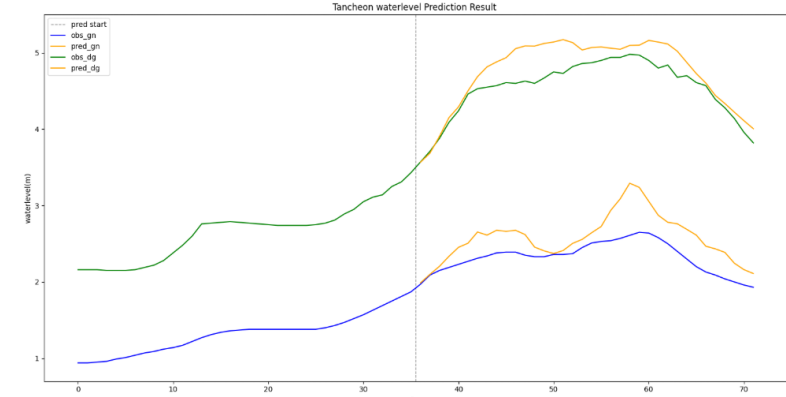
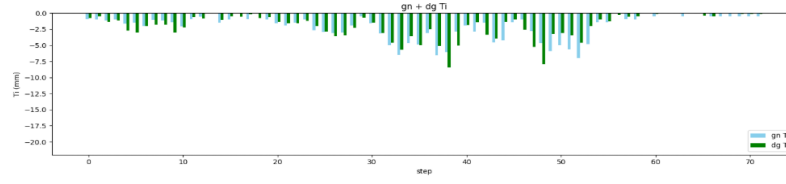


궁내교 + 대곡교

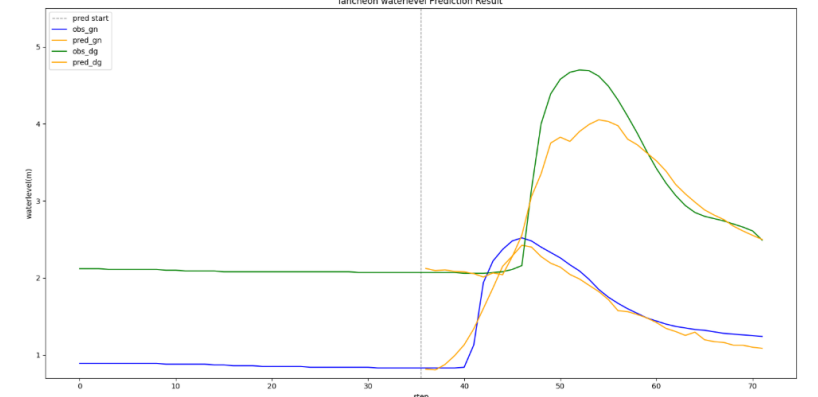
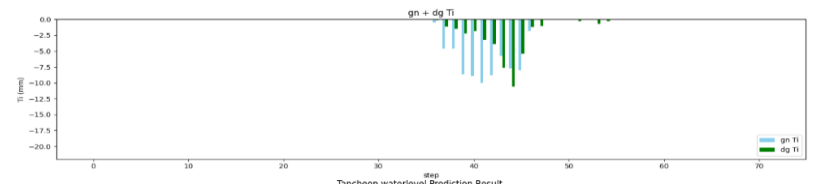
● 궁내교
● 대곡교
● 예측
* model_(gn/dg)_none



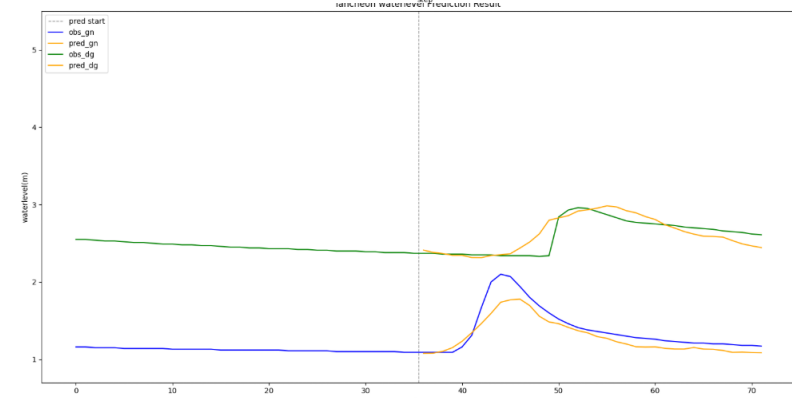
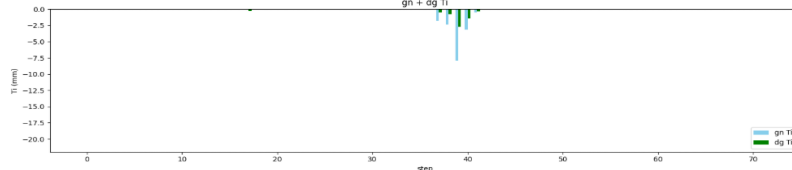
testdata_200802



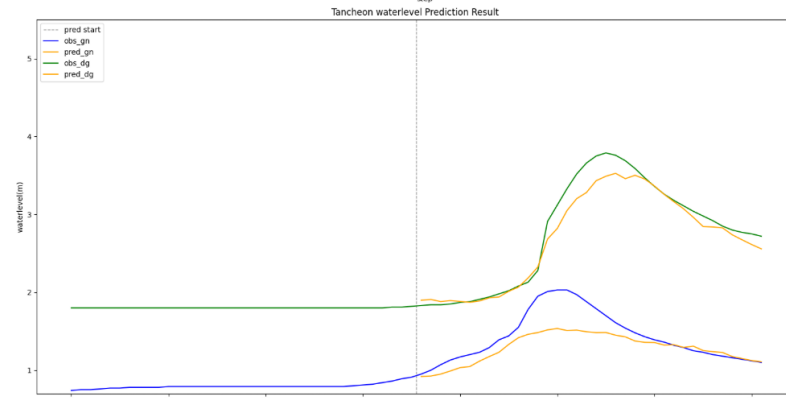
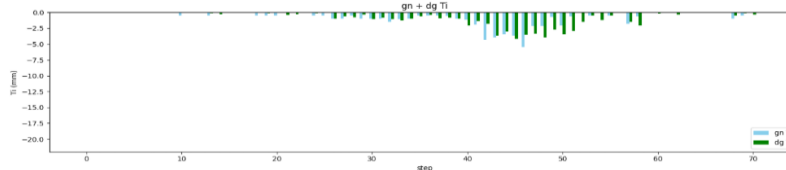
testdata_220712



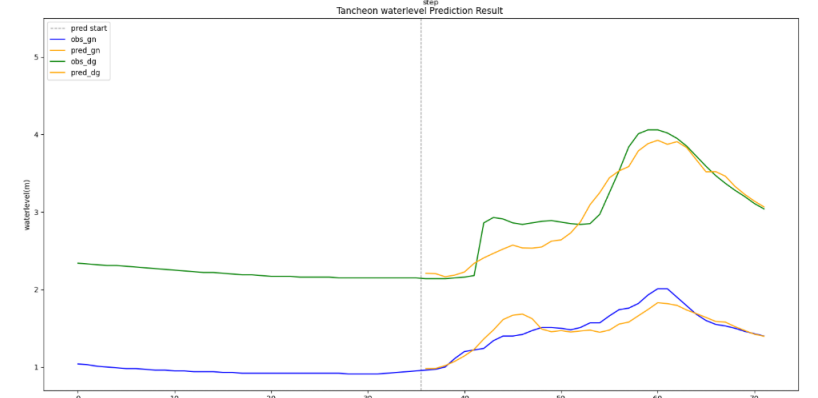
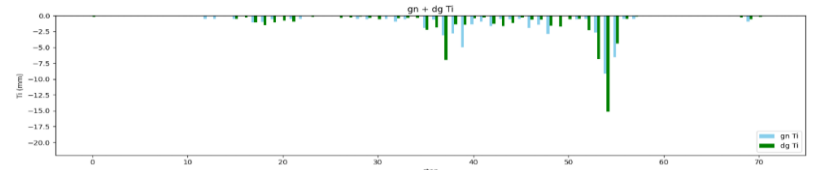
testdata_230711



testdata_240723



testdata_250716



testdata_250814

2020년 이후 데이터 학습

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0005
 - Loss_function = MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 40개 中 34개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2020 ~ 20240920 (24개)
- 검증 : 20240921 ~ 250919 85번 (7개)
- 테스트 : 250919 86번 ~ 251006 (3개)

2020년 이후 데이터 학습

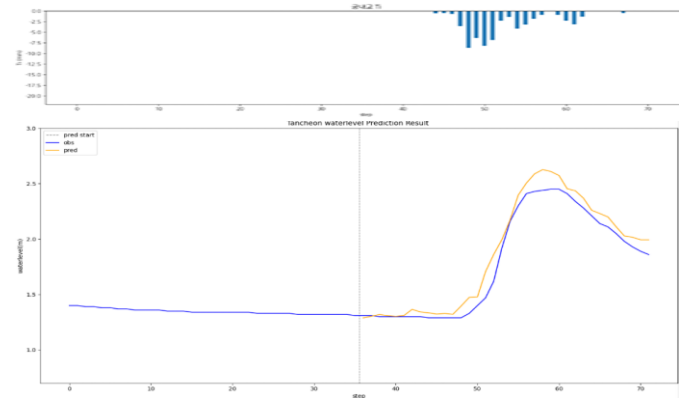
- » 궁내교 : 최근 패턴 집중 → 소폭 개선
- » 대곡교 : 학습 데이터 감소 → 일반화 성능 저하
- » 지점별 데이터 분리 ???

기본 모델 대비 평균 R2 : 궁내교 ▲ 개선 (+ 2.5 %) , 대곡교 ▲ 개선 (+ 0.5 %)

Test data	궁내교				대곡교			
	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.9513	0.0102	0.0814	0.1013	0.8845	0.0704	0.1797	0.2653
22.07.12	-0.8548	0.0735	0.2146	0.2712	-0.1561	0.1619	0.3550	0.4023
23.07.11	0.8751	0.0349	0.1217	0.1868	0.7304	0.2453	0.3701	0.4953
24.07.23	0.8136	0.0150	0.0610	0.1227	0.6077	0.0186	0.0981	0.1365
25.07.16	0.8213	0.0179	0.0927	0.1338	0.8647	0.0624	0.1862	0.2498
25.08.14	0.7835	0.0152	0.0984	0.1235	0.8840	0.0402	0.1572	0.2006

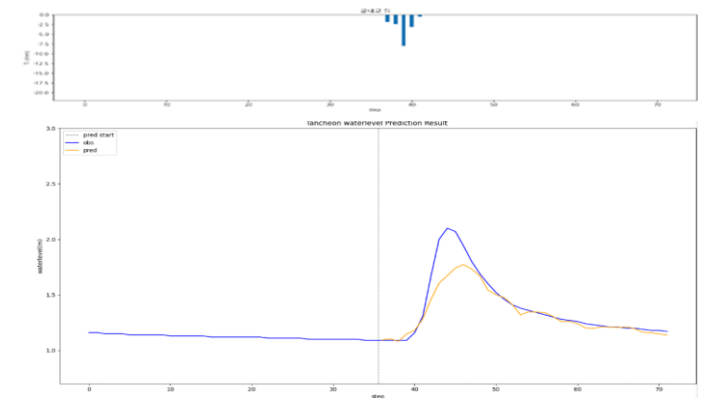
testdata_200802

R2 : 0.9513



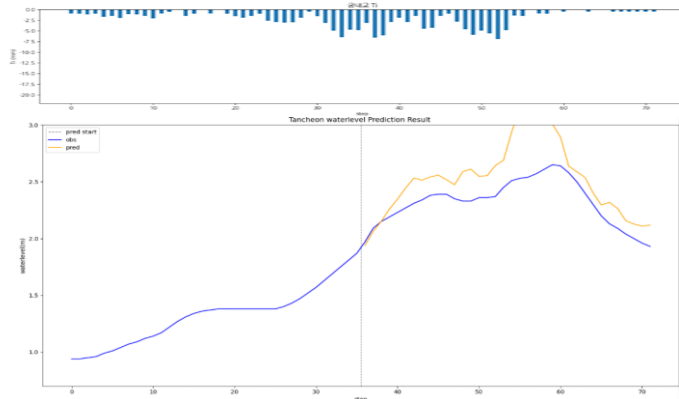
testdata_240723

R2 : 0.8136



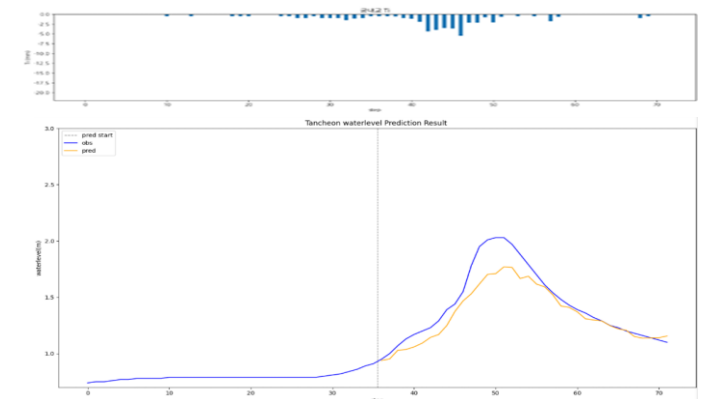
testdata_220712

R2 : -0.8548



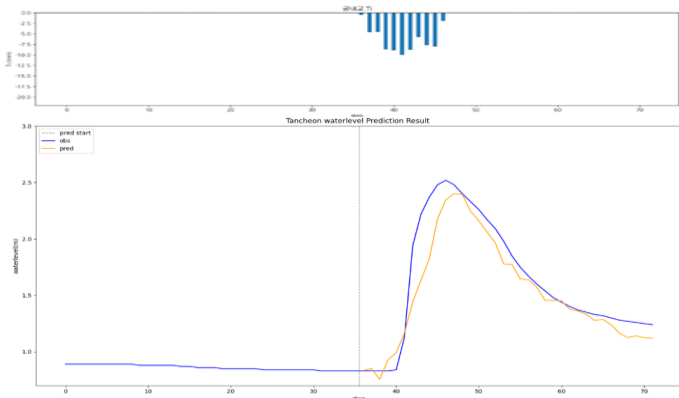
testdata_250716

R2 : 0.8213



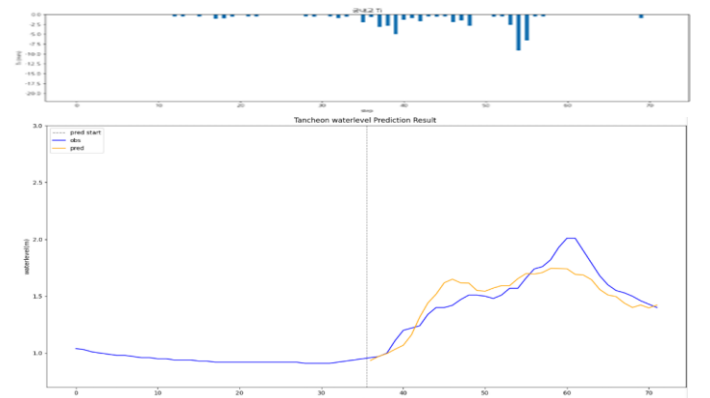
testdata_230711

R2 : 0.8751



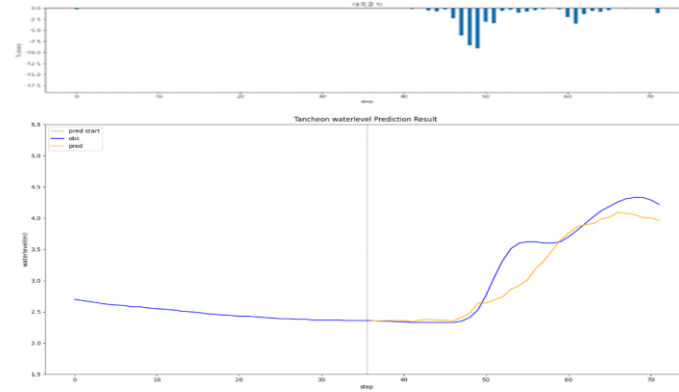
testdata_250814

R2 : 0.7835



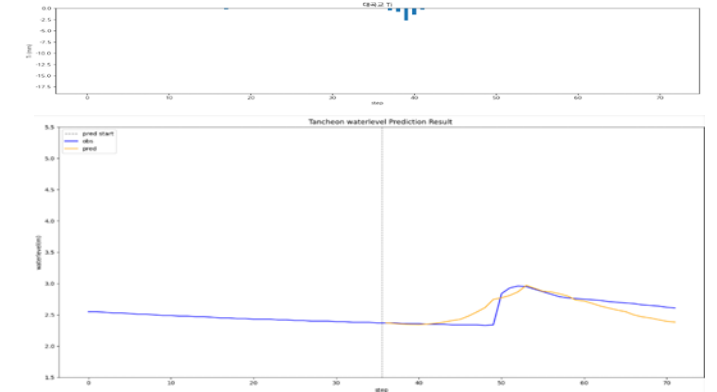
testdata_200802

R2 : 0.8845



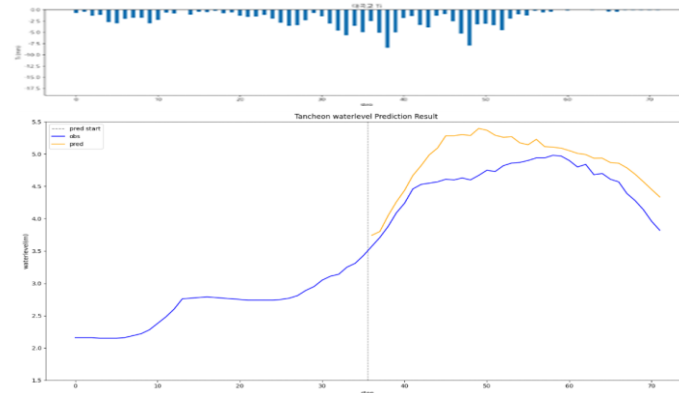
testdata_240723

R2 : 0.6077



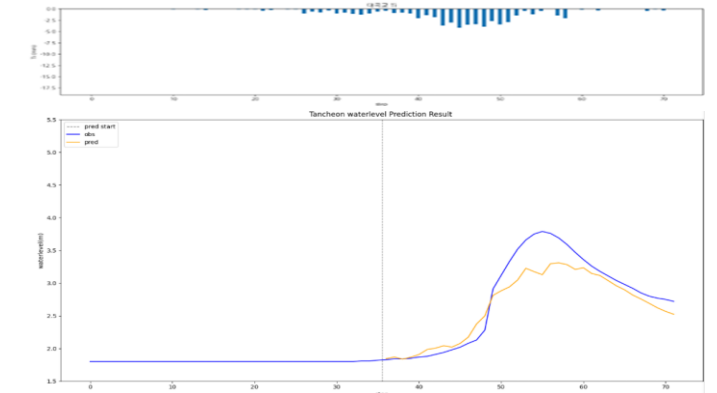
testdata_220712

R2 : -0.1561



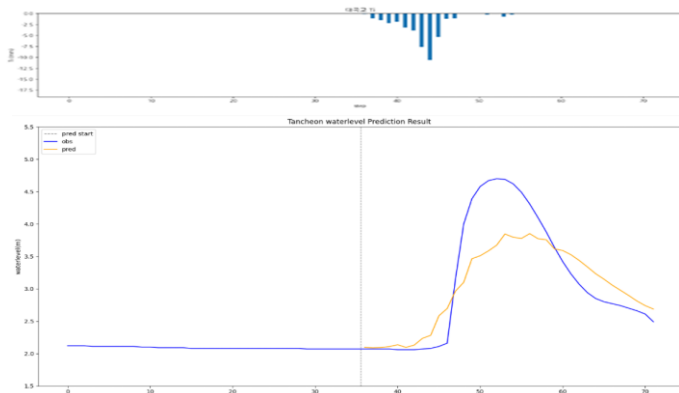
testdata_250716

R2 : 0.8647



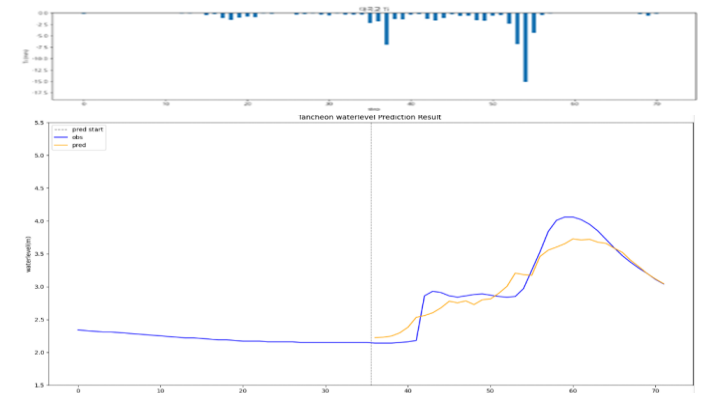
testdata_230711

R2 : 0.7304



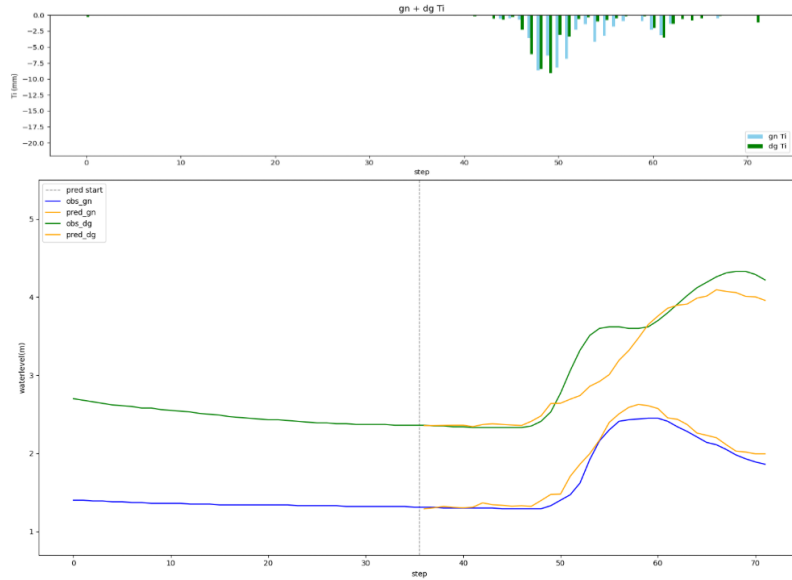
testdata_250814

R2 : 0.8840

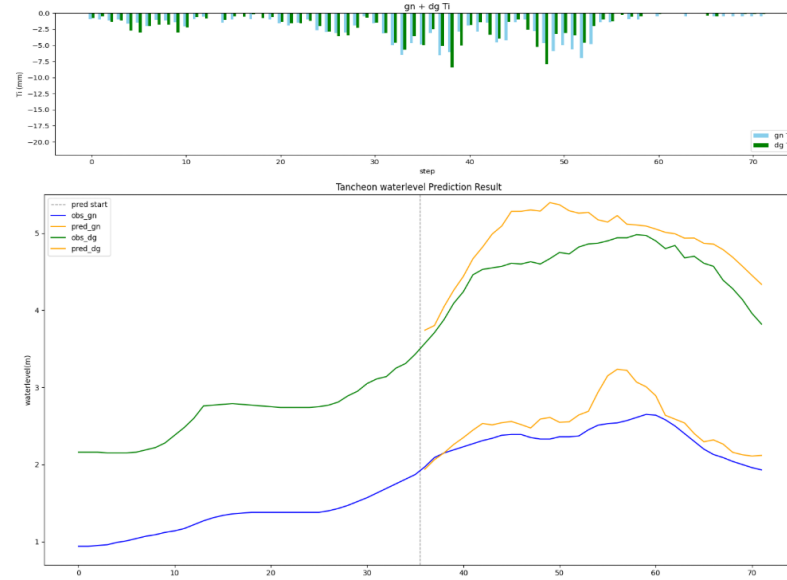


궁내교 + 대곡교

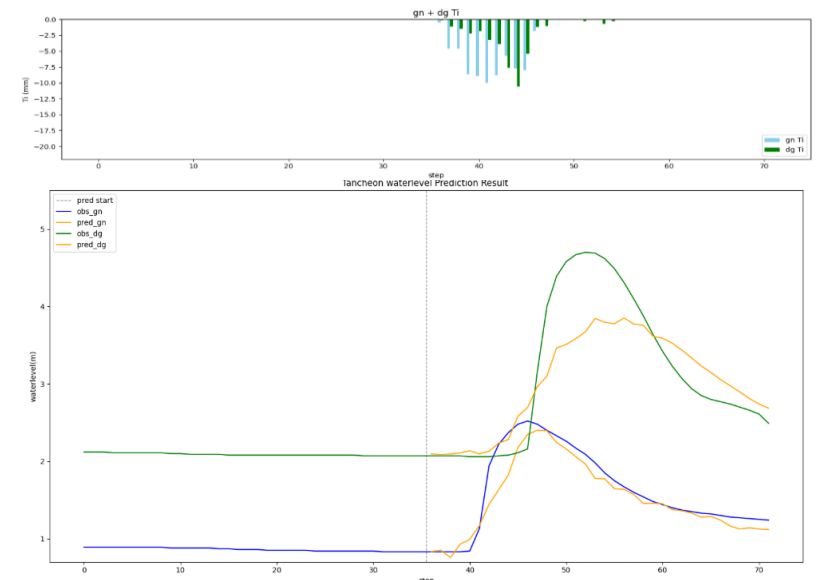
● 궁내교
● 대곡교
● 예측
* model_(gn/dg)_20



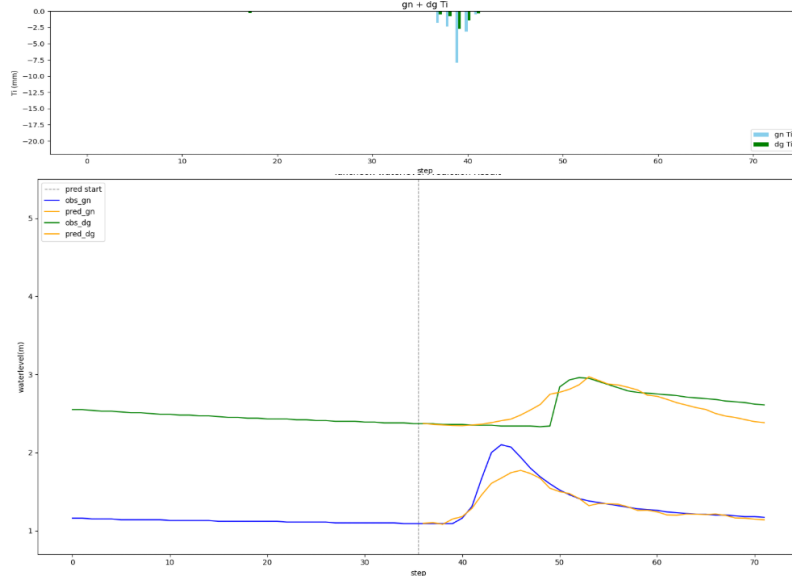
testdata_200802



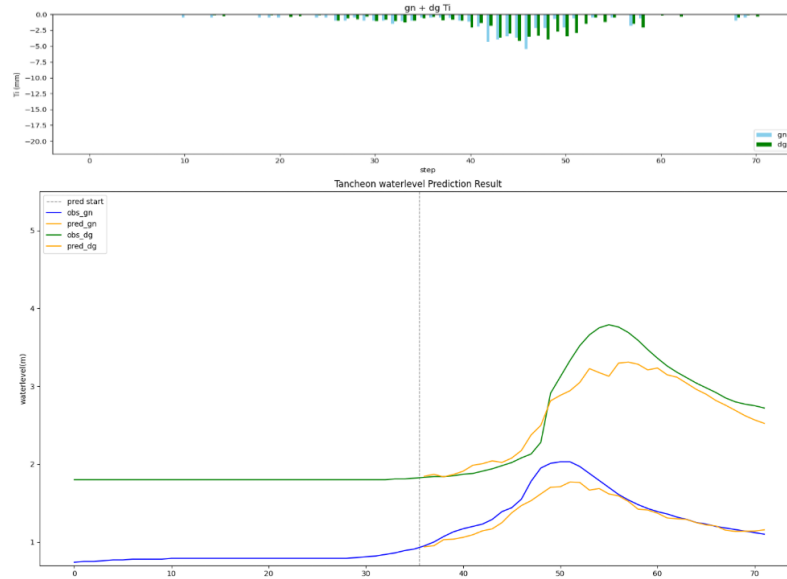
testdata_220712



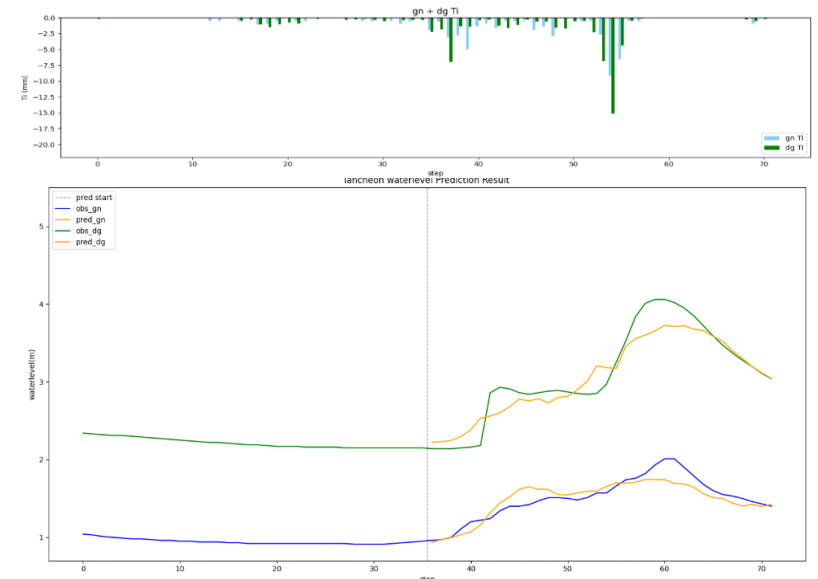
testdata_230711



testdata_240723



testdata_250716



testdata_250814

3시간 예측

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0005
 - Loss_function = MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 67개 中 61개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

3시간 예측

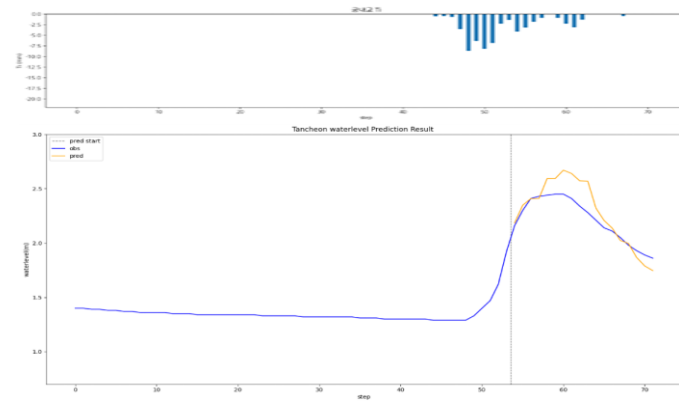
- » 이벤트별 변동성 가장 큼
- » 대곡교에서 -4.21 R² 발생
- » R² 기준으로는 손해

기본 모델 대비 평균 R2 : 궁내교 ▼저하 (- 13.4%) , 대곡교 ▼저하 (-26.5%)

궁내교					대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.5417	0.0184	0.1049	0.1359	0.4341	0.0495	0.1670	0.2225
22.07.12	0.8669	0.0084	0.0795	0.0921	0.9659	0.0042	0.0452	0.0651
23.07.11	0.9006	0.0031	0.0452	0.0563	0.7131	0.1341	0.3316	0.3663
24.07.23	0.5351	0.0014	0.0333	0.0384	-4.2174	0.0347	0.1570	0.1863
25.07.16	0.6605	0.0135	0.0984	0.1165	0.7288	0.0379	0.1619	0.1947
25.08.14	0.2037	0.0287	0.1493	0.1695	0.8953	0.0140	0.0854	0.1186

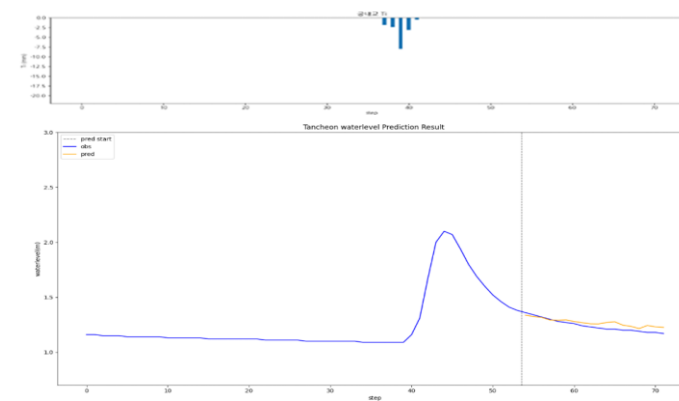
testdata_200802

R2 : 0.5417



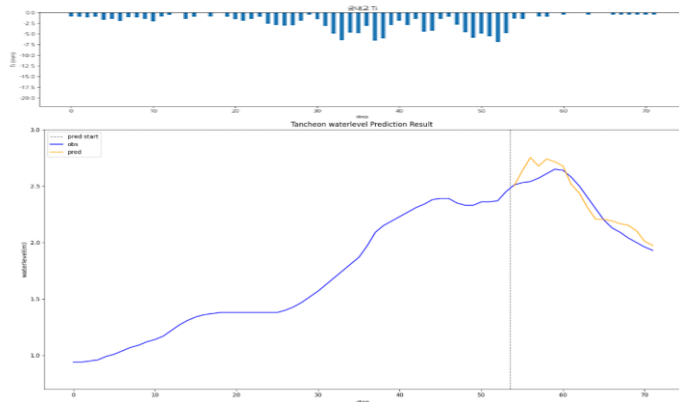
testdata_240723

R2 : 0.5351



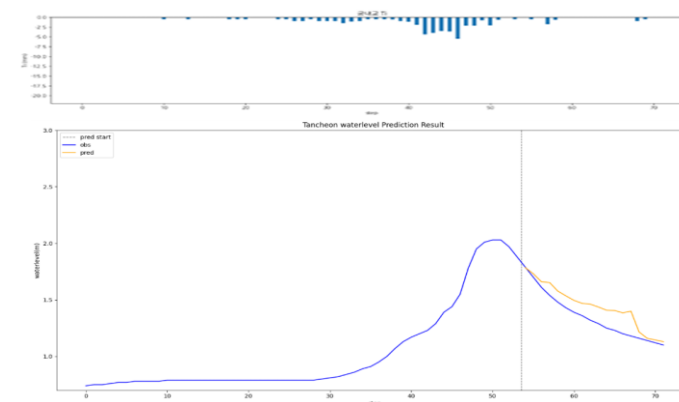
testdata_220712

R2 : 0.8669



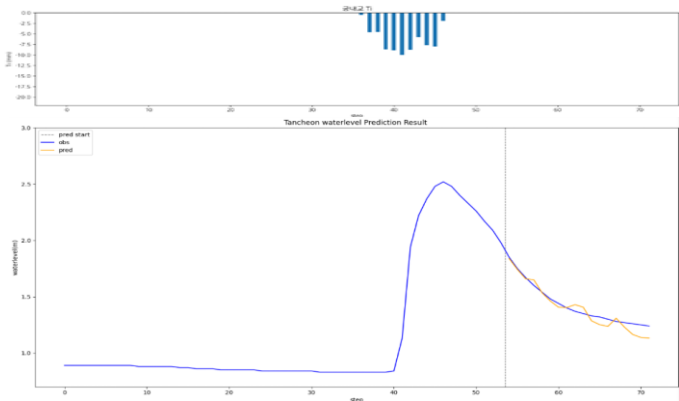
testdata_250716

R2 : 0.6605



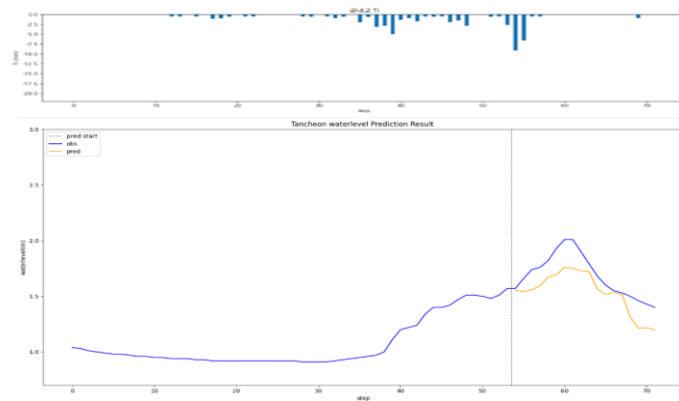
testdata_230711

R2 : 0.9006



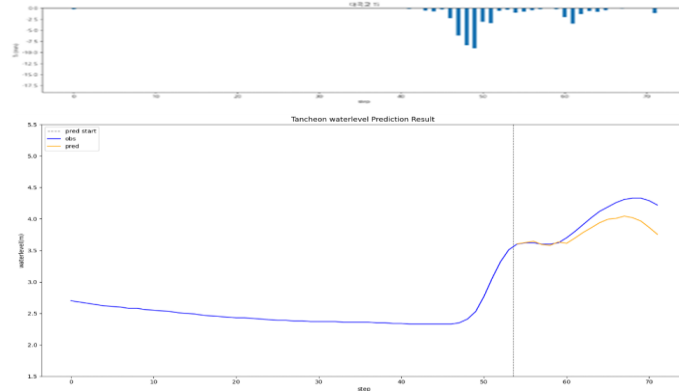
testdata_250814

R2 : 0.2037



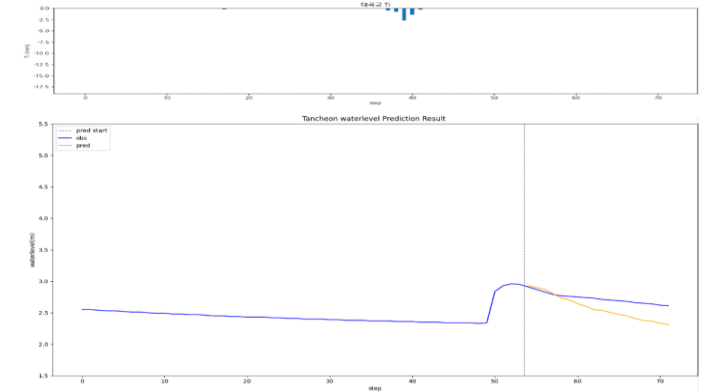
testdata_200802

R2 : 0.4341



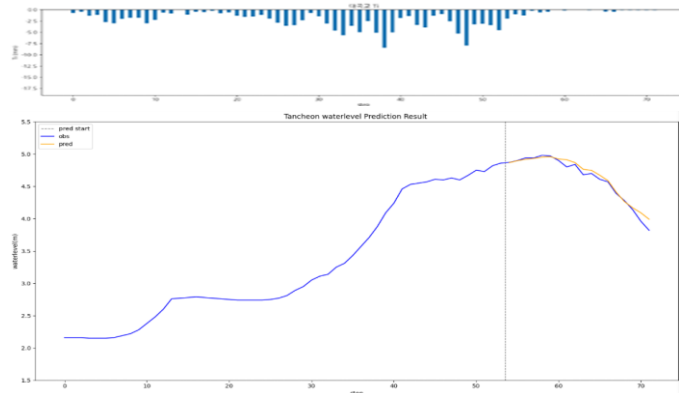
testdata_240723

R2 : -4.2174



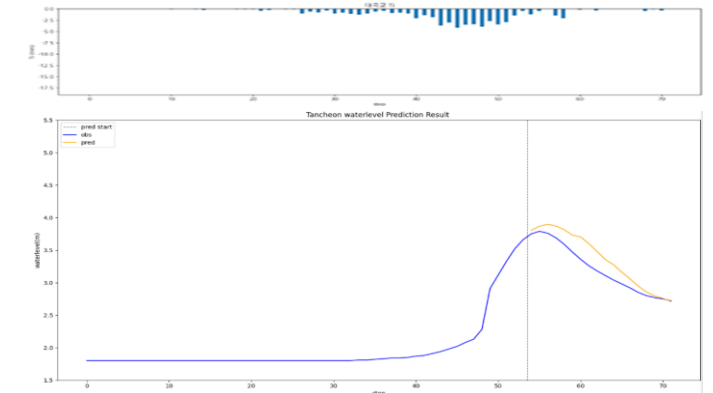
testdata_220712

R2 : 0.9659



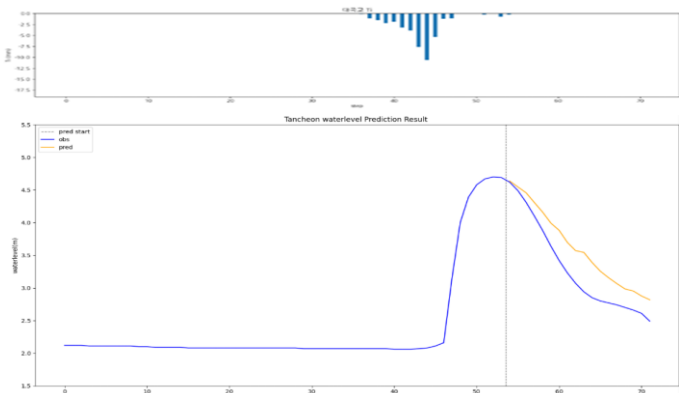
testdata_250716

R2 : 0.7288



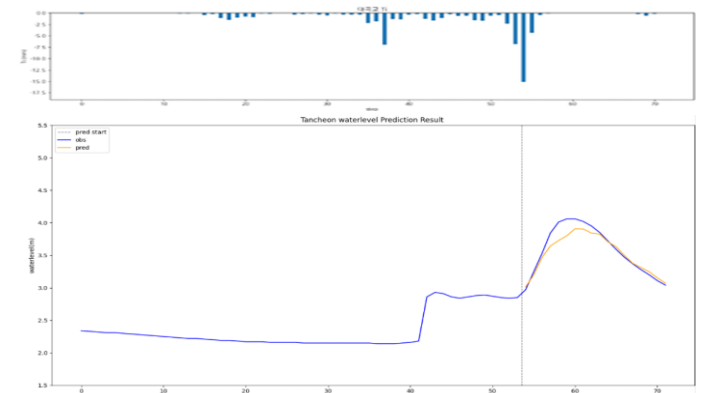
testdata_230711

R2 : 0.7131



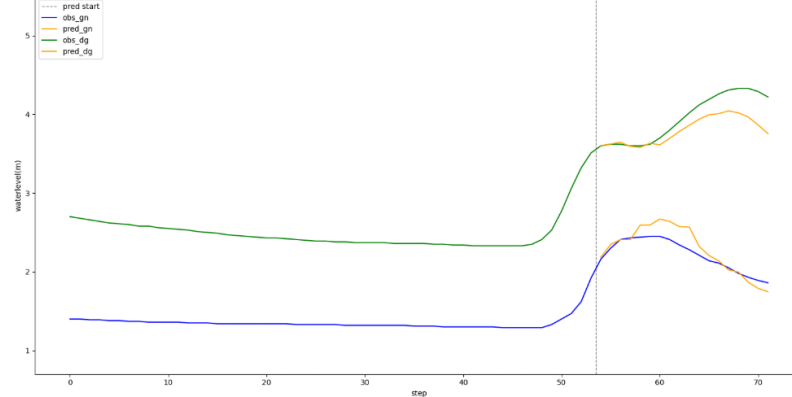
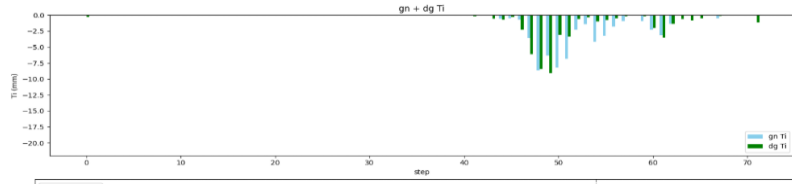
testdata_250814

R2 : 0.8953

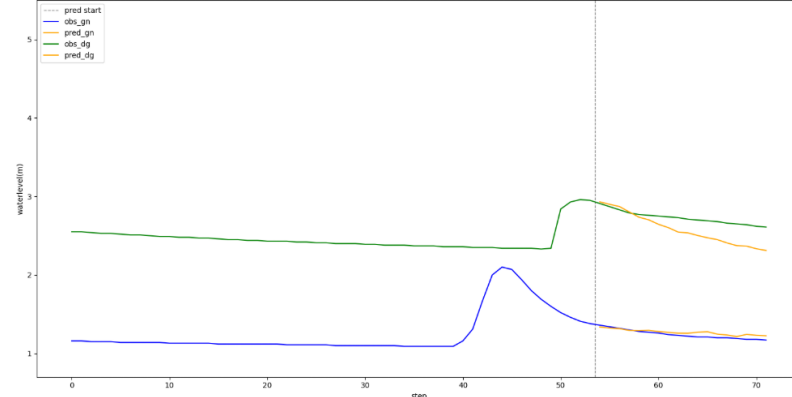
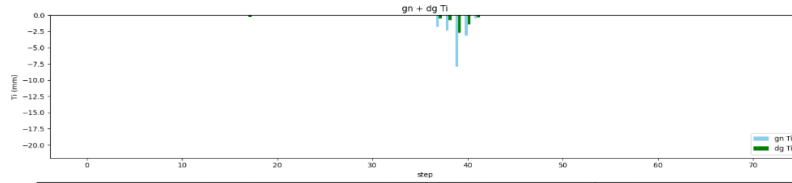


궁내교 + 대곡교

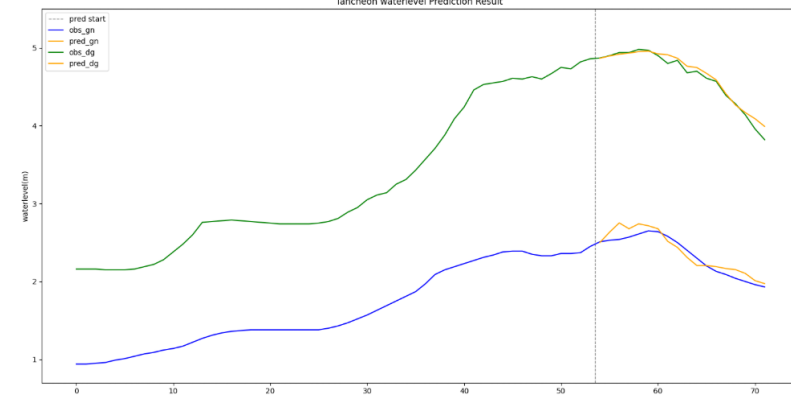
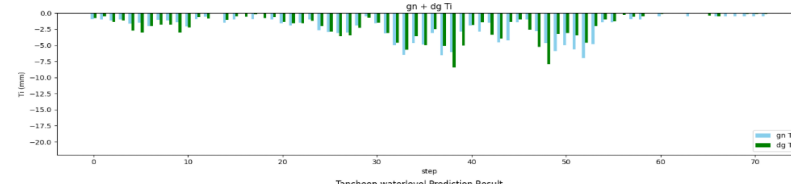
● 궁내교
● 대곡교
● 예측
* model_(gn/dg)_3



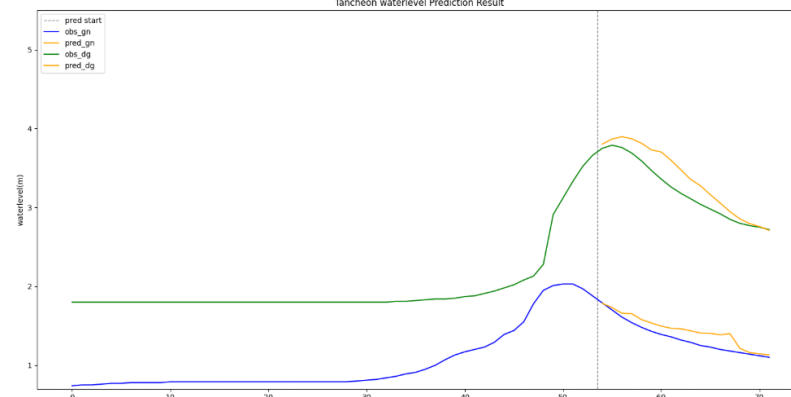
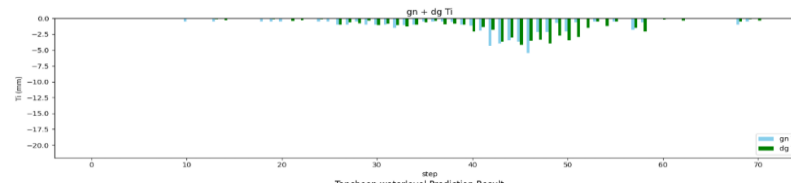
testdata_200802



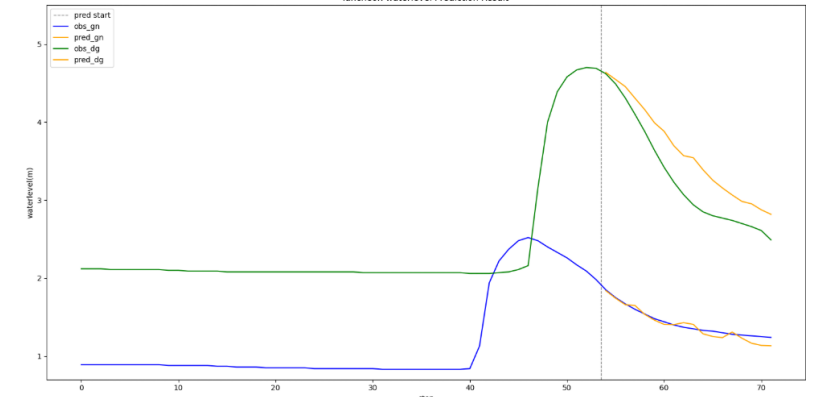
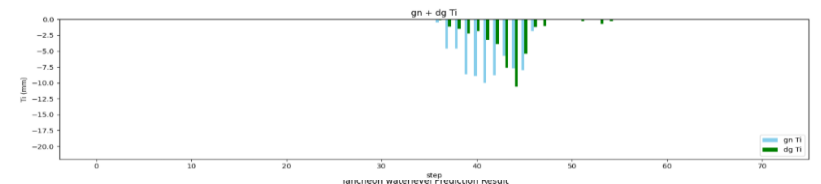
testdata_240723



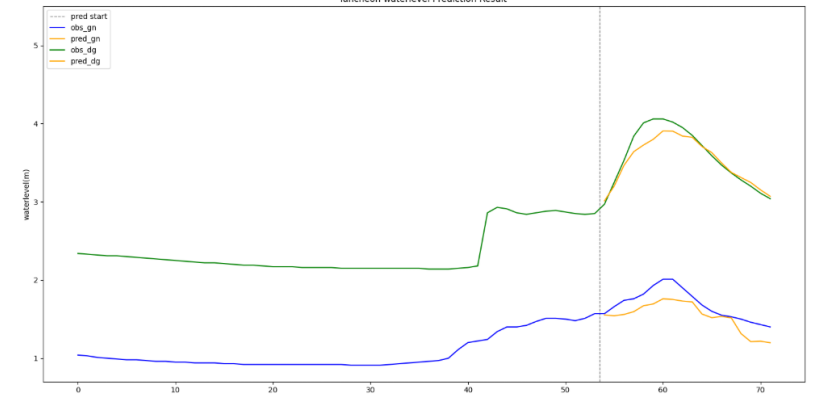
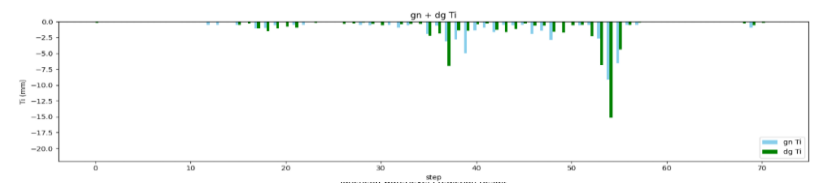
testdata_220712



testdata_250716



testdata_230711



testdata_250814

3시간 예측 (예측 시작 시점 -1시간)

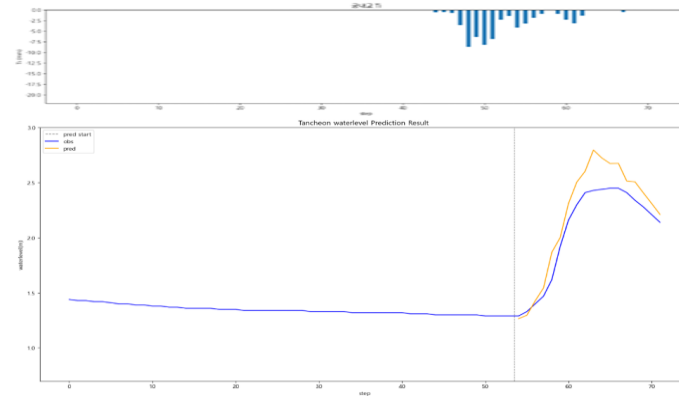
- » 이벤트별 변동성 가장 큼
- » R^2 기준으로는 손해

기본 모델 대비 평균 R^2 : 궁내교 ▼저하 (- 26.6%) , 대곡교 ▼저하 (-25.6%)

궁내교					대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.8214	0.0315	0.1508	0.1774	0.9113	0.0216	0.1275	0.1470
22.07.12	-0.3291	0.0215	0.1190	0.1467	-0.1308	0.0159	0.1159	0.1264
23.07.11	0.9196	0.0107	0.0819	0.1035	0.7288	0.1252	0.3182	0.3538
24.07.23	0.7568	0.0126	0.0911	0.1122	0.5679	0.0126	0.0911	0.1122
25.07.16	0.9456	0.0045	0.0528	0.0671	0.5949	0.0571	0.2051	0.2390
25.08.14	-0.9700	0.0607	0.2157	0.2464	0.8942	0.0257	0.1387	0.1604

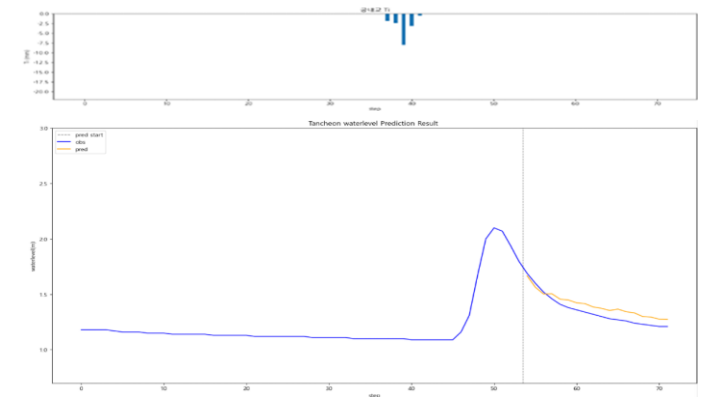
testdata_200802

R2 : 0.8214



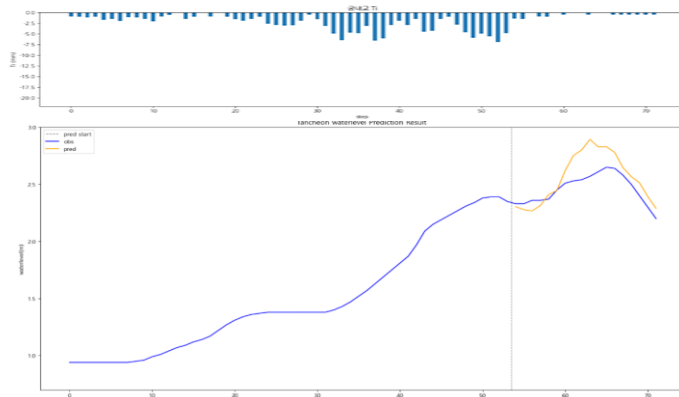
testdata_240723

R2 : 0.7568



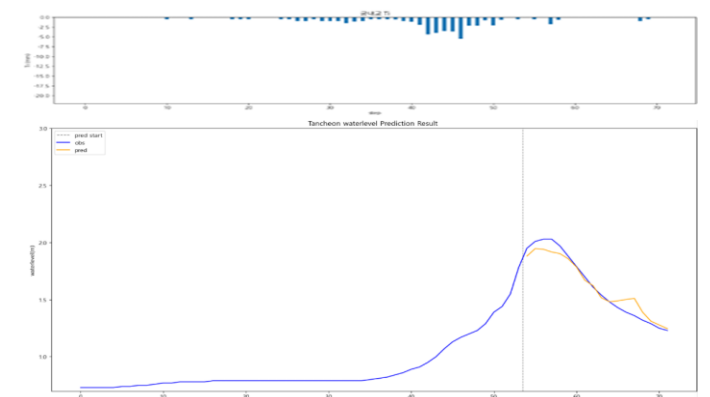
testdata_220712

R2 : -0.3291



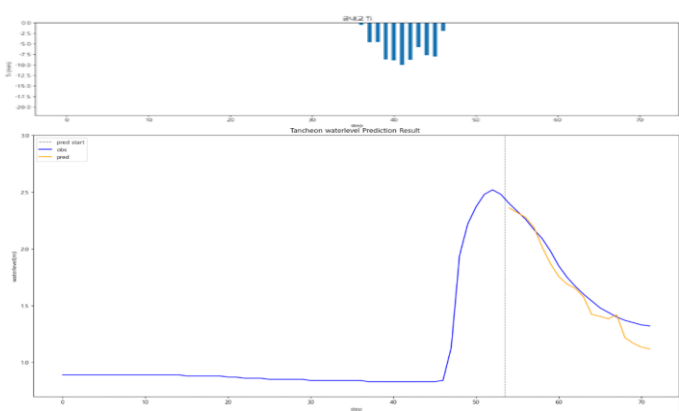
testdata_250716

R2 : 0.9456



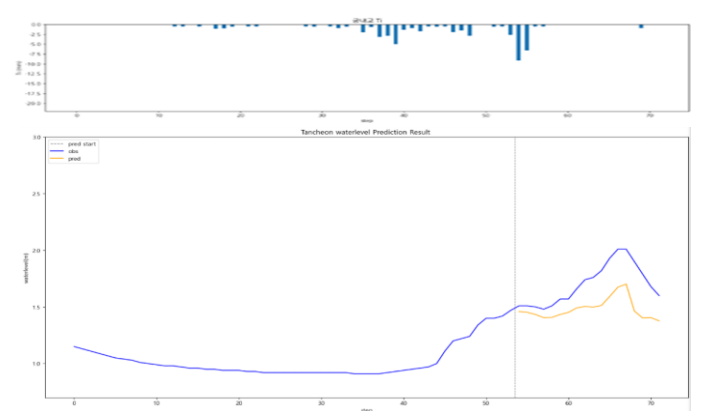
testdata_230711

R2 : 0.9196



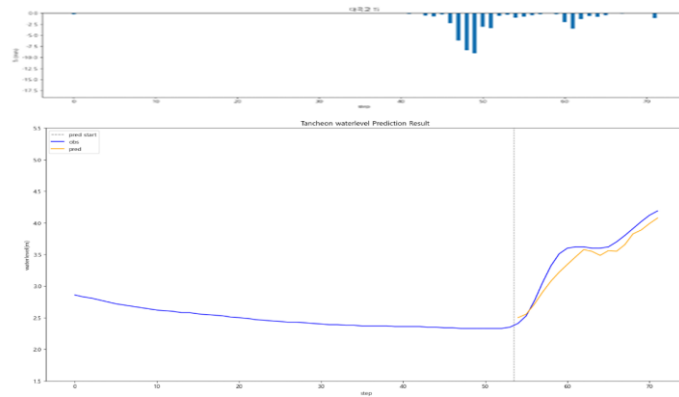
testdata_250814

R2 : -0.9700



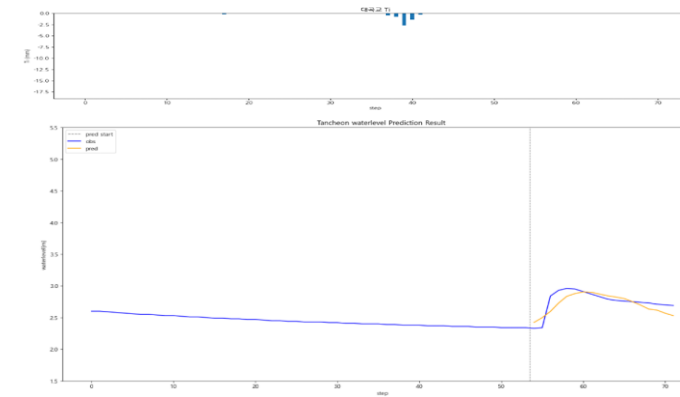
testdata_200802

R2 : 0.9113



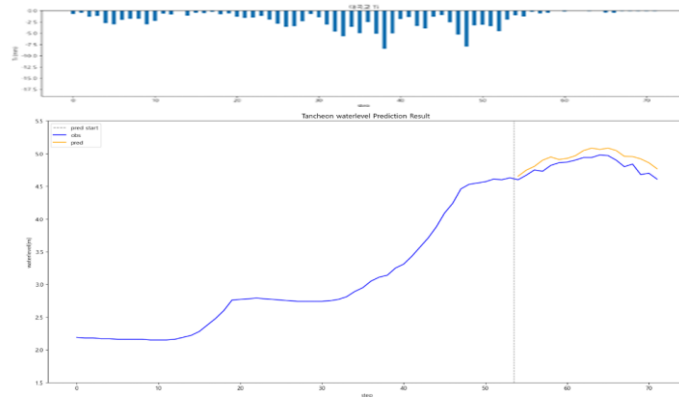
testdata_240723

R2 : 0.5679



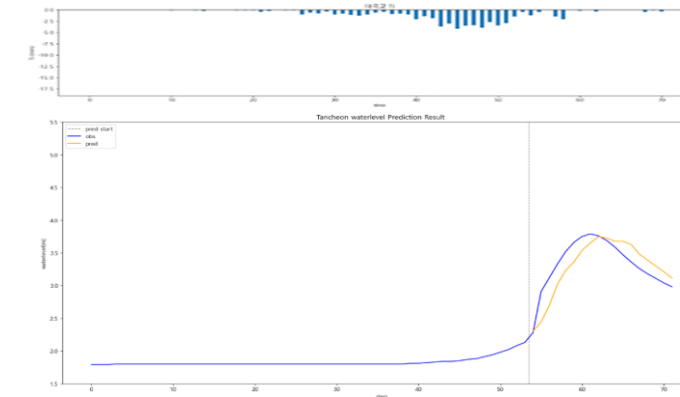
testdata_220712

R2 : -0.1308



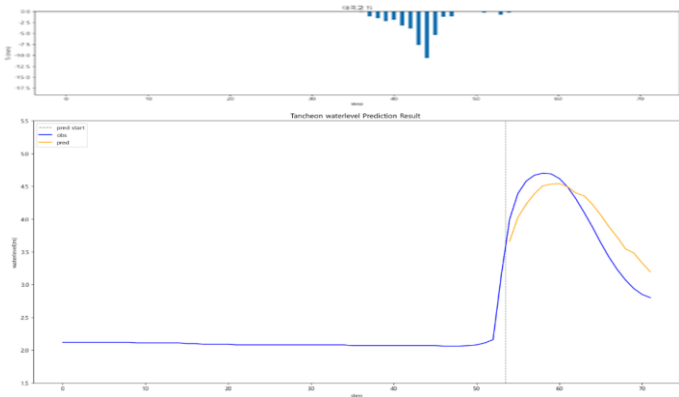
testdata_250716

R2 : 0.5949



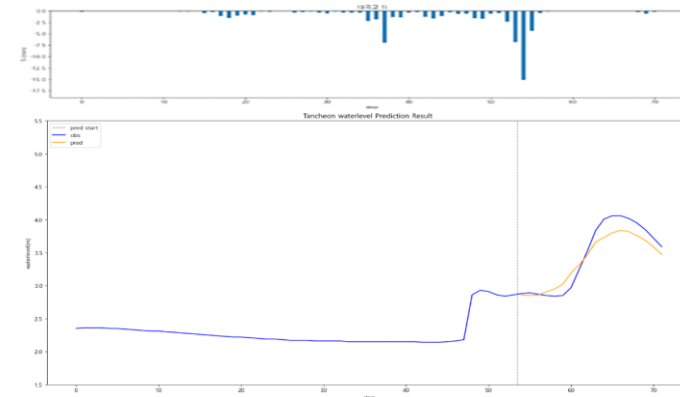
testdata_230711

R2 : 0.7288



testdata_250814

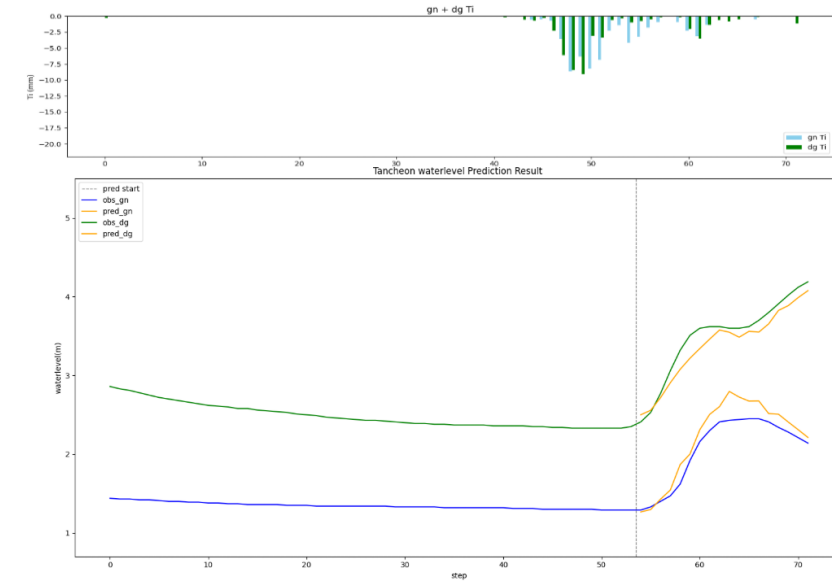
R2 : 0.8942



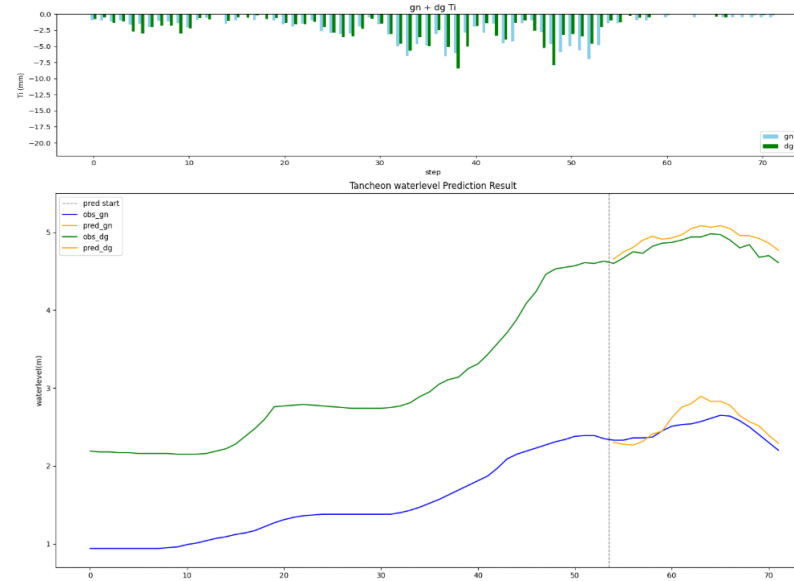
궁내교 + 대곡교

● 궁내교
● 대곡교
● 예측

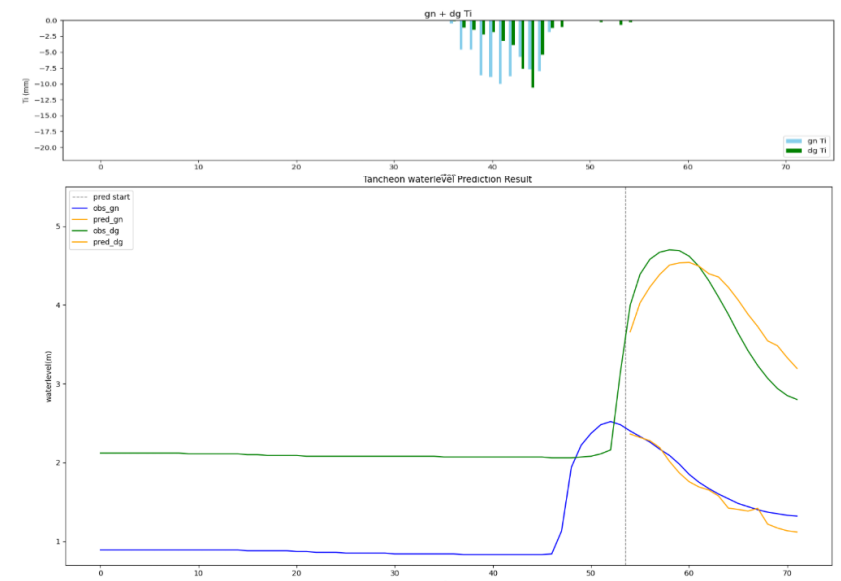
* model_(gn/dg)_3



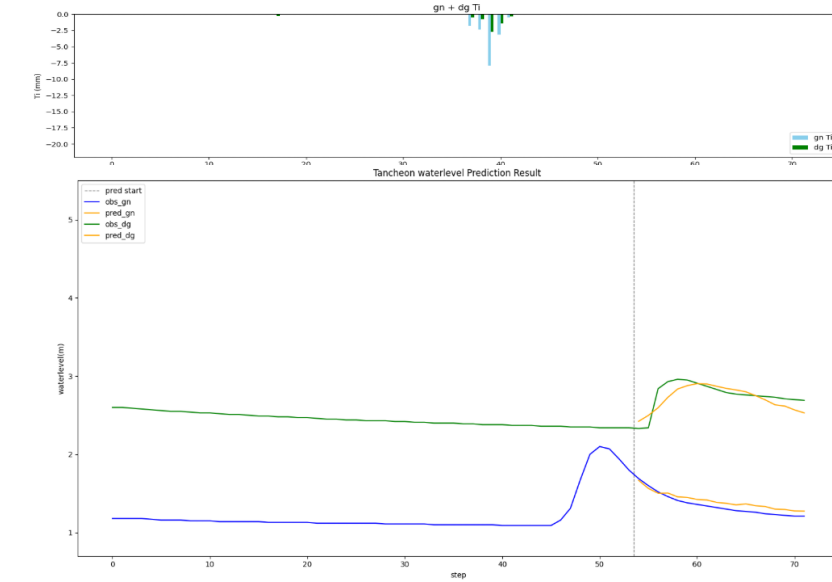
testdata_200802



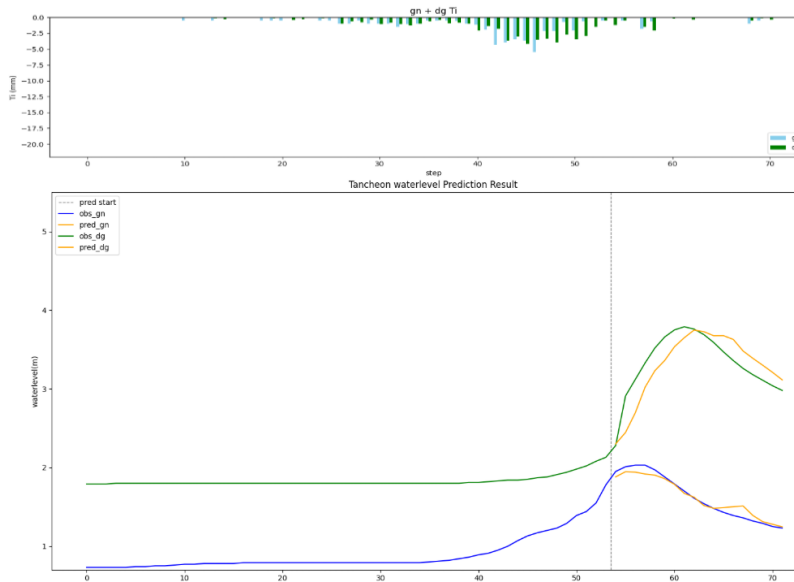
testdata_220712



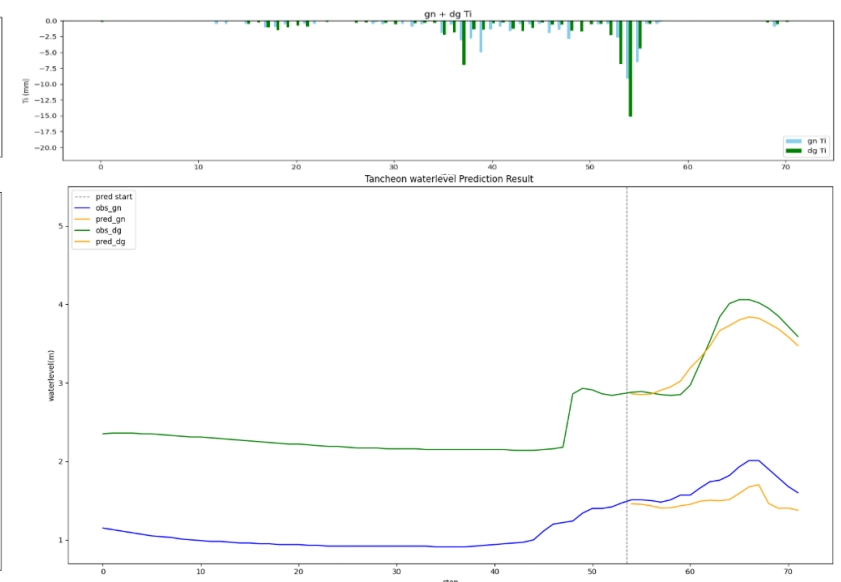
testdata_230711



testdata_240723



testdata_250716



testdata_250814

데이터 편집 (수위 $\times 0.98$, $\times 1.02$)

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0005
 - Loss_function = MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 67개 中 61개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

데이터 편집 (수위 X 0.98)

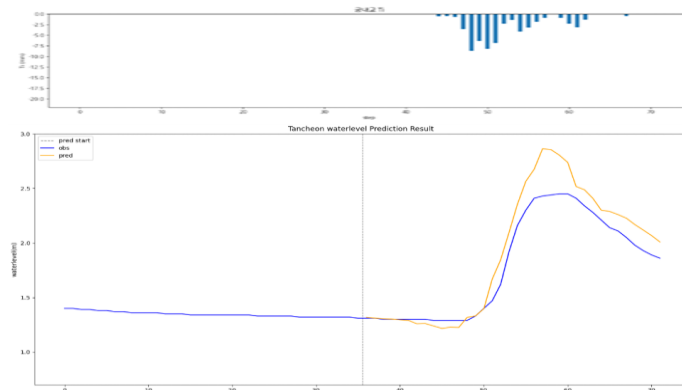
- » 편향 보정 효과는 있음
- » R² 기준으로는 손해

기본 모델 대비 평균 R2 : 궁내교 ▼저하 (- 6.2%) , 대곡교 ▼저하 (- 1.5%)

궁내교					대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.8492	0.8492	0.1350	0.1785	0.8579	0.0867	0.1858	0.2944
22.07.12	0.2864	0.0283	0.1397	0.1682	0.7718	0.0319	0.1560	0.1787
23.07.11	0.8814	0.0331	0.1257	0.1821	0.8246	0.1596	0.3170	0.3995
24.07.23	0.8244	0.0142	0.0692	0.1191	0.5221	0.0227	0.1072	0.1507
25.07.16	0.7627	0.0238	0.1369	0.1542	0.9736	0.0121	0.0856	0.1103
25.08.14	0.4144	0.0412	0.1441	0.2031	0.7718	0.0791	0.2257	0.2813

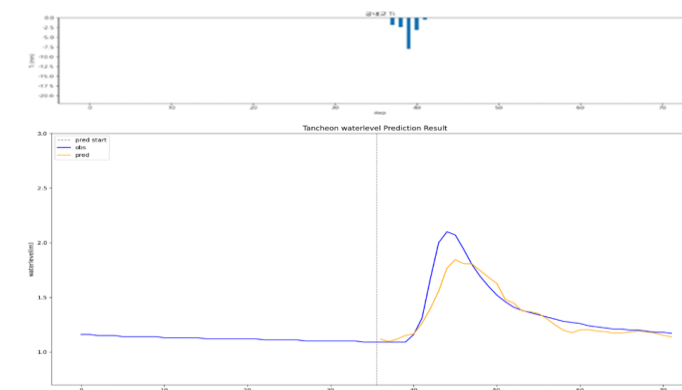
testdata_200802

R2 : 0.8492



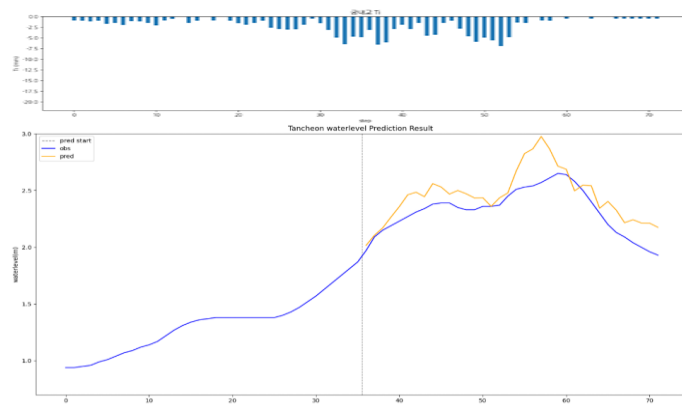
testdata_240723

R2 : 0.8244



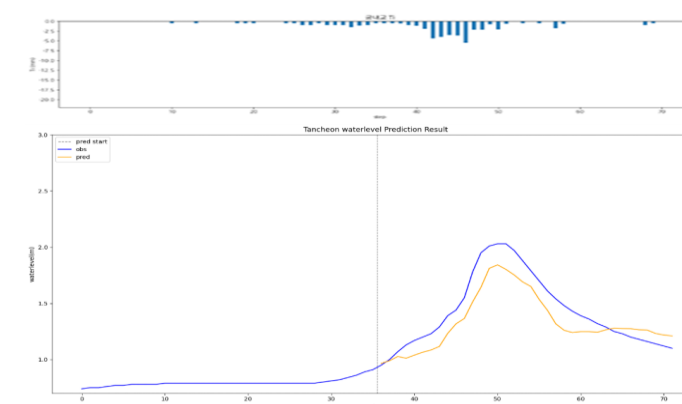
testdata_220712

R2 : 0.2864



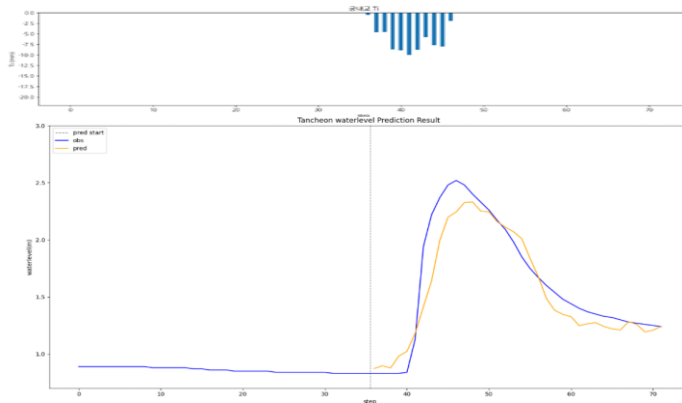
testdata_250716

R2 : 0.7627



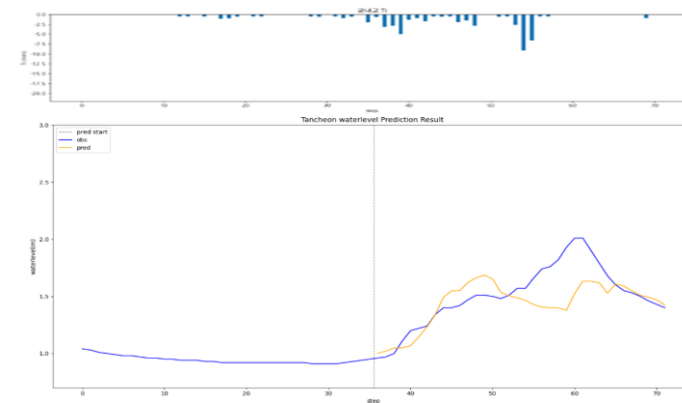
testdata_230711

R2 : 0.8814



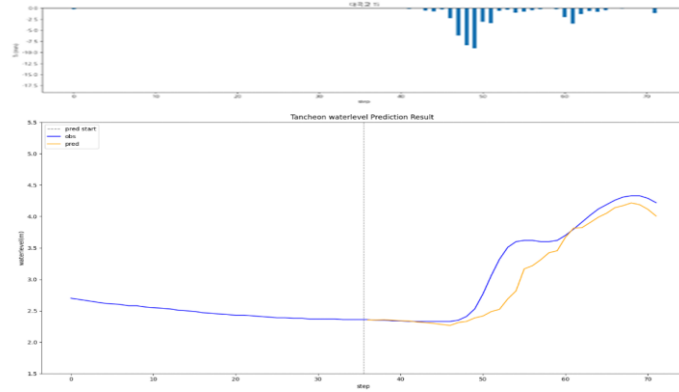
testdata_250814

R2 : 0.4144



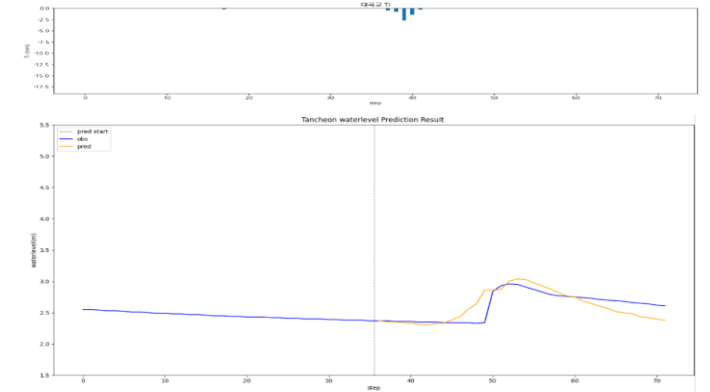
testdata_200802

R2 : 0.8579



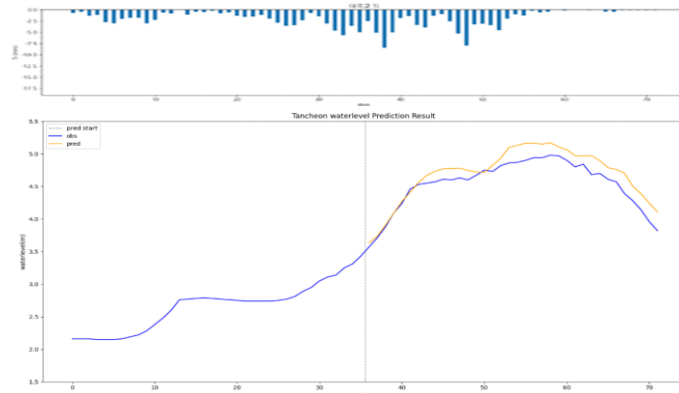
testdata_240723

R2 : 0.5221



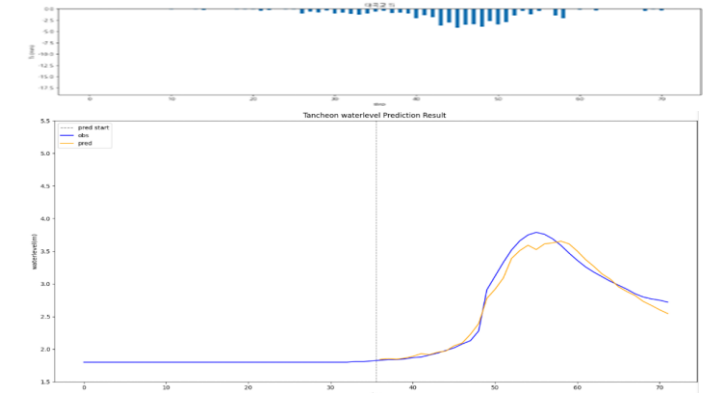
testdata_220712

R2 : 0.7718



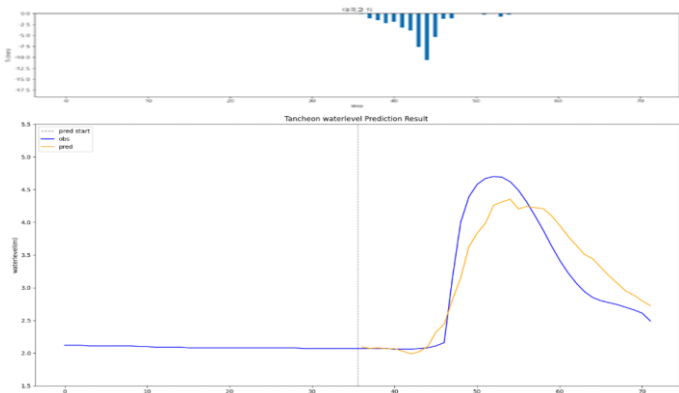
testdata_250716

R2 : 0.9736



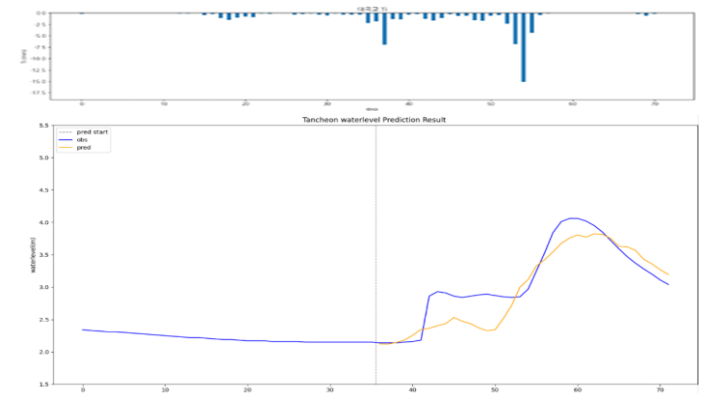
testdata_230711

R2 : 0.8246



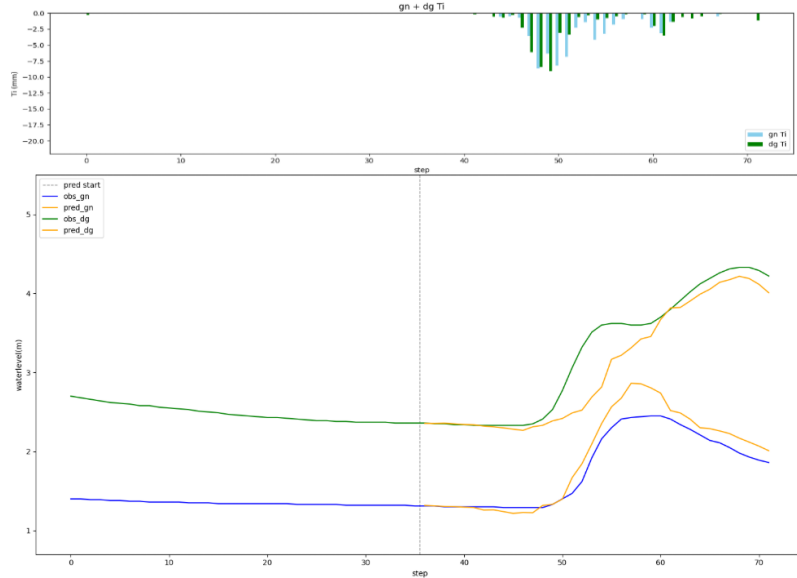
testdata_250814

R2 : 0.7718

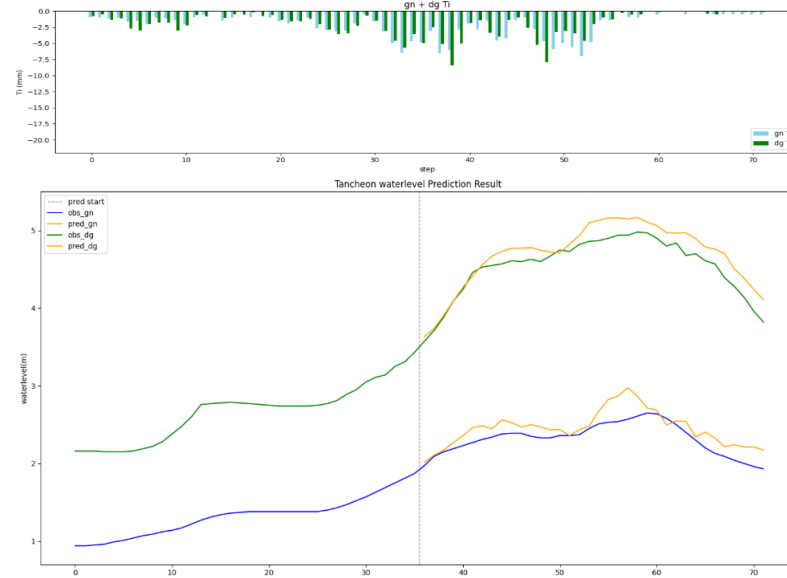


궁내교 + 대곡교

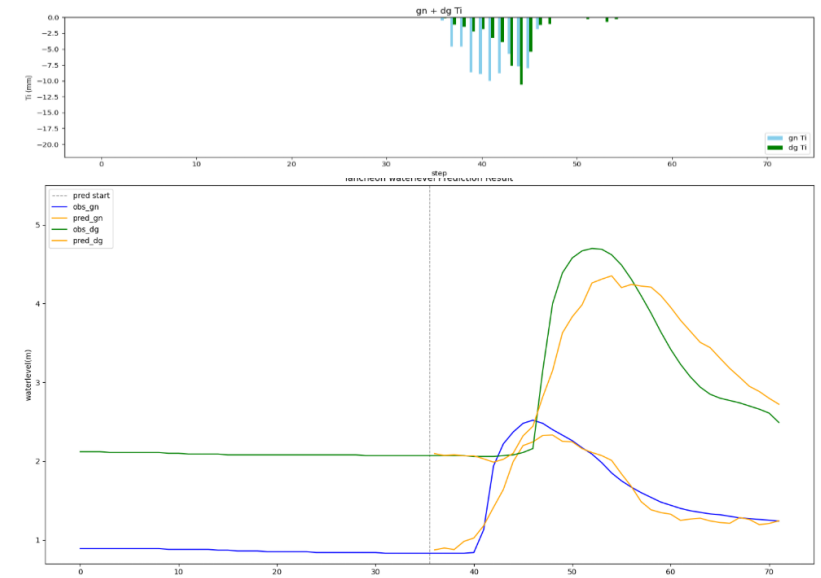
● 궁내교
● 대곡교
● 예측
* model_(gn/dg)_0.98



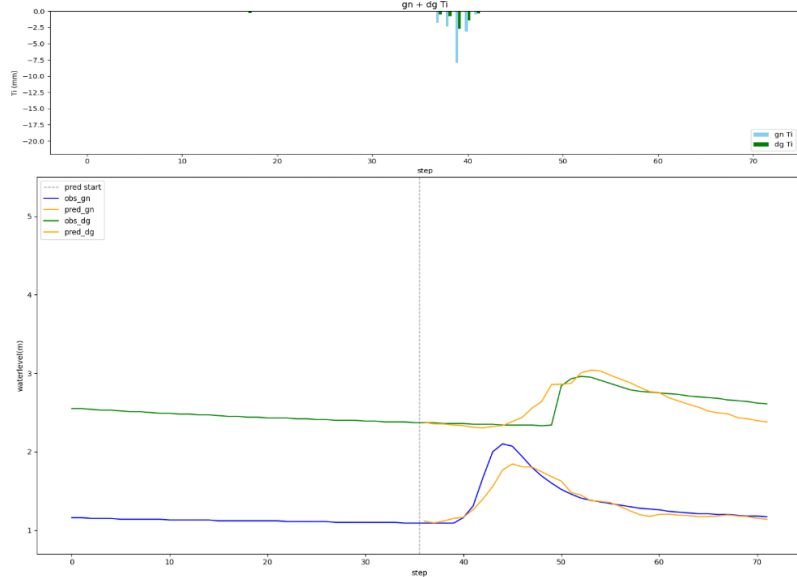
testdata_200802



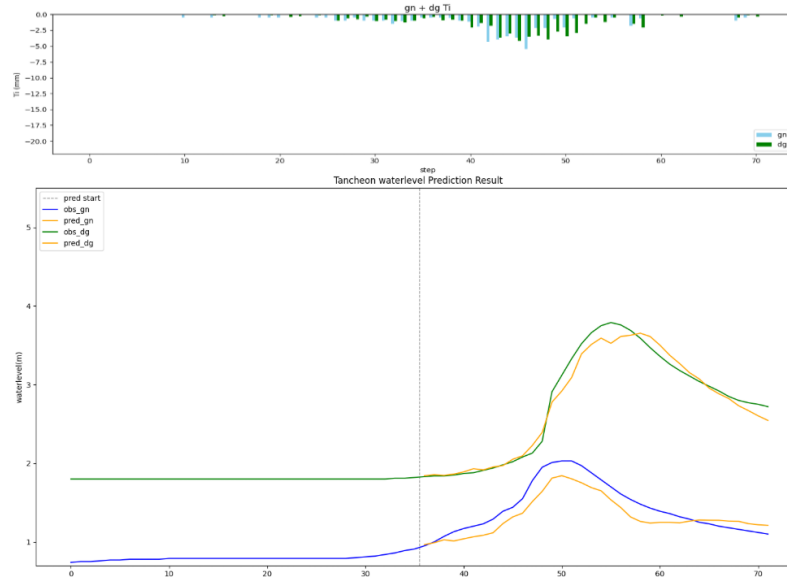
testdata_220712



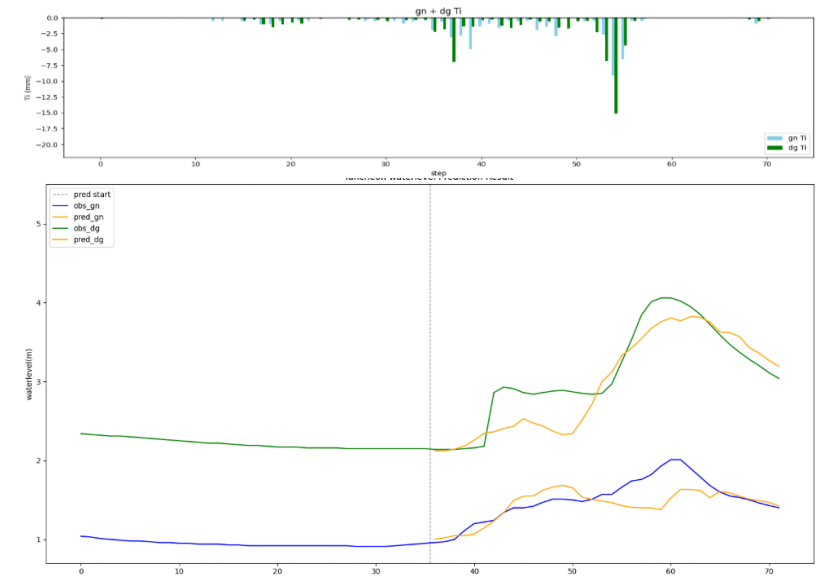
testdata_230711



testdata_240723



testdata_250716



testdata_250814

데이터 편집 (수위 X 1.02)

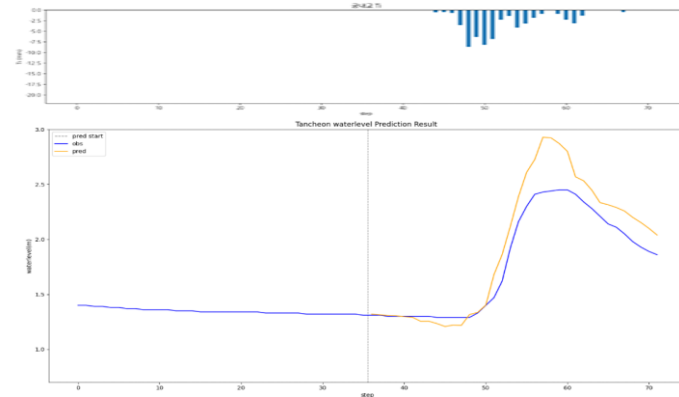
- » 과대보정 → 분산 증가
- » peak 근처에서 불안정

기본 모델 대비 평균 R2 : 궁내교 ▼저하 (- 11.1%) , 대곡교 ▼저하 (- 6.1%)

궁내교					대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.7922	0.0439	0.1592	0.2096	0.8688	0.0800	0.1656	0.2829
22.07.12	0.0156	0.0390	0.1680	0.1976	0.5974	0.0563	0.2138	0.2374
23.07.11	0.8994	0.0281	0.1186	0.1677	0.8086	0.1741	0.3343	0.4173
24.07.23	0.8400	0.0129	0.0691	0.1137	0.4905	0.0242	0.1086	0.1556
25.07.16	0.8186	0.0181	0.1220	0.1348	0.9726	0.0126	0.0917	0.1124
25.08.14	0.4433	0.0392	0.1474	0.1981	0.7639	0.0819	0.2331	0.2862

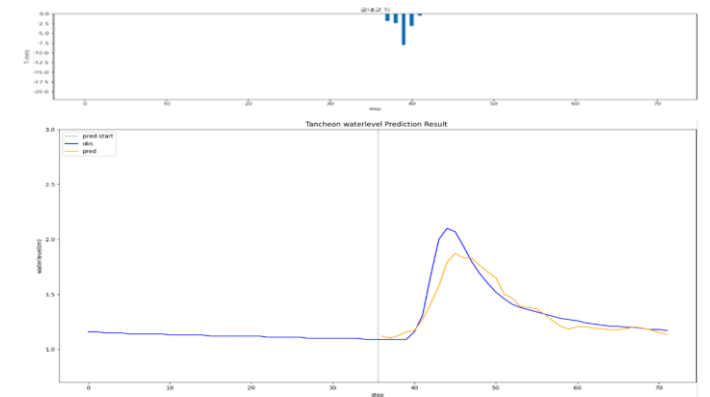
testdata_200802

R2 : 0.7922



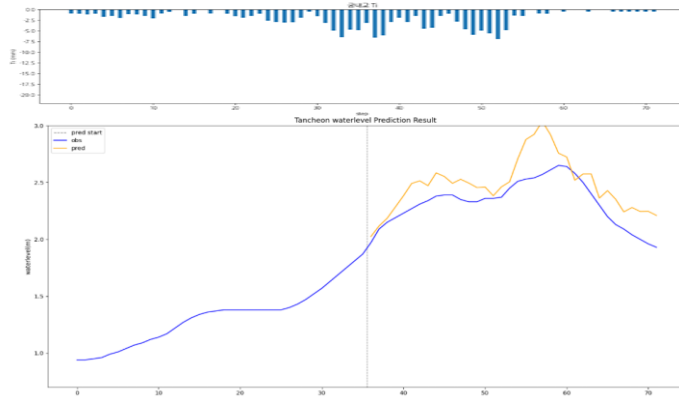
testdata_240723

R2 : 0.8400



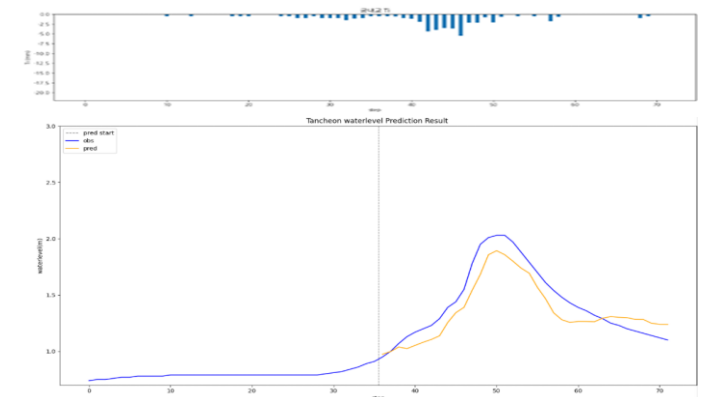
testdata_220712

R2 : 0.0156



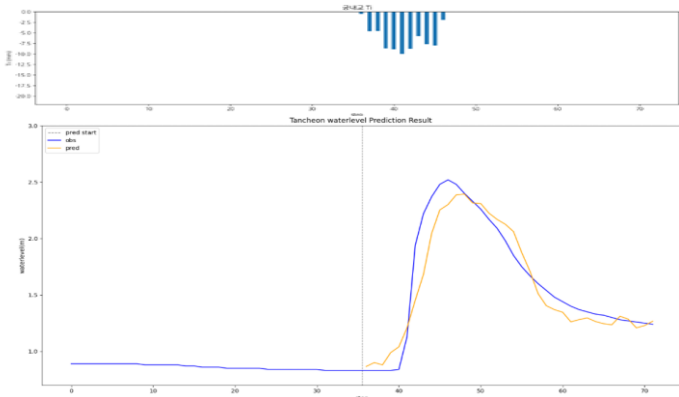
testdata_250716

R2 : 0.8186



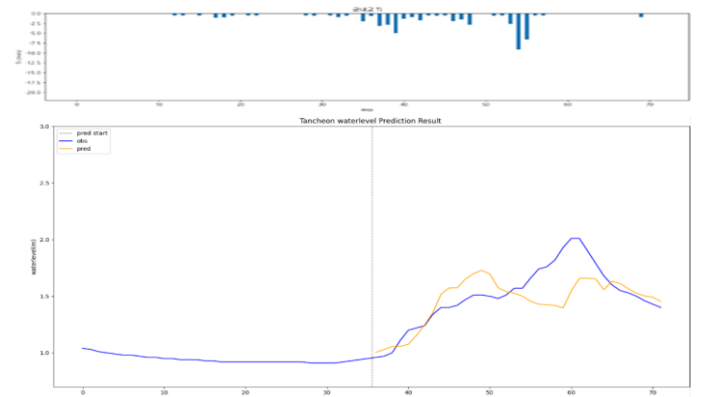
testdata_230711

R2 : 0.8994



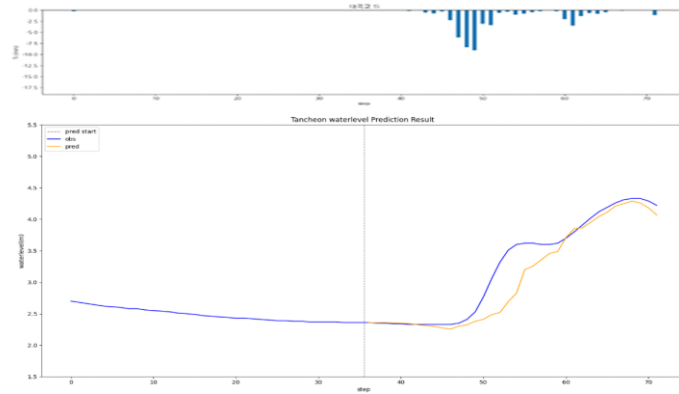
testdata_250814

R2 : 0.4433



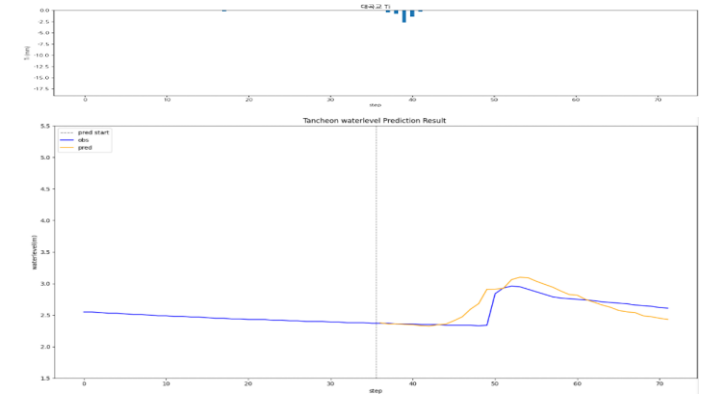
testdata_200802

R2 : 0.8688



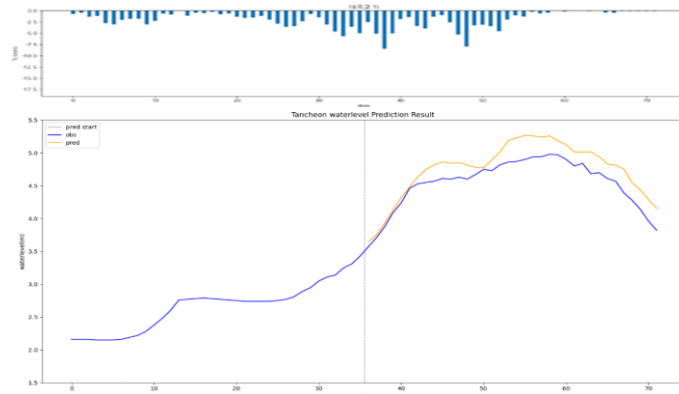
testdata_240723

R2 : 0.4905



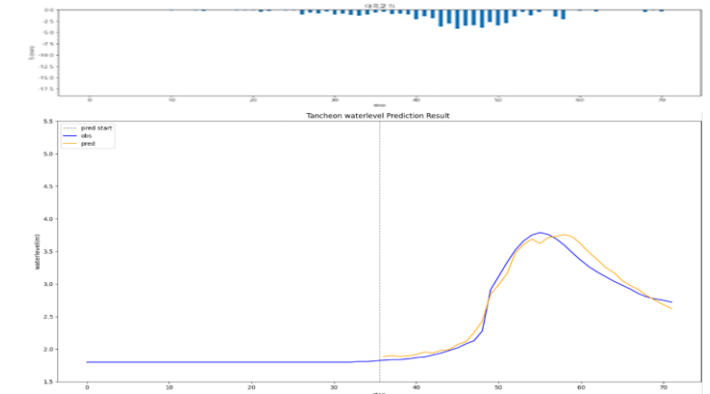
testdata_220712

R2 : 0.5974



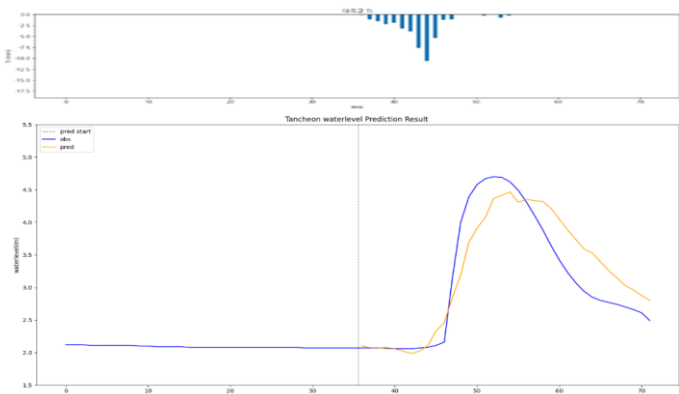
testdata_250716

R2 : 0.9726



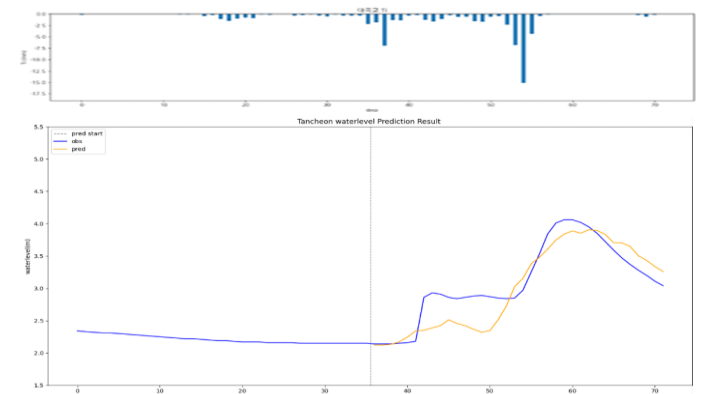
testdata_230711

R2 : 0.8086



testdata_250814

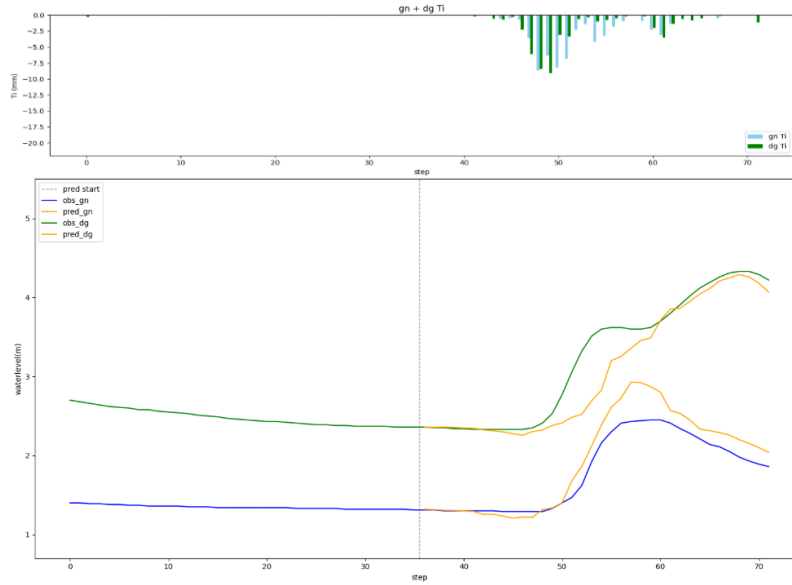
R2 : 0.7639



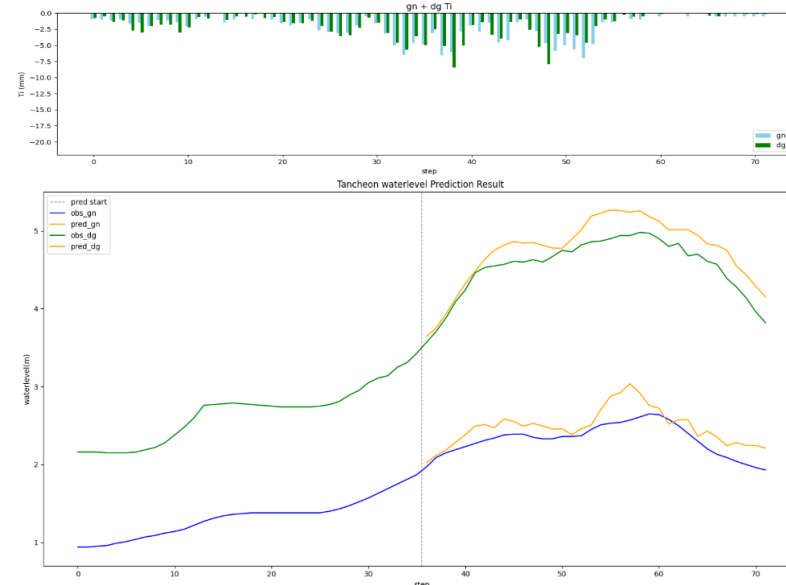
궁내교 + 대곡교

● 궁내교
● 대곡교
● 예측

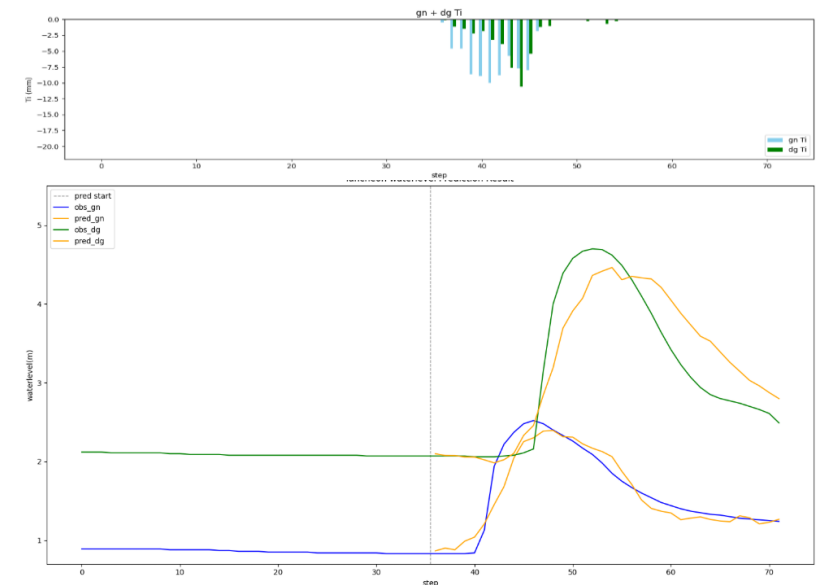
* model_(gn/dg)_1.02



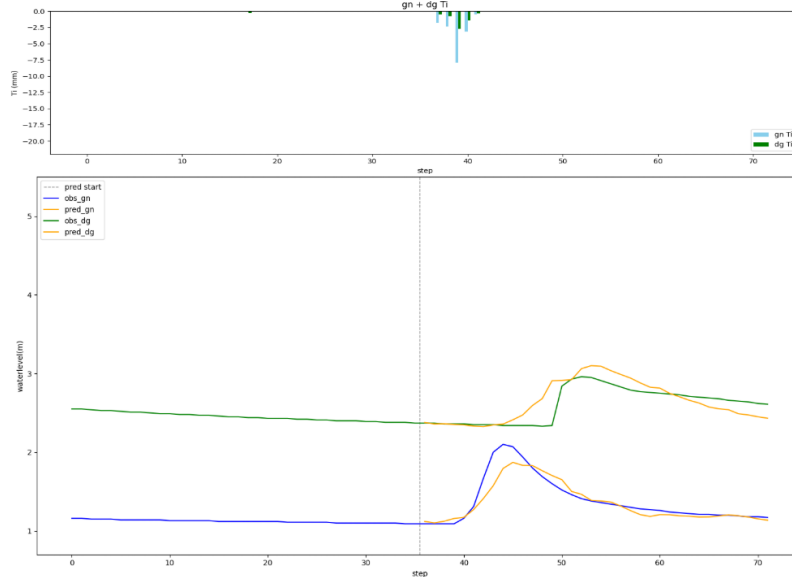
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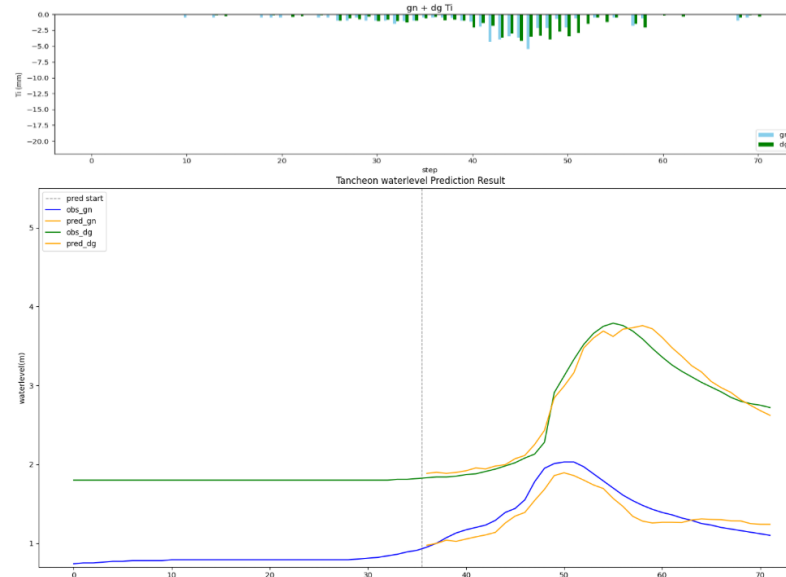
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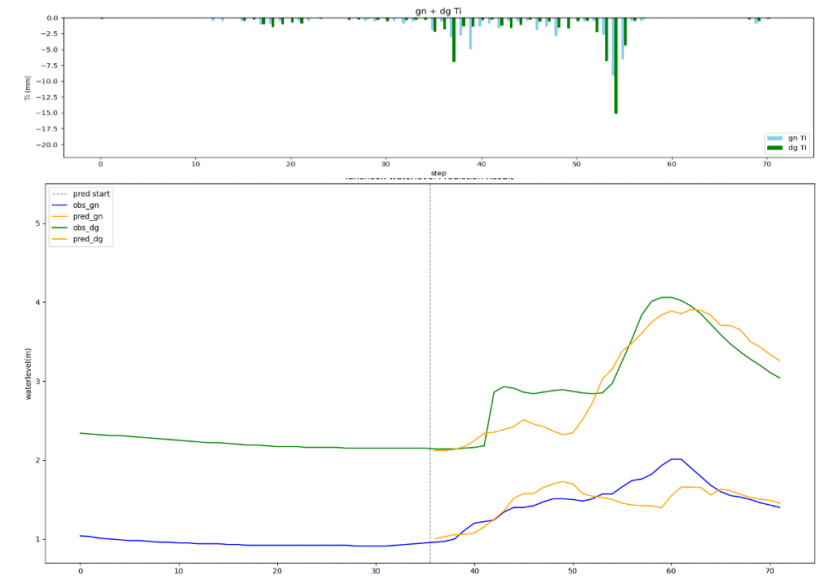
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testdata_240723



testdata_250716



testdata_250814